



M P V P N[®]

User's Manual

Version 10.1.2

• MPVPN User Manual 10.1.2

Table of Contents

Overview.....	5
Chapter 1: Initial Setup.....	13
Quick Install Instructions.....	15
Chapter 2: Interfaces	18
LAN.....	18
WAN.....	22
Chapter 3: System	31
General.....	31
Users.....	34
Configuring LDAP Server to work with FatPipe...	36
Active Directory Services.....	38
Unit Failover.....	41
DHCP Server.....	48
Syslog.....	51
NetFlow.....	54
Auto Configuration.....	55
Maintenance.....	56
Chapter 4: Load Balancing	58
Algorithms.....	58
SmartDNS™	61
Statistics.....	74
TCP Congestion Control.....	80
Site Load Balancing.....	80
Server Load Balancing.....	80
Chapter 5: Routing	81
Application Profile.....	81
Network Objects.....	82
Inbound Policy.....	85
Outbound Policy.....	100
Scheduler.....	107
Global Outbound Policy.....	115
Static Routes.....	115
Quality of Service (QoS).....	117
Global Quality of Service (QoS).....	117
VPN.....	117

MPSec	117
Compression	126
IPv6in4 tunnel	126
IPv6 Static Routes	128
Advanced Options	130
Chapter 6: Tools	132
Speed Chart	132
QoS Statistics	133
MPSec QoS Statistics	133
Diagnostics	133
Server Statistics	138
Session Details	138
Chapter 7: Quality of Service (QoS)	144
Configuration	145
QoS Statistics	147
Layer 7 QoS - Application Level QoS	150
Chapter 8: Site Load Balancing	153
Chapter 9: Server Load Balancing	158
Chapter 10: VPN	169
Chapter 11: VPN Site Failover	185
Chapter 12: WAN Optimization	190
TCP Congestion Control	192
Citrix and RDP Compression	194
MPSec Stats	195
Chapter 13: Web Content Blocking - SatBooster	197
Chapter 14 Central Manager	199
Chapter 15: Paging Software	215
Addendum A	221
Addendum B	229
FatPipe Virtual Appliance	229
Technical Support	236
FatPipe Product Warranty	237
FatPipe Networks End User	241
Software License Agreement	241

Overview

FatPipe® MPVPN is a high-speed router-clustering device from FatPipe Networks. It is the ultimate solution for companies that want the highest levels of WAN redundancy, reliability, and speed for data traffic directed from the network to the Internet as well as data traffic directed to servers hosted internally.

MPVPN aggregates any combination of DS3, T1, E3, E1, DSL, OCN, ISDN, wireless, 3G, 4G, and cable lines. It enables dynamic data transmission over multiple paths for the combined speed of the connections and redundancy, providing you the confidence that your data connectivity is insured regardless of individual router failure. MPVPN works with all existing hardware and applications. No BGP programming is required. FatPipe MPVPN is available with a variety of throughput options.

FatPipe can accommodate small companies and branch offices as a Branch in a Box™ with its lower throughput versions starting at 5 Mbps, as well as enterprise level customers who require speeds up to 40 Gbps.

You can access the User Manual, FatPipe MPVPN's configuration, and the FatPipe website from the configuration interface of MPVPN. The interface also has links to the feature set, sales and support contact information, and Frequently Asked Questions.

Chapter 1: Initial Setup

This chapter provides you with the information required to setup the cable connections and the initial configuration for FatPipe MPVPN. In this chapter you will learn how to:

- Install the MPVPN unit
- Connect MPVPN to your network

Chapter 2: Interfaces

This chapter explains how to setup the necessary networking parameters for FatPipe MPVPN to work with your existing networking environment. In this chapter you will learn how to:

- Setup the IP Address, Subnet Mask, and Default Gateway of each networking interface

-
- Setup IPv6 LAN and WAN settings
 - Configure Ethernet MAC address
 - Set Spillover Priority
 - Activate VLAN
 - Enable or Disable DHCP Relay for LAN connection
 - Configure Weighted Load Balancing (please also refer to Chapter 4: Load Balancing)
 - Enable or disable access to services running on the MPVPN unit
 - Check the status of each WAN connection

Chapter 3: System

This chapter explains how to set general user settings, save a configuration file backup, and establish unit failover. Along with user accounts, date and time, and SNMP settings, you can also choose to configure the high availability option using an additional standby MPVPN unit at your site. This is called Unit Failover. In this chapter you will learn how to:

- Set user privileges and passwords
- Setup Unit Failover between two MPVPN units
- Set system date and time
- Backup and restore the system configuration
- Reset the system configuration to default settings
- Enable SNMP access to MPVPN for monitoring the performance of your network
- Configure the built-in DHCP Server to assign IP addresses to devices on your local area network (LAN)
- Export traffic statistics using NetFlow protocol
- Send event messages to a Syslog Server
- Configure a list containing Hostnames of LAN devices and their respective IP address.

Chapter 4: Load Balancing

MPVPN dynamically load balances inbound and outbound IP traffic for the highest levels of reliability and redundancy of WAN/Internet connections. Use the management interface to setup the appropriate Load Balancing option, Route Test configuration and SmartDNS. In this chapter you will learn how to:

- Choose the appropriate Load Balancing option
- Set your Route Test configuration
- Configure SmartDNS for inbound load balancing and redundancy
- Configure Site Load Balancing
- Configure Server Load Balancing
- Configure TCP Congestion Algorithm (please refer to Chapter 12: WAN Optimization)

Chapter 5: Routing

You can setup and schedule Inbound and Outbound Policies, Static Routes, Quality of Service (QoS) Rules, and enable OSPF. In this chapter you will learn how to:

- Configure Inbound Policy to allow connections to internal servers
- Configure Outbound Policy to specify rules for outbound connections
- Configure Global Outbound Policy (please refer to Chapter 14: Central Manager)

- Configure Static Routes for additional routed subnets
- Schedule Policy Routing Rules for different times and days of the week using the Scheduler
- Configure Quality of Service (QoS) rules for use with Outbound Policy
- Configure Global QoS (Please refer Chapter 14: Central Manager)
- Enable and configure OSPF
- Configure VPN tunnels
- Configure MPSEC for VPN redundancy
- Configure Compression and Caching of different protocols for WAN Optimization (please refer to Chapter 12: WAN Optimization)
- Configure IPv6in4 tunnel
- Configure IPv6 static routes
- Enable Advanced Routing options

Chapter 6: Tools

Use FatPipe MPVPN's remote management interface to monitor the performance of your network. You can check the status of routers and Internet connections using FatPipe MPVPN's Diagnostic Tools and view the speed of connections using the Speed Chart. In this chapter you will learn how to:

- View the WAN's performance by using the Speed Chart
- Check the status of routers and connections using MPVPN's Diagnostic Tools
- View your WAN's performance with System Statistics
- View QoS Statistics for traffic going through MPVPN
- View Server Load Balancing Statistics

Chapter 7: Quality of Service (QoS)

You can optimize the efficiency of your network and prioritized data flow up to 10 levels in relation to priority, latency, and packet loss using FatPipeQoS. It gives you the ability to assign bandwidth parameters to business applications, guaranteeing the minimum quality and bandwidth as you define it. You can also classify packets based on the application they belong to. Application Rules supply the patterns used by the Layer 7 classifier as an extension of outbound Policy Routing rules. It allows the user to classify traffic based on application-specific information regardless of port numbers used by transport protocols. QoS is an add-on feature. Please refer to the contact information in the back of the manual or contact your local FatPipe representative for purchasing information. In this chapter you will learn how to:

- Setup and configure QoS
- Create one or more Application Rules
- Create a MPSecQoS

Chapter 8: Site Load Balancing

MPVPN units can be configured to automatically load balance site traffic to one or more remote sites, where inbound connectivity to Internet accessible servers is critical. Site Load Balancing also allows for Site Failover. This technology utilizes FatPipe Site Load Balancing, which is an add-on feature. Please refer to the contact information on the back of the manual or contact your local FatPipe representative for purchasing information. In this chapter you will learn how to:

- Configure FatPipe Site Load Balancing between two or more units residing at different sites

Chapter 9: Server Load Balancing

Server Load Balancing is a very fast and reliable solution offering high availability, load balancing and proxying for TCP and HTTP-based applications. It is particularly suited for web sites crawling under very high loads while needing persistence or Layer7 processing. Supporting tens of thousands of connections is clearly realistic with today's hardware. Its mode of operation makes its integration into existing architectures very easy and riskless, while offering the possibility not to expose fragile web servers to the Internet. In this chapter you will learn how to:

- Setup and configure Server groups and Servers.

Chapter 10: VPN

MPVPN can be setup as a VPN end-point. FatPipe VPN is an add-on feature. Please refer to the contact information in the back of the manual or contact your local FatPipe representative for purchasing information. In this chapter you will learn how to:

- Setup and configure VPN settings

Chapter 11: VPN Site Failover

MPVPN can be configured to provide failover of VPN tunnels from one site to another. Please refer to the contact information in the back of the manual or contact your local FatPipe representative for purchasing information. In this chapter you will learn how to:

- Setup and configure VPN Site Failover settings

Chapter 12: WAN Optimization

MPVPN units can be configured to automatically perform WAN Optimization on site to site traffic. FatPipe WAN Optimization comprises caching and compression technologies. Caching significantly helps reduce redundant data on the WAN thus making it faster. Along with caching, FatPipe WAN Optimization uses stream based and packet based compression technologies to maximize reduction of redundant or repetitive data and accelerate the overall performance of stream based traffic. WAN Optimization supports caching and stream based compression for HTTP, HTTPS, FTP, CIFS, RDP and ICA (Citrix). Packet based compression can be applied to other stream based protocols for optimization. In this chapter you will learn how to:

- Configure UDP Aggregation
- Configure HTTPs Acceleration

- Setup TCP Congestion Control
- MPSec Stats

Chapter 13: Web Content Blocking-SatBooster

Web Content Blocking is an add-on feature that helps block specific content on web pages. This feature called SatBooster helps companies use Satellite links optimally without compromising on the core content. In this chapter you will learn how to

- Configure Blocking file types
- Configure lossy compression
- Configure Chat Attachment Blocking

Chapter 14: Central Manager

Central Manager allows you to configure and manage all you MPVPN devices from one central location. You can configure and manage your branch MPVPN from a Central Location through our centralized GUI. This Chapter explains how to configure the Central Manager at your Head Quarters so that remote branch units can be managed. Global Policy Routing Rule allows you configure Outbound Policy routing rules for all remote location and apply them. Global Policy Routing rules are created in the central location and then applied to one or all remote locations. In this chapter you will learn how to:

- Configure Central Manager to manage remote MPVPN devices
- Configure Global Outbound Policy
- Configure Global Quality of Service

Chapter 15: Paging Software

FatPipe provides monitoring software that can be used to continuously test the status of your unit. This monitoring software, called Paging Software, will send you an alert if a failure occurs on the WAN. In this chapter you will learn how to:

- Install the Paging Software
- Setup and configure the Paging Software

Chapter 1: Initial Setup

FatPipe MPVPN comes in a 1U, 2U, 4U or desktop form factor. Each form factor has Ethernet interfaces located at the back of the chassis (see Figure 1.1). The LAN interface is used to connect to your LAN. The other interfaces are used to connect to your WAN routers. Each of the Ethernet interfaces must be configured to match the IP addresses of your network by using FatPipe MPVPN's remote management interface, also known as FatPipe MPVPN GUI - Graphical User Interface.

IMPORTANT: PLEASE REFER TO THE PREINSTALL WORKSHEET INCLUDED IN YOUR CUSTOMER PACKET THAT CAME WITH THIS PRODUCT.

IF YOU WANT A FATPIPE TECHNICAL SUPPORT ENGINEER TO ASSIST YOU WITH INSTALLATION, YOU MUST FILL OUT THE PREINSTALL WORKSHEET AT LEAST 72 HOURS PRIOR TO INSTALLATION AND EMAIL IT TO: FATPIPE TECHNICAL SUPPORT AT support@fatpipeinc.com OR CALL (800) 724-8521 Ext:3



Figure 1.1

Unpack MPVPN from its shipping box. You will receive a unit with power cord(s) supplied. (Dual power supply units will have two power cords). To install MPVPN you will need one Ethernet network cable for each interface you will use. You may also need an Ethernet crossover cable to use in between the LAN interface and a computer for initial configuration.

MPVPN can be configured and managed remotely through a browser-based management application. You must use an up-to-date Internet browser with the latest Java Virtual Machine (JVM) installed to access the remote management interface.

Important:

- Internet Explorer should install the JVM automatically. Other browsers may not install the JVM by default. Please make sure your browser has the latest JVM installed. Visit www.sun.com to find information on installing JVM.

- If you will be accessing the remote management interface from behind a firewall, make sure TCP port 5001 is allowed for outbound connections. Also make sure Java applets are allowed through the firewall.

Quick Install Instructions

The following section is a quick overview of the installation process. We recommend that you refer to the rest of the manual for detailed descriptions of the various menu items and screens.

Select a PC on your LAN to configure the MPVPN appliance. This PC will be referred to as the Management PC. Any PC on the LAN can be used to manage the MPVPN appliance once initial configuration is complete.

Connect the MPVPN unit to a UPS outlet. Power the unit on. It takes less than a minute to boot up.

Connect the LAN interface to your local network and the WAN interfaces to your WAN routers. Initial configuration must be done through the LAN interface.

Configure the Management PC with IP address 192.168.0.10, Subnet Mask 255.255.255.0, and Gateway 192.168.0.1.

Point the web browser on your Management PC to <http://192.168.0.1> and this will bring up the initial interface page of MPVPN.

At your first login, enter "Administrator" as the username (it is case-sensitive). The unit ships with no password. Simply click the Login button to authenticate and bring up the remote management interface. Click on System from the main menu and click on the Users tab and select "Administrator" from the user list. Click on the Edit button to set the login password. Be sure to remember this password, as you will not be able to access the MPVPN without it. You may also want to add additional users at this time.

Configure all the active WAN interfaces with IP Address, Subnet Mask, and Default Gateway settings. For more details, see "Chapter 2: Interfaces" in this manual. If any of your WAN IPs are assigned using DHCP or PPPoE, you can select those options instead.

Configure the LAN interface: Click on the Interface button in the main menu. Click on the LAN tab and then the Add button to add a new IP alias. We recommend keeping the default 192.168.0.1 IP address, assuming it does not conflict with anything on your network. Click on

the OK button to return to the LAN page. Click on the SAVE button to save the changes.

At this point your MPVPN unit should be setup for Internet access. All you need to do is set your Default Gateway of your LAN to point to the LAN IP of the MPVPN unit.

The Home Page**This page displays**

Version of the firmware running on the unit,
Serial number of the unit
Licensed throughput
Licensed Add-ons

Helpful Tips:

- Once MPVPN is in place, we recommend that you reboot your routers and firewalls to clear their ARP caches. This will assure proper network communication between MPVPN and your other network devices.
- If you are using public IPs on the LAN side of MPVPN in a pass-through configuration, (see Inbound Policy), it may not be necessary to change your network's Default Gateway. MPVPN uses Proxy ARP to automatically forward packets destined for any of the WAN routers. This makes MPVPN completely transparent to internal devices accessing the Internet.

Chapter 2: Interfaces

The Interface section is where you configure settings for the LAN and WAN interfaces of MPVPN.

LAN

To access and set LAN parameters, click on the Interfaces button in the main menu and click on LAN tab (see Figure 2.1).

Home

Interfaces

LAN

WAN 1

WAN 2

WAN 3

WAN 4

WAN 5

System

Load Balancing

Routing

Tools

LAN

☒ Enable Proxy ARP

Ethernet

MAC [a0:36:9f:54:fb:f4] SET

Link Speed / Duplex Mode

Auto Negotiation

Current Negotiation : NO LINK

DHCP Relay

☐ Enable DHCP Relay

Reporting IP Address : 0.0.0.0

This IP Address is used for sending local syslog and snmp through VPN or GRE tunnel.

IPv4 IPv6

LAN Aliases

IP Address	Subnet Mask
192.168.0.1	255.255.255.0

Add Edit

Figure 2.1 - LAN Interface

Enable Proxy ARP

This will enable or disable Proxy ARP on the LAN side. When this option is enabled, MPVPN will respond to ARP requests for any IPs that belong to any of the WAN subnets. If you disable this option, you will not be able to communicate with devices directly connected to the WAN that are in the same subnet as where you are coming from.

You should only disable Proxy ARP if you have devices on the LAN side that have IPs from one of the WAN subnets. The default option is to have Proxy ARP enabled.

Ethernet MAC Address

This "Set" option allows you to set custom or default Ethernet MAC address of the LAN interface.

Link Speed / Duplex Mode

This option allows you to manually configure Ethernet link speed and the duplex mode. The default value is set to "Auto-negotiation."

VLAN

A Virtual Local Area Network (VLAN) may be defined as a group of LANs that have different physical connections, but which communicate as if they are connected on a single network segment. VLAN is a broadcast domain formed by switches. VLANs allow you to create multiple separated networks with only a single switch.

VLANs increase overall network performance by grouping users and resources that communicate most frequently with each other. To activate VLAN, click on the Active checkbox and enter a Valid VLAN ID (Range 0 to 4096).

Enable DHCP Relay

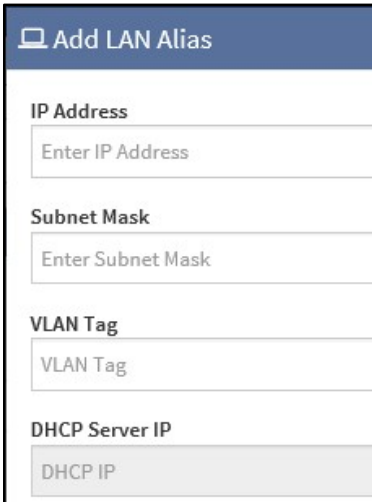
This option allows you to relay DHCP requests from a LAN segment to a DHCP server on the WAN side.

Reporting IP address

The reporting IP address field is used for sending local Syslog and SNMP messages through VPN or GRE tunnel.

IPv4

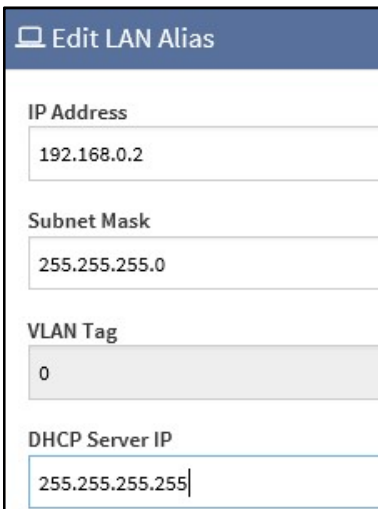
To view the IPv4 LAN configuration, click the IPv4 tab (see Figure 2.1). To add an IPv4 address, click the IPv4 tab and click on the Add button. Specify the IP Address and Subnet Mask for each IP subnet connected to the LAN interface to configure one or more IPs on the LAN interface (see Figure 2.2). Also specify the VLAN ID of each subnet if any.



The screenshot shows a web interface titled "Add LAN Alias" with a blue header bar. Below the header, there are four input fields stacked vertically. The first field is labeled "IP Address" and contains the placeholder text "Enter IP Address". The second field is labeled "Subnet Mask" and contains the placeholder text "Enter Subnet Mask". The third field is labeled "VLAN Tag" and contains the placeholder text "VLAN Tag". The fourth field is labeled "DHCP Server IP" and contains the placeholder text "DHCP IP".

Figure 2.2 – Add IPv4 LAN IP Address, Subnet Mask and VLAN ID

Click on the SAVE button to make the changes permanent. To edit LAN information, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 2.3).



The screenshot shows a web interface titled "Edit LAN Alias" with a blue header bar. Below the header, there are four input fields stacked vertically. The first field is labeled "IP Address" and contains the value "192.168.0.2". The second field is labeled "Subnet Mask" and contains the value "255.255.255.0". The third field is labeled "VLAN Tag" and contains the value "0". The fourth field is labeled "DHCP Server IP" and contains the value "255.255.255.255".

Figure 2.3 – Edit IPv4 LAN IP and Subnet Mask

IPv6

Click the IPv6 tab to view the IPv6 address configuration (see Figure 2.4).

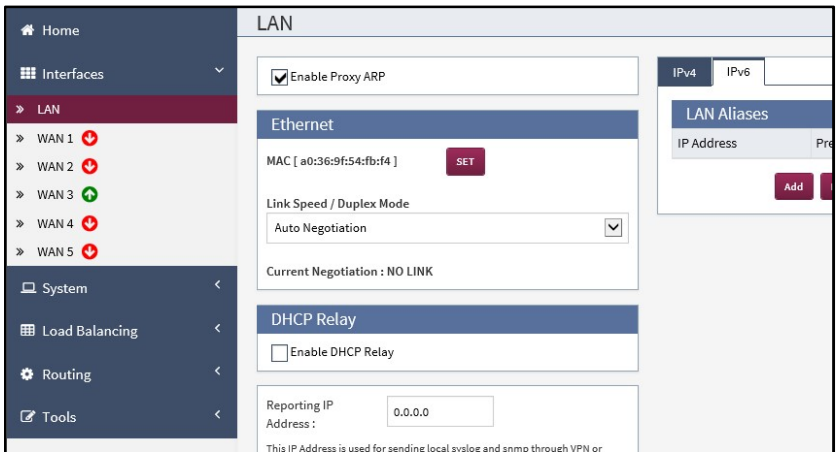


Figure 2.4 - IPv6 LAN Interface

To add an IPv6 address, click the IPv6 tab, and then click the Add button. Specify the Scope, IP and prefix length (see Figure 2.5).

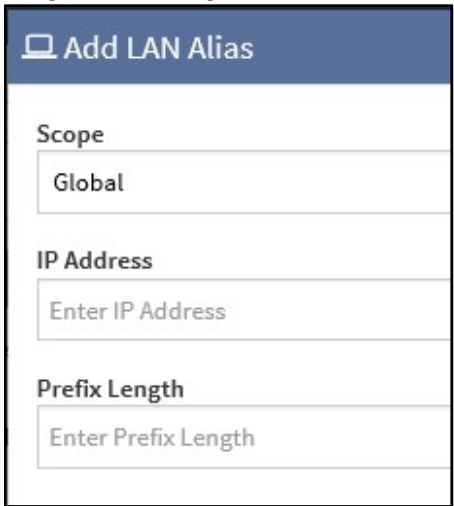


Figure 2.5 - Add IPv6 LAN IP and prefix length

Click on SAVE button to make the changes permanent.

To edit LAN information, select it from the list and click the Edit button, click the SAVE button to make the changes permanent (see Figure 2.6).



Edit LAN Alias	
Scope	Global
IP Address	3001::1
Prefix Length	64

Figure 2.6 - Edit IPv6 LAN IP and prefix length

To delete a LAN IP, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

WAN

To configure each WAN interface in your network, click on Interfaces button in the main menu and click WAN1, WAN2, or WAN3 tab.

ISP Name

Give name of the ISP to the WAN interface

Route Test

When Usage of an interface is set to "Backup," you can select when to perform the route test for that interface. It is set to "Always" by default. This means FatPipe will always check the line for Internet connectivity, even if the line is not actively being used for outbound sessions. If you choose the option "On Primary Failure," then FatPipe will not check for connectivity on that line unless all interfaces with Usage set as Primary are down.

Link Stabilizing Factor

This is the number of consecutive Route Test failures or successes that must occur before Line Status is changed. If the Line Status is UP, the status will change to DOWN only after this number of consecutive Route Test

failures.If the Line Status is DOWN, the status will change to UP only after this number of consecutive Route Test successes.See Chapter 4, Route Test.

Ethernet MAC Address

This "Set" option allows you to set custom or default Ethernet MAC address of the WAN interface.

Link Speed / Duplex Mode

This option allows you to manually configure Ethernet link speed and the duplex mode. The default value is 'Auto-negotiation.'

VLAN

A Virtual Local Area Network (VLAN) may be defined as a group of LANs that have different physical connections, but communicate as if they are connected on a single network segment. VLAN is a broadcast domain formed by switches. VLANs allow you to create multiple separated networks with only a single switch.

VLANs increase overall network performance by grouping users and resources that communicate most frequently with each other. To activate VLAN, click on the Active checkbox and enter a Valid VLAN ID (Range 0 to 4096).

Weight

This setting is for use with the Weighted Load balancing algorithm. Values configured here will be assigned as the Weight for that WAN interface.

Spillover Priority Level

Spillover priority level allows you to assign different priorities to WAN connections to prevent line saturation. Traffic is sent over the lines with the highest priorities set by you. Traffic is sent over the lower priority lines only after at least 90% throughput of higher priority lines is reached. You have the option of marking a line as 'backup'. Traffic will be sent out of a 'backup' link only if all the other links are down. '1' has the highest priority and decreases as the numeric value increases, depending on the number of WAN interfaces you have.

This algorithm provides a solution for users that are charged for line usage that is proportionate to the traffic they generate. You will normally want to use this type of feature as a backup for when your network is carrying a high load. By assigning lower priority to such a line, you will achieve optimal usage and minimize cost.

Services

FatPipe is a secure system with most services disabled except those needed to provide Remote Management, SSH, DNS, IPSEC, SNMP and Site Load Balancing. Although these services present minimal risk, you can enable or disable these features as desired. You can block Ping (ICMP ECHO) requests to the WAN interface IP. These options do not affect traffic routed through MPVPN.

Watch Parameters - When enabled, FatPipe monitors the link conditions like latency, jitter and packet loss and allows redirecting traffic to alternate links if a pre-defined threshold is crossed even if the link is UP. This is achieved by configuring the thresholds using Outbound Policy Routing Rules. FatPipe pings a pre-defined IP (default is 8.8.8.8 but can be changed by FatPipe Support Team) to determine the above factors.

Bandwidth (kbps)

Upload - This setting is for use with Quality of Service (QoS). You should specify the maximum outbound bandwidth available for your WAN line in Kbps (Kilobits per second). For example, if you have 1.5Mbps of bandwidth outbound, you would enter 1500.

Download - This setting is for use with Quality of Service (QoS). You should specify the maximum bandwidth available inbound for your WAN line in Kbps (Kilobits per second). For example, if you have 1.5Mbps of bandwidth inbound, you would enter 1500.

Enable Bridging with LAN

In situations where we cannot split a network to create a separate small transport subnet, this option enables you to bridge the LAN with the WAN interface of that network.

WAN IP List

Defines all the subnets that are on the WAN side of the bridged interface, this helps the FatPipe route traffic accordingly.

IPv4 WAN Settings

Select "Obtain an IP address automatically using DHCP" to have WAN IP settings assigned dynamically by a DHCP server (see Figure 2.7).

To connect to your ISP using PPPoE, select "Connect using PPPoE" (see Figure 2.8 and Figure 2.9).

To connect a 3G/4G line, plug a 3G USB Modem to any of the USB interfaces on the MPVPN device. The USB Modem will be automatically detected. Select "Connect using 3G/4G device."

Choose the device model from the "Detected 3G/4G USB Modem" Dropdown. IMEI/ESN and Model Name of the USB Modem will be displayed. This information cannot be modified. The APN and Phone Number will also be displayed. This information can be modified. Click SAVE to make the changes permanent. (See Figure 2.10) Select "Specify an IP address" to assign IP Address, Subnet Mask, and Default Gateway settings to each WAN interface. The Default Gateway is typically the IP address of your WAN router (see Figure 2.11).

The screenshot shows the configuration page for WAN 3. The left sidebar contains a navigation menu with options: Home, Interfaces (selected), LAN, WAN 1, WAN 2, WAN 3 (active), WAN 4, WAN 5, System, Load Balancing, Routing, Tools, and EnterpriseView. The main content area is titled 'WAN 3' and includes a 'Site:' dropdown. The 'Line Status' is 'UP' with a green arrow. Below this, there are fields for 'ISP Name' (ABCDE) and 'ISP Notes' (Title). The 'WAN IP Settings' section has radio buttons for 'Obtain an IP address automatically using DHCP' (selected), 'Connect using PPPoE', 'Connect using 3G / 4G device', and 'Specify an IP address'. Under 'Specify an IP address', there are tabs for 'IPv4' and 'IPv6'. The 'IPv4' tab is active, showing fields for 'IP Address', 'Subnet Mask', and 'Default Gateway'. Below these are 'Bandwidth (Kbps)' settings for 'Upload' (50000) and 'Download' (50000), and 'Line Type' settings with 'Public' selected. At the bottom, there is a 'Services' section. On the right side, there are three panels: 'Route Test' with 'Perform' set to 'Always' and 'Link Stabilizing Factor' set to '1'; 'Ethernet' with 'MAC' [30:85:a9:a7:56:92] and 'Current Negotiation: 1000baseTX-FD'; and 'VLAN' with 'Enable' unchecked and 'ID' set to '0'. At the bottom right, there is a 'Type' section with 'Weight' set to '1' and 'Usage' set to 'Primary'.

Figure 2.7 – Connecting automatically using DHCP

The screenshot shows the configuration page for WAN 3, similar to Figure 2.7, but with different settings. The 'WAN IP Settings' section has 'Connect using PPPoE' selected. Under 'Dynamic PPPoE', there are fields for 'Login' (Admin), 'Password' (****), 'Local IP' (10.0.0.31), and 'Remote IP' (10.0.0.254). The 'Static PPPoE' section is also visible. The 'Line Type' settings are the same. The 'Services' section at the bottom has checkboxes for 'Ping' (checked), 'Remote Management' (checked), 'DNS' (unchecked), 'IPSEC' (checked), 'SNMP' (unchecked), 'Site Load Balancing' (unchecked), 'SSH' (checked), and 'WAN Metrics*' (unchecked). On the right side, the 'Route Test' panel has 'Link Stabilizing Factor' set to '1' and 'Link Stabilizing Factor' set to 'Down'. The 'Ethernet' panel has 'Link Speed / Duplex Mode' set to 'Auto Negotiation'. The 'VLAN' panel has 'Enable' unchecked and 'ID' set to '0'. The 'Type' panel has 'Weight' set to '1', 'Usage' set to 'Primary', and 'Spillover Priority' set to '1'. At the bottom right, there are 'Save' and 'Refresh' buttons.

Figure 2.8 – Connecting using Dynamic PPPoE

The screenshot shows the WAN 3 configuration page. The 'Line Status' is 'UP'. Under 'WAN IP Settings', 'Obtain an IP address automatically using DHCP' is selected. Under 'Services', 'Ping', 'Remote Management', 'DNS', and 'IPSEC' are checked. The 'Route Test' section shows 'Perform' set to 'Always'. The 'Ethernet' section shows 'MAC' as '30-83-00-00-00-00' and 'Link Speed / Duplex Mode' set to 'Auto Negotiation'. The 'VLAN' section has 'Enable' checked and 'ID' set to '0'. The 'Type' section has 'Weight' set to '1', 'Usage' set to 'Primary', and 'Spillover Priority' set to '1'. At the bottom, there are 'Save' and 'Refresh' buttons.

Home Interfaces

LAN

WAN 1

WAN 2

WAN 3

WAN 4

WAN 5

System

Load Balancing

Routing

Tools

WAN 3

Line Status: UP

ISP Name: ABCDE ISP Notes: Title

WAN IP Settings

Obtain an IP address automatically using DHCP

Connect using PPPoE

Connect using 3G / 4G device

Specify an IP address

IPv4 IPv6

Dynamic PPPoE Login

Static PPPoE Local IP

Admin 11.11.11.20

Password 11.11.11.20 Remote IP 11.11.11.20

Bandwidth (Kbps)

Upload: 5000 Download: 5000

Services

Ping Remote Management DNS IPSEC

SNMP Site Load Balancing SSH WAN Metrics*

Route Test

Perform: Always Link Stabilizing Factor: 1 Link Stabilizing Factor Down: 1

Ethernet

MAC: 30-83-00-00-00-00 Link Speed / Duplex Mode: Auto Negotiation

VLAN

Enable ID: 0

Enable Bridging with LAN**

Type

Weight: 1 Usage: Primary Spillover Priority: 1

Save Refresh

*A public IP is pinged to identify Latency, jitter & Packet Loss on that link.
**You will need to clear ARP on connected devices when enabling/disabling bridge mode.

Figure 2.9 – Connecting using Static PPPoE

The screenshot shows the WAN 3 configuration page. The 'Line Status' is 'UP'. Under 'WAN IP Settings', 'Obtain an IP address automatically using DHCP' is selected. Under 'Services', 'Ping', 'Remote Management', 'DNS', and 'IPSEC' are checked. The 'Route Test' section shows 'Perform' set to 'Always'. The 'Ethernet' section shows 'MAC' as '30-83-00-00-00-00' and 'Link Speed / Duplex Mode' set to 'Auto Negotiation'. The 'VLAN' section has 'Enable' checked and 'ID' set to '0'. The 'Type' section has 'Weight' set to '1', 'Usage' set to 'Primary', and 'Spillover Priority' set to '1'. At the bottom, there are 'Save' and 'Refresh' buttons.

Home Interfaces

LAN

WAN 1

WAN 2

WAN 3

WAN 4

WAN 5

System

Load Balancing

Routing

Tools

WAN 3

Line Status: UP

ISP Name: ABCDE ISP Notes: Title

WAN IP Settings

Obtain an IP address automatically using DHCP

Connect using PPPoE

Connect using 3G / 4G device

Specify an IP address

IPv4 IPv6

IP Address Subnet Mask Default Gateway

Bandwidth (Kbps)

Upload: 5000 Download: 5000

Services

Ping Remote Management DNS IPSEC

SNMP Site Load Balancing SSH WAN Metrics*

Route Test

Perform: Always Link Stabilizing Factor: 1 Link Stabilizing Factor Down: 1

Ethernet

MAC: 30-83-00-00-00-00 Link Speed / Duplex Mode: Auto Negotiation

VLAN

Enable ID: 0

Enable Bridging with LAN**

Type

Weight: 1 Usage: Primary Spillover Priority: 1

Save Refresh

*A public IP is pinged to identify Latency, jitter & Packet Loss on that link.
**You will need to clear ARP on connected devices when enabling/disabling bridge mode.

Figure 2.10 – Connect using 3G/4G device

Figure 2.11 - Specify IP Address

Note:Line Status will indicate UP when the WAN connection is functioning and available for data communication. Line Status will read DOWN when the WAN connection is unavailable.

IPv6 WAN Settings

By default the “Obtain an IP address automatically using DHCP” and “Connect using PPPoE” options are disabled for IPv6 WAN settings. Click on IPv6 tab to assign IPv6 Address, prefix length, and Default Gateway settings to each WAN interface. The Default Gateway is typically the IPv6 address of your WAN router (see Figure 2.12 and Figure 2.13).

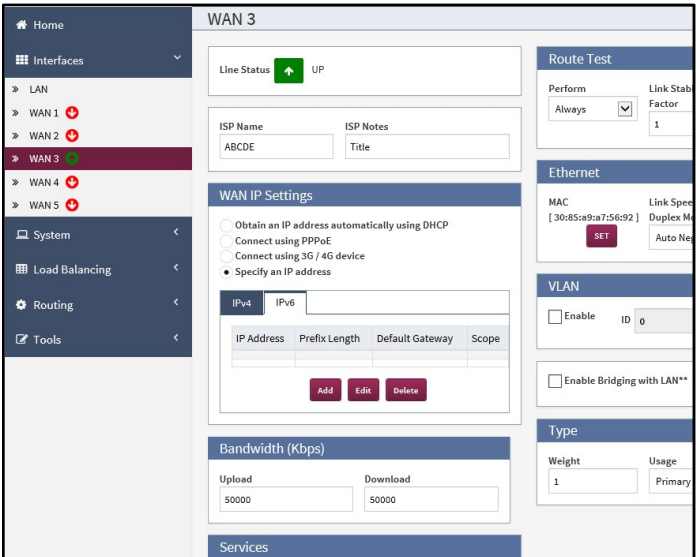


Figure 2.12 – Specifying IPv6 Address

Add WAN IP

Scope

Global

IP Address

Enter IP Address

Prefix Length

Enter Prefix Length

Default Gateway

Enter Default Gateway

Figure 2.13 – Add IPv6 WAN IP and prefix length

Enable Bridging with LAN

In situations where we cannot split a network to create a separate small subnet, this option enables you to bridge the LAN with the WAN interface of that network.

WAN IP List

Defines all the subnets that are on the WAN side of the bridged interface, this helps the FatPipe route traffic accordingly.

Chapter 3: System

This section allows you to configure basic parameters of your MPVPN unit. Under the System menu, you can setup failover between multiple MPVPN units at the same location (Unit Failover). The System section is also where you can set user privileges and user passwords.

General

To configure system settings click on System in the main menu and click on the General tab (see Figure 3.1). You can set a Host Name and Domain Name to identify the system.

The screenshot displays the 'General' configuration page. On the left is a dark sidebar menu with options: Home, Interfaces, System, General (selected), Users, Active Directory Services, Unit Failover, SNMP, DHCP Server, Syslog, NetFlow, HostName, Static ARP, Auto Configuration, Maintenance, and Load Balancing. The main content area is titled 'General' and contains several configuration sections: 'Host Name' (text box with 'host'), 'Domain Name' (text box with 'domain'), 'Session Timeout' (text box with '120'), 'TCP Timeout (min)' (text box with '3'), 'UDP Timeout (min)' (text box with '3'), and 'Login Banner' (empty text box). On the right side, there are three additional panels: 'Date / Time Properties' with a date picker set to 12/21/2016, a 'Use NTP' checkbox, and a 'Time Zone' dropdown set to '(GMT-07:00) Mountain Time (US & C...); 'Backup and Restore' with 'Backup Settings*' and 'Restore Se...' buttons; and 'ARP' with 'View ARP Table' and 'Clear ARP Ta...' buttons.

Figure 3.1 – General settings

Date and Time Properties

You can set the date, time and time zone for the system (see Figure 3.2, Figure 3.3).

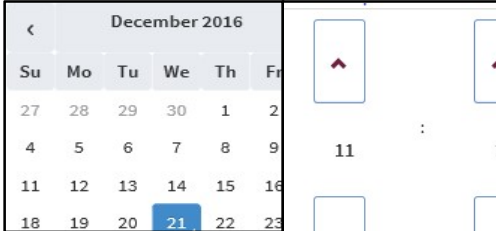


Figure 3.2 - Set date

You can set date and time using the NTP. Check the Use NTP checkbox and click the Set button to synchronize with external time servers (see Figure 3.4).

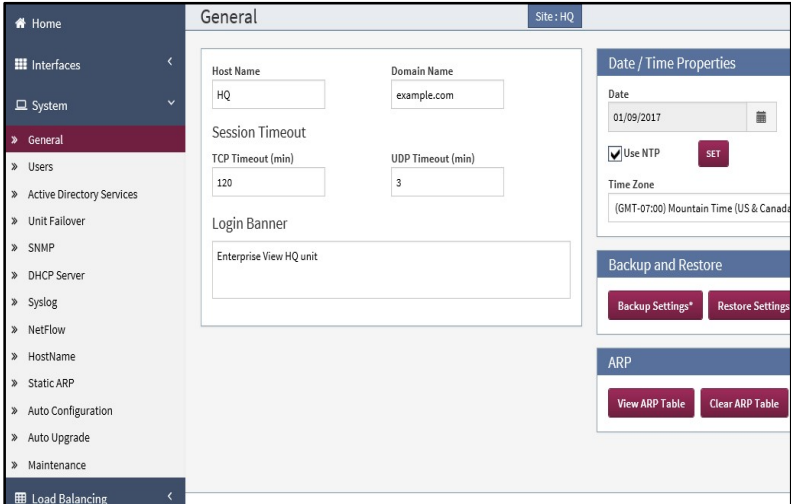


Figure 3.4 - Set the date and time using a NTP server

Uncheck "Use Custom Time Server" to use the default time servers. If you want to use a different set of time servers other than the default ones, check "Use Custom Time Server" and add the time server (see Figure 3.5).

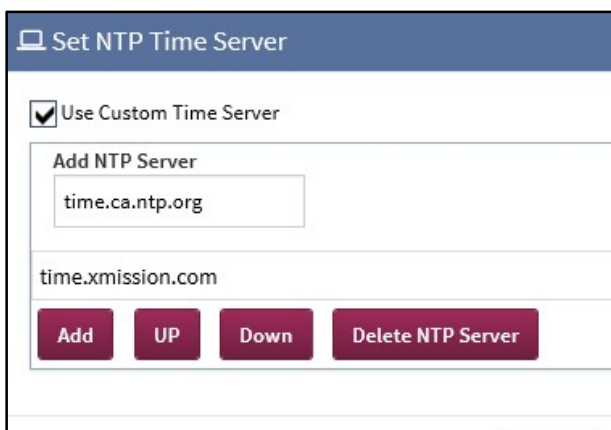


Figure 3.5 - Set the NTP Time Server

Session Timeouts

You can specify TCP and UDP idle timeouts for connections routed through MPVPN. The defaults are 120 minutes (2 hours) for TCP and 3 minutes for UDP. It is not recommended that you change these settings, except under rare circumstances.

Backup and Restore

You can backup or restore configuration settings. If you click on the Backup Settings button you will be prompted to save a backup configuration file in a new popup window. If you click on the Restore Settings button, in a new popup window you will be prompted to import a previously saved backup configuration file. If you click on the Restore Defaults button, you will be prompted to restore the system back to factory defaults.

View ARP Table

Use to view the ARP Table of the FatPipe MPVPN Unit(See Figure 3.1).

Clear ARP

Use this to clear the system's ARP cache.

Login Banner

You can specify a message that will be displayed on the Remote Configuration login page.

Users

To manage user accounts, click on System button in the main menu and click the Users tab (see Figure 3.6).

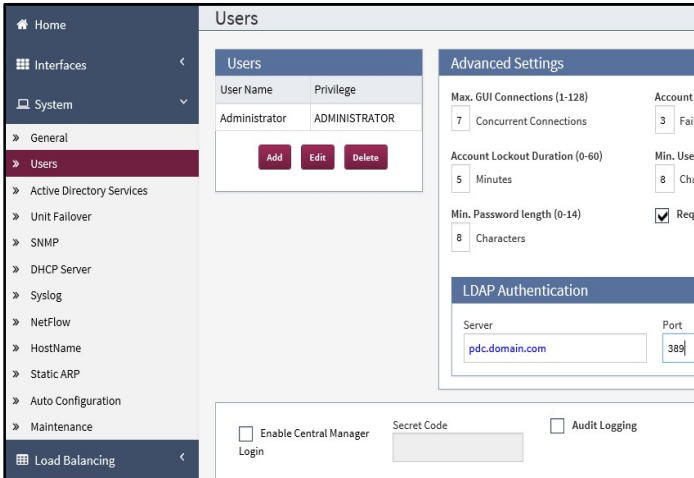


Figure 3.6 - List of Users

Click on the Add button and specify the username, password and set the privileges (see Figure 3.7).

The screenshot shows the 'Add User' form. It has a title bar with a user icon and the text 'Add User'. The form contains several sections: 'User Name' with a text input field containing 'testuser'; 'Set Privilege' with two radio buttons, 'User' (selected) and 'Administrator'; 'Set Password' with a 'Password' field and a 'Confirm Password' field, both containing masked characters (dots).

Figure 3.7 - Add New User Account

Click on SAVE button to make changes permanent.

Note: Users with Administrator privileges are allowed to make changes to user accounts.

Advanced Settings

You can specify advanced settings that are applied to all new logins and account creations. This is made up of the following policies: (see Figure 3.6)

Maximum GUI Connections

Sets the limit on the number of concurrent connections that are allowed to the remote management interface.

Account Lockout Threshold

Specifies the number of failed login attempts allowed before locking out the user.

Account Lockout Duration

Specifies the number of minutes before a user can attempt to login again after being locked out.

Minimum User Name Length

Specifies the minimum number of characters required for usernames for new user accounts.

Minimum Password Length

Specifies the minimum number of characters required for passwords for new user accounts.

Note: Configuring a value of zero allows you to create user accounts without any password.

Require Mixed Passwords

Will enable complex password checking. Passwords for new user accounts must contain a mix of letters, numbers, and special characters when this is enabled.

Enable Central Manager Login

This provides access to the Central Manager Software. The FatPipe Central Manager is a software tool used separately to manage multiple FatPipe boxes via one interface. Contact your account manager for more information.

To edit user information, select it from the list and click the Edit button, click on SAVE button to make the changes permanent (see Figure 3.8).

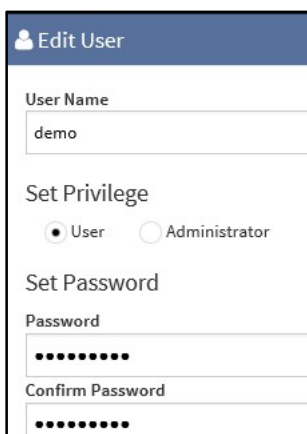


Figure 3.8 - Edit Username or Password

To delete a user account, select it from the list and click the Delete button, click on SAVE button to make changes permanent.

Configuring LDAP Server to work with FatPipe

To utilize the LDAP authentication service on the FatPipe, check the LDAP check-box on the GUI login screen and enter your LDAP username and password.

To specify the address of the LDAP server, open the FatPipe GUI and go to the System page, and then click on the Users tab. At the bottom right corner, there are two text-boxes for the LDAP server configuration. Enter the LDAP server's IP address in the Server text box. Port information is optional. If you leave it blank, it defaults to port 389. However, if you use a different port number, you will need to change the port number in the FatPipe GUI (see Figure 3.6).

Secure Encrypted Connection

FatPipe uses Transport Layer Security technology to ensure confidential data exchange between the LDAP server and the FatPipe in order to protect sensitive information, including the usernames, passwords and privileges. FatPipe offers two secure protocols: TLS and SSLv3. TLS is an upgraded version of SSLv3. TLS is used by default. The SSLv3 is only used when TLS is not functional on the server.

To use either of the two, the server (LDAP) must be configured with a valid certificate (RSA -public key cryptography).

Note: The LDAP server needs to be configured so it does not verify authenticity of the FatPipe.

By default, TLS works on the same port as the non-encrypted or non-secure connection allowed by the LDAP. This port is generally 389 (reserved) and this is the port used by both non-encrypted (unsecured) and secure (TLS) connections. However, if the server does not support TLS, FatPipe creates a secure connection using SSLv3 protocol. The port used by this protocol is 636 and the LDAP server should be configured to listen to this port.

Setting User Privileges

In order for a user with a LDAP account to be authenticated to theFatPipe, the user's LDAP record must contain the attribute "FatPipeUser". Allowed values for this attribute are "Administrator" and "User", which correspond to the two levels of user privileges in the FatPipe GUI.

Distinguished Name of the LDAP Server

An LDAP server contains a directory tree, which reflects various geographic, and/or organizational boundaries. LDAP deployments today tend to use Domain Name System (DNS) names for structuring the top-most levels of the hierarchy. Deeper inside the directory might appear entries representing people, organizational units, printers, documents, groups of people, or anything else that represents a given tree entry (or multiple entries). The root of the directory tree has a special name by which LDAP queries it and known as "Distinguished Name" or dn.

The LDAP server configured must have a distinguished name or dn set in "dc=" format where "dc" means "domain component." It is required by SSL for securing a connection.

The Fully Qualified Distinguished Name or FQDN of any server is generally something of this form "example.com". So the dn is 'dn="dc=example, dc=com"'. This just has to be set, and it can be any name.

The LDAP will search for the requesting user from this search base, or if you have multiple search bases, the first dn with a valid "dc=" statement will be searched. The chart below shows a very common example of an LDAP hierarchy (see Figure 3.9).

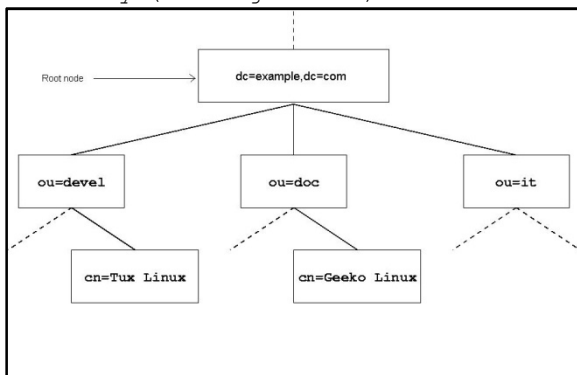


Figure 3.9 - LDAP hierarchical chart

Server

Enter the server host name or IP address.

Port

Enter the port number. If the LDAP server is specified, the default port number is 389.

Active Directory Services

ADS service on FatPipe can be used to make work easy for Administrators and users of FatPipe's. Instead of creating a separate user name and password, administrators can provide rights to a user for logging into the FatPipe using their default sign on for their Computer.

To configure ADS, click on System button in the main menu and select Active Directory Services tab. (see Figure 3.10)

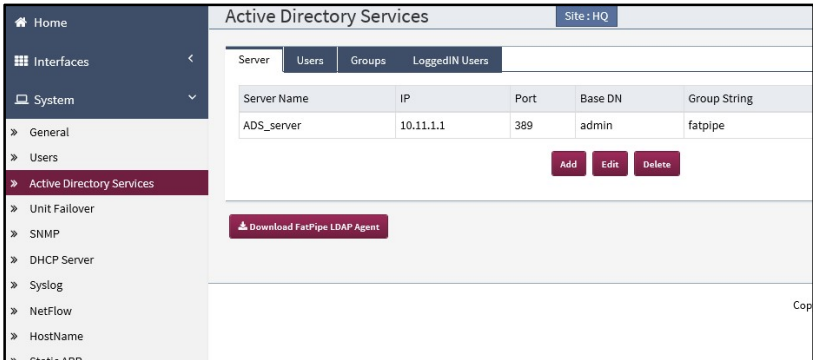


Figure 3.10 – Active Directory Services

To add information about ADS server to FatPipe, select Add button and a window will pop out to put the details. (see Figure 3.11) Click on save button to make the changes permanent.

The 'Add ADS Server' dialog box contains the following fields and options: Server Name (ADS_server), Server IP (10.11.1.1), Server Port (389), Base DN (admin), Group String (fatpipe), an unchecked 'Auto deploy' checkbox, and a checked 'Use FatPipe Window' checkbox.

Figure 3.11 – Add ADS server details

Server Name

Add name you wish to assign to the ADS server on FatPipe.

Server Port

Port number ADS server is on.

Group String

Any name given to the group using FatPipe.

Server IP

IP address of the ADS server.

Base DN

Place where server will search for users

Use FatPipe Windows API

Select this option to tell the administrator about logged in users.

To edit any details in ADS server information, select it from the list and hit on Edit button (see Figure 3.12), click on save button to make the changes permanent.

Server Name ADS_server	Server IP 10.11.1.1
Server Port 389	Base DN admin
Group String fatpipe	☐ Auto deploy ☒ Use FatPipe Window

Figure 3.12 - Edit ADS server details

Users

To Display the list of all the users pointed to access FatPipe, select Users tab in Active Directory Services. (see Figure 3.13) This displays user names and the group in ADS they belong to.

Home Interfaces System » General » Users » Active Directory Services » Unit Failover » SNMP » DHCP Server » Syslog » NetFlow » HostName	Active Directory Services Site: HQ	
	Server Users Groups LoggedIN Users	
	User Name ADS Groups	
	Update	
	Download FatPipe LDAP Agent	
	Copyright	

Figure 3.13 - List of Users in ADS

Groups

To get the list of different groups that can accessFatPipe. Select Active Directory Services from System button and click on Groups tab to have them displayed (see Figure 3.14)

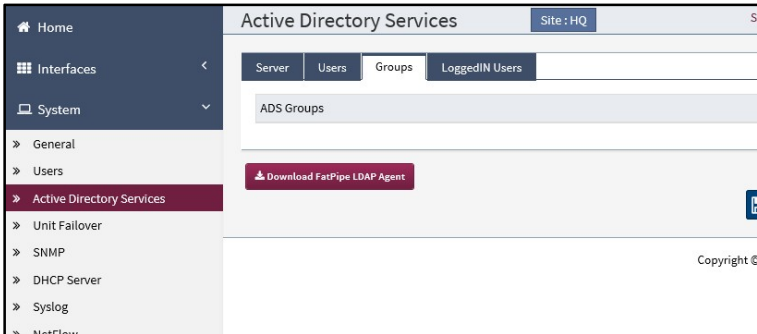


Figure 3.14 - Groups in ADS

Logged IN Users

The number of users that are logged into FatPipe will be displayed here. User Name, IP address, Logged in Time and Date will be displayed (see Figure 3.15). This will make it easy for the administrator to track who is logging into the FatPipe.

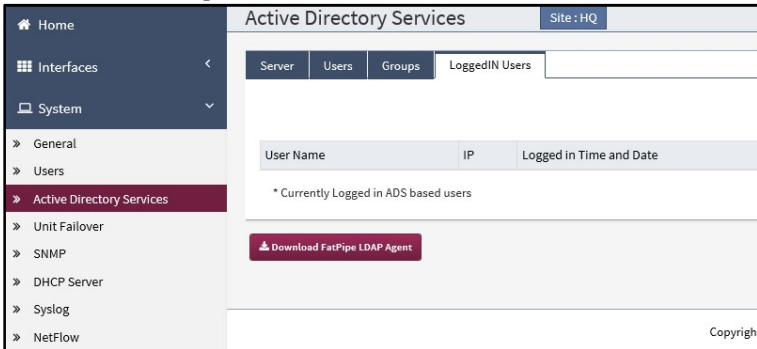


Figure 3.15 - Logged in Users in ADS

Download FatPipe LDAP Agent

FatPipe LDAP agent .exe file is used to gather information from the ADS server. Select the System button, choose Active Directory Services and click on Download FatPipe LDAP Agent. This will download .exe file in the computer. Run it and follow steps to get the ADS list of users from server.

Unit Failover

To configure MPVPN units to automatically failover in case of hardware failure, click on System button in the main menu and click the Unit Failover tab (see Figure 3.16). This helps to maintain a reliable and redundant

connection to the Internet. At least two units are required to implement Unit Failover. At any given time, one will be in an Active state and the other will be in a Standby state. Only the Active unit will route traffic.

The screenshot shows the 'Unit Failover' configuration page. On the left is a sidebar menu with options: Home, Interfaces, System, General, Users, Active Directory Services, Unit Failover (selected), SNMP, DHCP Server, Syslog, NetFlow, HostName, Static ARP, Auto Configuration, Maintenance, Load Balancing, Routing, and Tools. The main content area is titled 'Unit Failover' and contains several sections:

- Enable Unit Failover:** A checkbox labeled 'Enable Unit Failover' is checked, and 'Stateful Failover' is unchecked.
- Local Unit:** A section with 'Group ID' set to 10 and 'Access IP/Mask' set to 192.168.10.1.
- Peer Unit:** A section with 'Access IP' set to 192.168.10.2.
- Heartbeat:** A section with radio buttons for 'Ethernet' (selected) and 'Serial', and a dropdown menu for 'Interface' set to 'eth6'.
- Role:** A section with radio buttons for 'Primary' (selected) and 'Backup'.
- Email Alert Settings:** A section with fields for 'Sender e-mail' (admin@fatpipeinc.com), 'Receiver e-mail' (sales@fatpipeinc.com), 'SMTP Server' (23.68.92.12), and 'Port' (25), with a 'Send Mail' button.
- Peer Units:** A table with columns 'IP Address' and 'Serial Number'.

Figure 3.16 – Unit Failover

Initial Setup

The physical setup consists of splitting the Ethernet connections from each router to the corresponding WAN interfaces of the two MPVPN units. This will require the use of a separate switch (or hub) for each router. For example, to setup the hardware for WAN1, you would connect a cable from the router to a switch, and then connect a cable from each of the WAN1 interfaces to the switch. This will allow communication between the router and both WAN1 interfaces. You would do the same between your LAN interfaces and your internal device (firewall or router).

To enable Unit Failover, select the Failover checkbox and enter the failover information as described below.

Local Unit

The Group ID uniquely identifies the failover group. This only needs to be changed if you have more than one pair of MPVPN units using Unit Failover on the same network. Valid range is 1-255. Both of your failover units must use the same Group ID.

Access IP/Mask uniquely identifies each unit in a private subnet common to both units and will be used to access

the unit when in Standby mode (when all other IPs are deactivated). You must use IP/Mask format (e.g.: 192.168.0.10/24).

Email Alert Settings (optional)

Email Alert Settings allows you to specify email information so an email can be sent whenever failover occurs. This email will be sent from a unit that goes from Standby to Active state.

Heartbeat

This option indicates the medium through which the heartbeat packets between the two units are exchanged (failover option). Choose LAN to exchange the heartbeat packets over the LAN interface. Choose Serial to exchange the heartbeat packets over the Serial interface. It is necessary to have a null modem cable for connecting the serial ports between the two units.

Role

Role indicates the preferred role of each unit. One unit will be set as Primary and the other as Backup. The role only applies when both units are powered on at the same time. The unit marked as Primary will go to the Active state and the unit marked as Backup will go to the Standby state.

State

State shows the current failover status of the unit you are logged into, either Active or Standby.

Note: Units will be in Standby mode if doesn't detect LAN connections and at least one WAN connection and UI will not be accessible.

Force to Standby

Force to Standby will allow you to force an Active unit to Standby mode, allowing the other unit to become Active.

Peer Units

The Peer Unit shows details about any unit that is detected as a backup to the one you are viewing. The IP address of the backup unit is the Access IP. Serial Number is the Serial Number of the Peer Unit. The State could be displayed as "Up," "Backup," or "Down." If it is marked as Down, it means the unit is no longer detected.

Note: At a minimum, you must specify a Group ID and an Access IP. Note that when you click SAVE, you may be disconnected. This occurs because each of the LAN and

WAN interfaces use a new virtual MAC address. Therefore, you may not be able to access the unit until the ARP cache has cleared on any devices between you and the MPVPN unit. You could either run a command to clear the ARP caches on those devices or simply reboot them. This only needs to be done when you enable or disable Unit Failover.

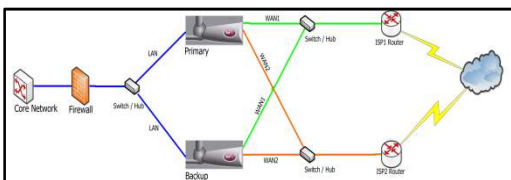


Figure 3.17 - Pictorial representation of Unit Failover

Steps to Configure Unit Failover

Physically layout Network as shown above.

Make sure both FatPipe units are powered off.

Configure your pc/laptop connected to the LAN switch with IP 192.168.0.10/24.

Power on one FatPipe unit.

Open a browser on your pc/laptop and connect to IP 192.168.0.1. (Default LAN IP)

Configure the FatPipe unit and then enable unit failover with access IP 172.16.1.1/24. Group ID should be set as 3 and role as Primary.

Save all changes

You will then need to log back into the GUI of the FatPipe.

Save a backup of the configuration to your local pc/laptop.

Power off the FatPipe unit.

Power on the secondary unit.

Open a browser and connect to 192.168.0.1.

Upload the saved configuration to the unit.

The FatPipe will reboot.

Log back into the GUI

Enable unit failover with access IP 172.16.1.2/24 and the same Group ID as the primary box (default 3). Select the role as Backup.

Save Changes.

The GUI will reset.

At this point, you can either leave this box on and power on the other device to come up in standby, or you can power off both boxes. If both boxes are powered off, power on the primary box. When the GUI of that box can be accessed, power on the other FatPipe unit which will now recognize the other FatPipe unit and go into a Standby mode.

Note: FatPipe recommends utilizing a Hub/Switch per Ethernet Interface used on the FatPipe as described in the above diagram.

Some customers would rather use a VLAN setup with one Switch - if this is the case, you will need to make sure VLAN tagging is not being used. Also, verify that Multicast UDP port 5002 and 5003 are being allowed as that is the port the FatPipe uses to maintain unit failover between the two devices. (Verification of ports should not be needed on a standard Layer 2 Switch)

Also, if using a switch check that portfast is enabled.

Stateful Failover

Enable Stateful Failover to ensure all sessions from the Active units are failed over transparently to the Standby unit in case of failure of Active unit. (See Figure 3.10).

SNMP

FatPipe products support SNMPv2 (Simple Network Management Protocol version 2) with MIB-II (Management Information Base II) compliance, to accommodate SNMP queries in addition to sending out SNMP traps. This allows you to use SNMP management software to monitor and gather statistics from FatPipe products and view and monitor system parameters of your FatPipe unit.

Please note that FatPipe SNMP is read-only. Write access is not currently supported. You can configure SNMP settings from within the web-based management application. Once SNMP is configured, you can monitor the FatPipe unit using any SNMP manager.

To configure SNMP settings, click on the System button in the main menu and click on the SNMP tab (see Figure 3.18).

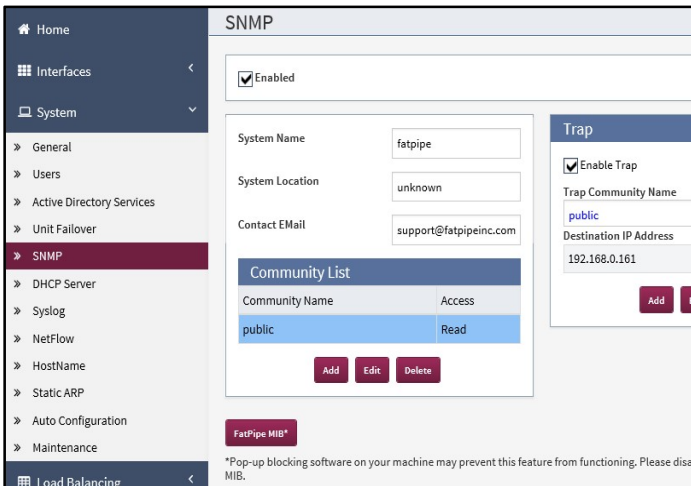


Figure 3.18 – SNMP

Community List

The Community List is a list of community names that will be used to access FatPipe SNMP information. The community List has a default community name, "Public," with only "Read" access available.

To add community names, click on the Add button (see Figure 3.19).



Figure 3.19 – Add Community Name

Click on the OK button to return to the SNMP page, click on SAVE button to make the changes permanent.

To edit the Community list, select it from the list and click the Edit button, click on SAVE button to make the changes permanent (see Figure 3.20).



Figure 3.20 - Edit Community Name

To delete a Community, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

Enable Trap

If this is enabled, MPVPN will send an SNMP trap to alert you when there is a physical link failure with any of your WAN lines. You must specify a community name and one or more IP addresses that will receive the trap.

FatPipe MIB

Click on this button to download FatPipe's custom MIB. This MIB can be imported into existing SNMP software's device list, and allows you to view almost all settings that you see in the remote management interface (GUI) from within an SNMP management application.

DHCP Server

FatPipe DHCP server allows you to configure the built-in DHCP Server to assign IP addresses to devices on your local area network (LAN).

To configure DHCP Server settings, click on System button in the main menu and click the DHCP Server tab (see Figure 3.21).

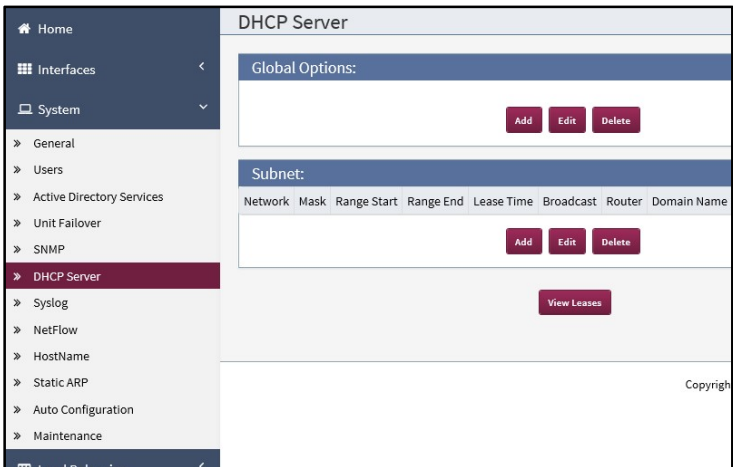


Figure 3.21 - DHCP Server

To add a new DHCP subnet, click on the Add button (see Figure 3.22), click on SAVE button to make the changes permanent.

The screenshot shows the 'Add/Edit DHCP Subnet' form. It includes input fields for Network (192.168.0.0), Mask (255.255.255.0), Default Gateway IP (192.168.0.254), Broadcast (192.168.0.255), Lease Time (Seconds) (10000), and Domain Name (corp.example.com). There is a note '(Set 0 for infinite lease)'. Below these are two sections: 'Range (Start:End)' with a value of 192.168.0.1-192.168.0.49 and 'Domain Name Servers' with values 8.8.8.8 and 4.2.2.2. Both sections have 'Add', 'Edit', and 'Delete' buttons. At the bottom is an 'Extra Option' section.

Figure 3.22 - Add DHCP subnet

Network

Any LAN network that needs DHCP IP assignment

Mask

Subnet mask of the above network that needs DHCP IP assignment

Range Start

The starting IP address for the DHCP range as defined in the network field

Range end

The last IP for the DHCP range as defined in the network field.

Lease time

The amount of time a DHCP client may have an IP address before it is required to renew the lease.

Broadcast

The broadcast IP corresponding to the above Network and Mask fields

Router

Gateway IP address that will be assigned to clients

Domain Name

IP addresses of the name server that is common to the entire organization

Domain Name servers

IP address of the preferred DNS servers in hierarchical order

To edit the existing DHCP subnet, click on the Edit button (see Figure 3.23), click on SAVE button to make the changes permanent.

Add/Edit DHCP Subnet

Network 192.168.0.0	Mask 255.255.255.0	Default Gateway IP 192.168.0.254	Broadcast 192.168.0.255
Lease Time (Seconds) 10000 (Set 0 for infinite lease)	Domain Name corp.example.com		

Range (Start:End)
192.168.0.1-192.168.0.49

Add **Edit** **Delete**

Domain Name Servers
8.8.8.8
4.2.2.2

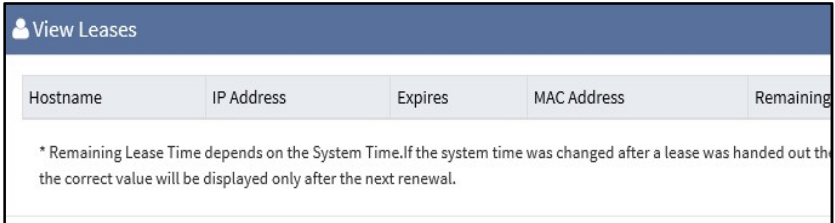
Add **Edit** **Delete**

Extra Option

Figure 3.23 - Edit DHCP subnet

View Leases

To monitor the IP addresses assigned to each client, click the View Leases button (see Figure 3.24).



Hostname	IP Address	Expires	MAC Address	Remaining
* Remaining Lease Time depends on the System Time. If the system time was changed after a lease was handed out the correct value will be displayed only after the next renewal.				

Figure 3.24 - View Leases

The Revoke button is used to cancel the lease for a specific LAN device, and releases the entry from the lease table. Use the Revoke button if the device no longer needs the leased IP address, because it has been removed from the network. If the lease table becomes full or nearly full, you can use the Revoke button to recover space in the table for new entries, by removing lease entries for hosts that no longer need a DHCP lease.

Click Refresh button to update the View Lease table (If any new host is added or renewed).

Syslog

Syslog is a standard for forwarding log messages in an IP network. In order to take advantage of this feature, a running syslog server on a host reachable from the FatPipe is necessary.

In the case you use FatPipe VPN and you wish to send logs to a remote syslog server through a VPN tunnel, you need to enter an IP for FatPipe to use as its reporting IP. (Normally, the source IP address for the logs sent from FatPipe is chosen automatically based on the routing information). Go to the Interfaces page and select the LAN tab. At the bottom of the page, there is a text box labeled "Reporting IP." Enter the IP address which you want the syslog packets to come from. Currently, this address is restricted to one of the LAN alias IP addresses. Click Save.

Refer to the reporting IP section in Chapter 2 under the LAN heading.

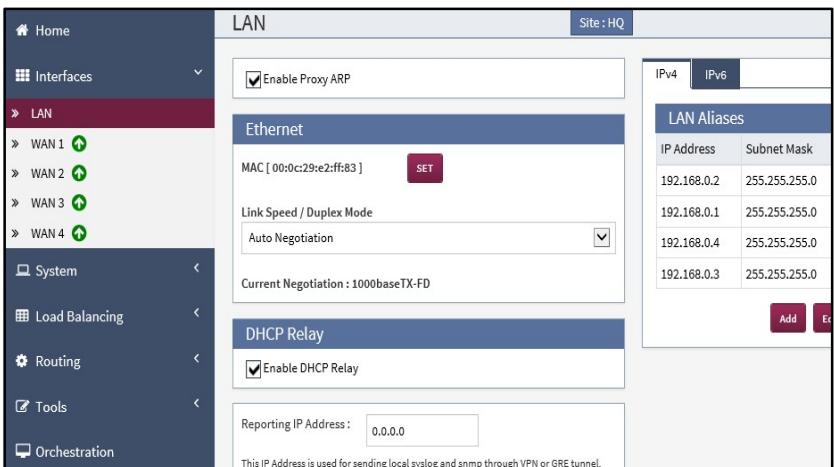


Figure 3.25 - Sending logs to a remote Syslog server through a VPN setup

To configure Syslog settings, click on the System button in the main menu and click the Syslog tab (see Figure 3.26).

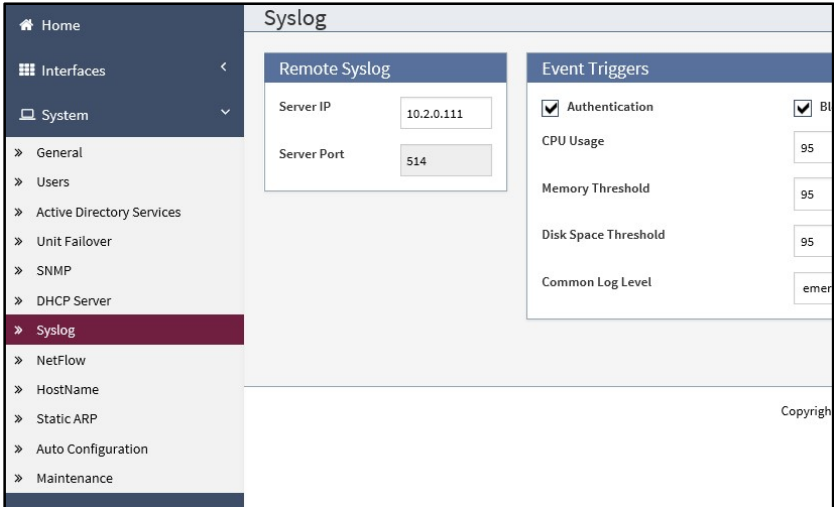


Figure 3.26 - Remote Syslog

Remote Syslog Server IP

The IP of the remote server in which the Syslog is configured.

Remote Syslog Server Port

The remote syslog server port number. By default, the syslog server uses UDP port 514 for communication. This is the default value for port number in your FatPipe device. If you did not change this value on your syslog server, leave the default value for port number.

Logging Events - Authentication

If this is enabled, a log message will be sent to a syslog server giving the information about the login and logout time of a user to the FatPipe GUI.

Blocked Packets

If this is enabled, a log message will be sent to a syslog server giving the information about the packets source, destination and type that are being dropped by the FatPipe device.

CPU Usage

A syslog alert is generated and sent when the cpu usage of the FatPipe device crosses the defined limit.

Memory Threshold

A syslog alert is generated and sent when the memory usage of the FatPipe device crosses the defined limit.

Disk Space Threshold

A syslog alert is generated and sent when the Disk Space utilization of the FatPipe device crossed the defined limit.

Common Log Level

Log messages are generated by the system and logged at the particular log level with emergency level being the lowest and debug being the highest.

Caution:At debug level, the system logs messages from every daemon to the best possible detail. This will cause disk space to be used up rapidly and use significant CPU resources. Debug option should be used only if the situation warrants.

NetFlow

FatPipeNetFlow allows you to export traffic statistics using NetFlow protocol.

The screenshot shows the NetFlow configuration page. On the left, a sidebar menu has 'NetFlow' highlighted. The main content area is titled 'NetFlow' and contains a configuration box. Inside this box, there is a checked checkbox for 'Enable Remote Netflow Reporting'. Below it are two radio buttons for 'Version 5' (selected) and 'Version 9'. Further down are three input fields: 'FatPipe Source IP' with the value '172.17.127.180', 'Remote Netflow Server IP' with the value '172.17.127.20', and 'Remote Netflow Server Port' with the value '2055'. The bottom right of the page shows a 'Copyright' notice.

Figure 3.27 – Remote NetFlow Reporting

Enable Remote NetFlow Reporting

If this is enabled, the source IP, Remote NetFlow server IP and port number can be configured.

FatPipe Source IP

The FatPipe LAN IP.

Remote NetFlow Server IP

The IP of the remote server in which the NetFlow is configured.

Remote NetFlow Server Port

The port number on which the remote NetFlow server is listening.

HostName

Configure a list of Hostname of devices in the LAN and their corresponding IP address. These Hostnames will be displayed while viewing the Packet Log Report for the LAN. (See Figure 3.28).

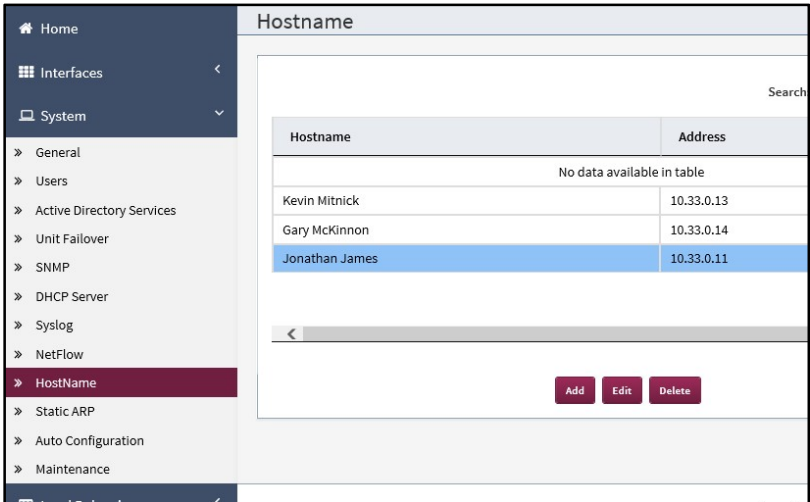


Figure 3.28 - HostNames

Auto Configuration

The Auto Configuration feature is used to configure MPSec tunnels, VPN, Web Filters or Policy Routing Rules (PRR). Auto Configuration is based on the configured Orchestration. When branch contacts the Orchestrator, it should validate the branch device. Information of branch will be provided to Orchestrator and the branch then asks for the templates and orchestrator provides them based on key and branch configures itself based on the template. To go to Auto Configuration, select the System button and click on Auto Configuration (see Figure 3.29)

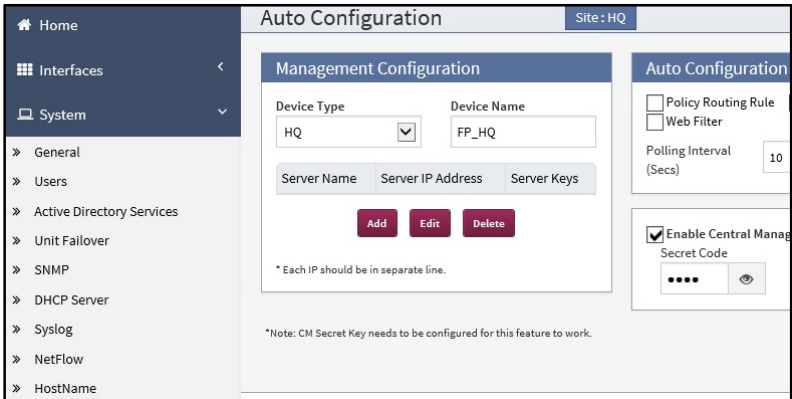


Figure 3.29 - Auto Configuration

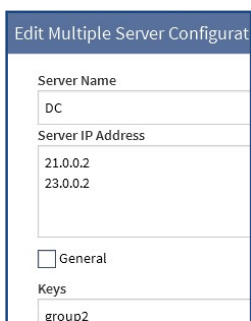


Figure 3.30 shows a screenshot of the 'Edit Multiple Server Configuration' dialog box. The dialog has a title bar 'Edit Multiple Server Configuration'. It contains three main sections: 'Server Name' with a text box containing 'DC'; 'Server IP Address' with a text box containing '21.0.0.2' and '23.0.0.2'; and a 'General' section with a checkbox labeled 'General' which is unchecked. Below the checkbox is a 'Keys' section with a text box containing 'group2'.

Figure 3.30 - Configure Orchestration

To configure Orchestration information (see Figure 3.30), click on Add in Auto Configuration page. Server Name can be any name to identify the Server, Server IP Address will be the IP address of Orchestrator. Multiple IP addresses can be added. Keys will be the key configured on Orchestrator and will be used to identify the unit.

Management Configuration takes information about FatPipe's located at remote location and builds tunnel or policies. Select Policy Routing Rules, MPSec, VPN or Web Filter to configure between two locations. Enable Central Manager Login and give a Secret Code to have access to branch units using the same code. To add a server, click on Add and provide the server name, IP address and key.

Maintenance

Shutdown or reboot MPVPN safely by clicking the corresponding button (see Figure 3.31). You will be prompted to confirm or cancel the operation.

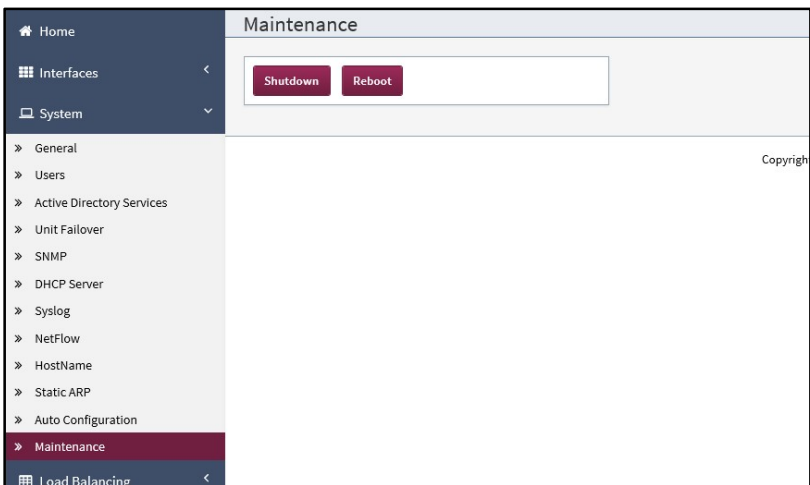


Figure 3.31 – Reboot/Shutdown

Chapter 4: Load Balancing

Algorithms

FatPipe MPVPN provides four methods of load balancing: Round Robin, Response Time, Fastest Route, and Weighted. To configure a specific load balancing algorithm, click on the Load Balancing button listed in the main menu and click the Algorithms tab (see Figure 4.1). You can also set Primary and Backup lines per WAN interface (see Chapter 2, WAN).

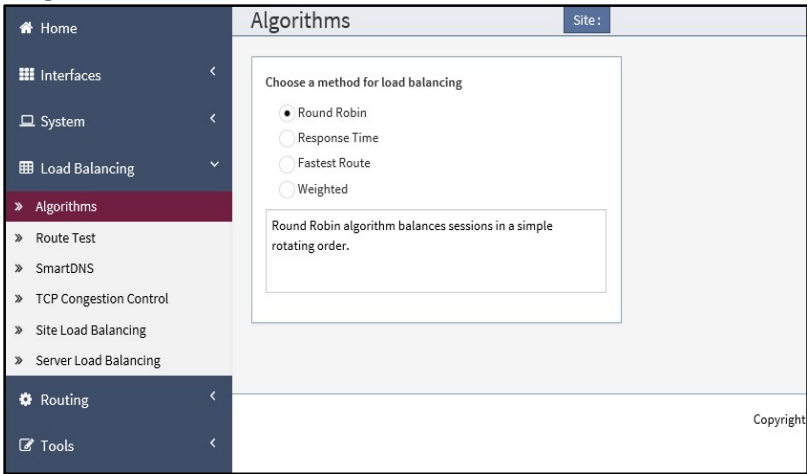


Figure 4.1 – Load Balancing

Round Robin

Configures FatPipe MPVPN to send sessions down the lines in rotating order. This method is recommended for similar speed connections to the Internet, even if the connections are not of the same ISP (e.g., combining two similar speed fractional T1s and a DSL line).

Response Time

Configures FatPipe MPVPN to balance traffic based on each line's average response time for Internet requests. This method is recommended for unequal speed connections. The fastest line will be used more often with Response Time.

Fastest Route

Configures FatPipe MPVPN to balance traffic on a per-destination basis. Each session will go over the fastest line for its destination. Choose this option when you want to make sure each session goes out the line with the fastest route for its destination. (There is slight overhead with this algorithm since SYN packets get sent out on all lines at the start of each session).

Weighted

Algorithm configures FatPipe MPVPN to balance traffic in proportion to the WAN weights defined by you. Each interface needs to be assigned a weight. (Default value for each interface is 1.) The ratio of these weights determines the ratio of downloaded traffic on the respective Internet lines, which the load balancing algorithm maintains.

For example, if weights for WAN1, WAN2, WAN3 are 1, 2, 3, respectively, and total download traffic amounts to 600kbps, the traffic will be balanced over respective lines as 100, 200, 300 kbps. Because FatPipe MPVPN balances sessions rather than packets, real world results will rarely achieve this ideal. In general, the greater the number of sessions, the closer the distribution of traffic will be to the specified weights.

Route Test

FatPipe MPVPN tests connections to the router, to the Internet Service Provider (ISP), and to a maximum of three user-specified sites on the Internet. Each site can be specified using a domain name or an IP address. The port number should be a valid listening TCP port at the site. The default is port 80 for HTTP (web servers).

To configure test sites, click on the Load Balancing button in the main menu and click on the Route Test tab (see Figure 4.2).

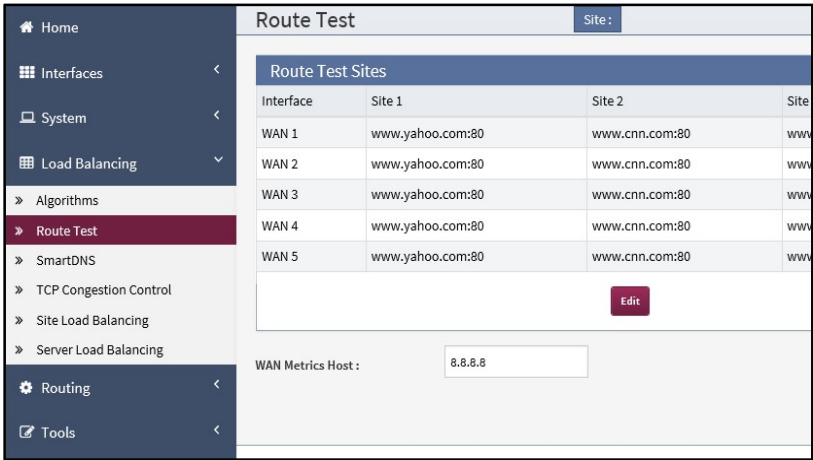


Figure 4.2 – Route Tests

To add sites, click the Add button (see Figure 4.3). Click the OK button to return to the route test page, click on SAVE button to make the changes permanent.



Figure 4.3 – Add Route Test

To edit a route test, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 4.4).

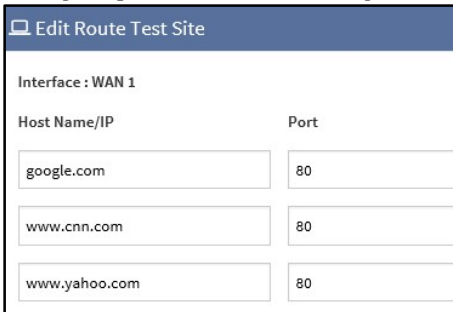


Figure 4.4 – Edit Route Test

To delete a route test, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

SmartDNS™

SmartDNS is a patented technology that provides inbound load balancing to public facing servers in the LAN.

The benefits of FatPipe's SmartDNS feature are:

- **Load Balancing:** SmartDNS balances load by advertising the different paths into a host on a LAN. The host appears to be a different IP address at different times, thus using all available lines. The IP addresses are resolved based on the selected interface-to-network mappings.
- **Speed:** Through load balancing, FatPipe SmartDNS speeds up the delivery of inbound traffic according to the interface-to-network mappings selected by the administrator.
- **Failover:** SmartDNS will dynamically sense when a failure occurs and will make adjustments to the DNS replies so it will not hand out IP addresses that are associated with connections that are down.

SmartDNS allows hosts on a network to have multiple IP addresses associated with them from different providers, and will hand out the IP addresses for these hosts using the interface-to-network mappings. SmartDNS uses the Line Status, determined by the Route Test function, to check when a WAN interface loses connectivity. If the Line Status is marked "Down" for that interface, SmartDNS will change the advertised paths to compensate for the WAN interface that is unavailable; advertising the pathways for whose interface is "Up" only.

Before moving DNS services to MPVPN, it is recommended to configure SmartDNS first and test resolution locally by querying the MPVPN directly.

SmartDNS Setup

To configure SmartDNS settings, click on the Load Balancing button in the main menu and click on the SmartDNS tab.

To create a Master Zone, click on the Add button and choose the option Master (see Figure 4.5). Click on the Next button to input Domain Name, Master Server, Email Address, Refresh, Retry, Expire, and TTL information for the Master Zone.

We recommend you keep record of the defaults (see Figure 4.6). Click on the Next button to manage your Zone Records: A, NS, MX, CNAME, PTR, TXT, SRV, SPF and DNAME.

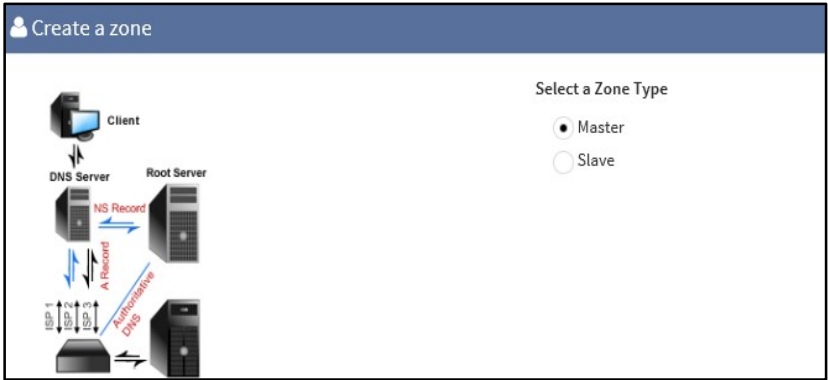


Figure 4.5 - select Master Zone option

Create a zone

Domain Name example.com.	Master Server ns1.example.com.
Records File db.example.com.	Email Address adm.example.com.

Zone Parameter(in Seconds):

Refresh 28800	Interval for slave servers to refresh the data.
Expire 604800	Interval for slave servers to retry refresh failure
Retry 7200	Time before slave servers expire data after refr

Figure 4.6 - Add a Master Zone

Click any one of the tabs, to manage your Zone Records: A, NS, MX, CNAME, PTR, TXT, SRV, SPF and DNAME. Click on the Create button and a new record is inserted, which will allow you to enter the Name, IP address and TTL information of the records (see Figure 4.7).

Create a zone									
A	NS	MX	CNAME	PTR	TXT	SRV	SPF	DNAME	DNSSEC
Name	Name Server	TTL							
@	ns1	10							
@	ns2	10							
@	ns3	10							

Figure 4.7 - Create Zone Records

Click on the Next button to view the zone and total record information (see Figure 4.8). Then click on the Finish button to return to the main screen. Click on SAVE button to save the changes.

Click 'Finish' button create a new Master Zone

Master Zone Name: example.com.
 Total A Records: 0
 Total NS Records: 3
 Total MX Records: 0
 Total CNAME Records: 0
 Total PTR Records: 0
 Total TXT Records: 0
 Total SRV Records: 0
 Total DNAME Records: 0
 Total SPF Records: 0

Figure 4.8 - Summary of the master zone configured

Note: On a Forward DNS zone (e.g., example.com), you will never specify PTR records. PTR records are only used in Reverse DNS zones (e.g., 3.2.1.in-addr.arpa).

To make changes to a zone, choose the zone from the zone list. The zone information window will be populated with details related to the selected zone. Make the changes if needed and click on the SAVE button to save the changes.

Click on the Create button to add a new record. To change an existing record, select it and click on the Edit button. Click on SAVE button to save the changes (see Figure 4.9, Figure 4.10).

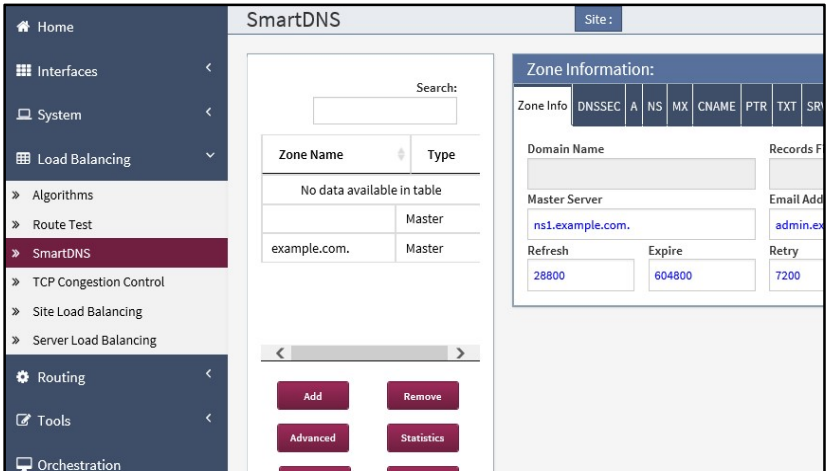


Figure 4.9 - Add records from SmartDNS page

Name
_sip._tls.example.com
Default
TTL
10
Priority
20
Weight
2
Port
8080
Target

Figure 4.10 - Create SRV record for master zone

Select the particular record information and click on the Delete button to delete the record.

Click on the Reset button to refresh the data for that particular zone to its original state. Any changes made that were not saved will be lost.

Click Zone Info tab to view the Zone Information (see Figure 4.11).

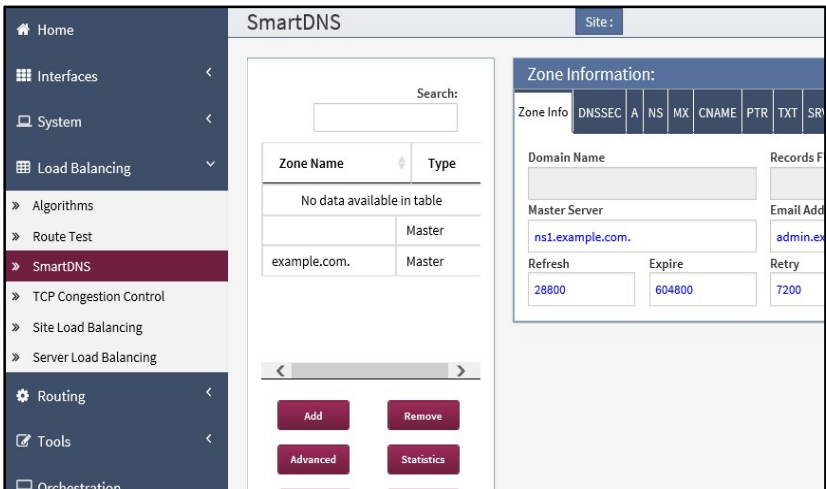


Figure 4.11 – SmartDNS with Zone Information for Master Zone

DNSSEC

The Domain Name System Security Extensions (DNSSEC) deals with cache poisoning and a set of other DNS vulnerabilities such as "Man in the Middle" attacks and data modification in authoritative servers. Its major objective is to provide the ability to validate the authenticity and integrity of DNS messages in such a way that tampering with the DNS information anywhere in the DNS system can be detected.

To secure a zone, select the zone and click on DNSSEC tab from the main page (see Figure 4.12).

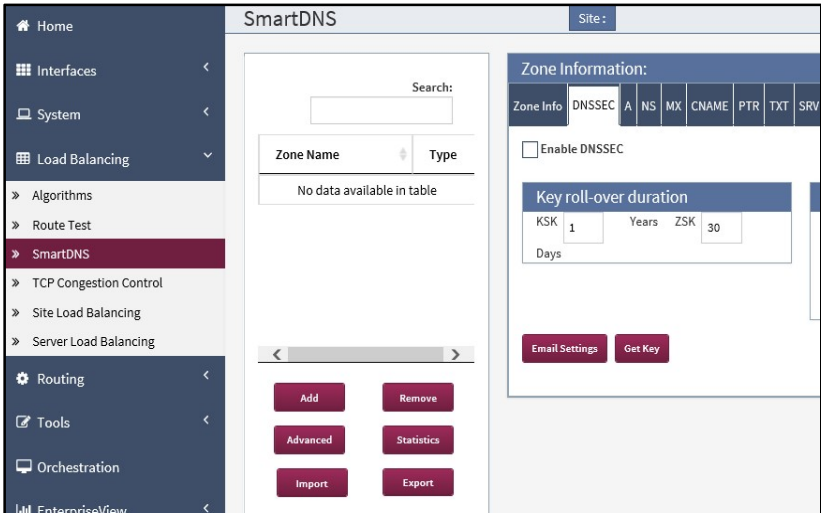


Figure 4.12 - SmartDNS with DNSSEC

To make that zone secure, click select the Enable DNSSEC checkbox. Enter the KSK rollover duration in years (by default it is 1 year). Enter ZSK rollover duration in days - usually it is 90 days. Enter a valid email address to notify the System Administrator about the rollover (see Figure 4.13) and then click on the Save button to generate the Key, the Signing Key and the Zone signing key.

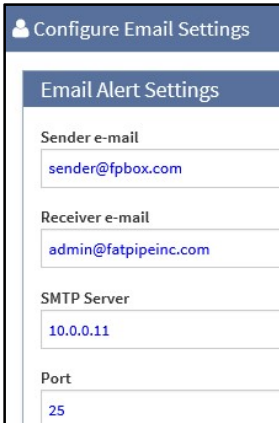


Figure 4.13 - Email settings

Once the Keys are generated, then the zone needs to be signed with these keys. To sign the zone, select the date and time and click the Sign Zone button. The zone signing

will happen at the date and time specified (see Figure 4.14, Figure 4.15)

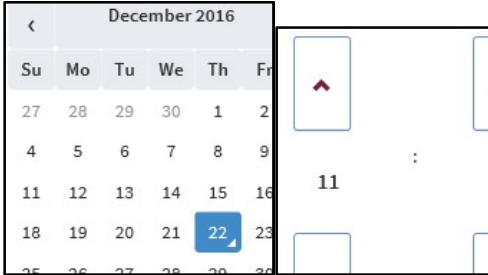


Figure 4.14 - Set date to sign the zone

Figure 4.15 - Set time to Sign the zone

Click on Get Key button to get the KSK for the zone that was generated (see Figure 4.16).



Figure 4.16 - Key generation

After the KSK duration is expired, a new KSK is generated and the zone needs to be resigned. To resign a zone, select the date; time and click Resign Zone (see Figure 4.17). FatPipeSmartDNS automatically resigns the ZSK rollover.

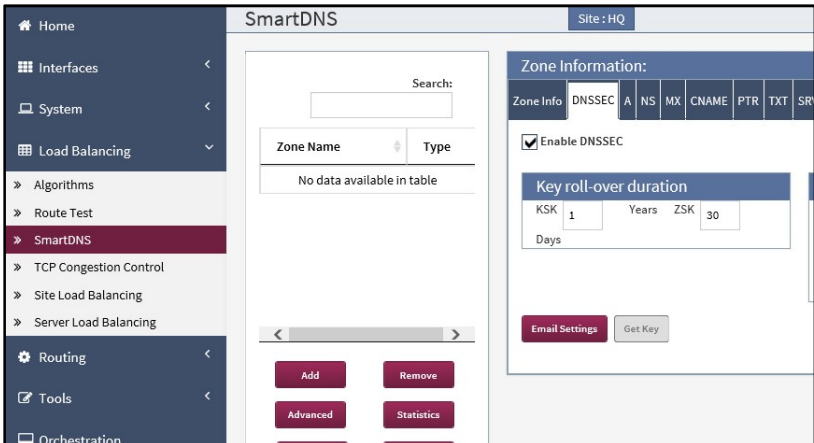


Figure 4.17 - Re-sign the zone

To do an unscheduled key rollover or an emergency key rollover if there is suspected compromise of the keys or loss of private key, click on Unscheduled Rollover button and this will regenerate KSK and ZSK for the zone. The zone has to be re-signed using the Re-sign Zone button after an Unscheduled Rollover.

To create a slave zone, click on the Add button and choose the option, Slave (see Figure 4.18). Click on the Next button to input the Domain Name Master Server IP address and Records File information (see Figure 4.19). To add a Master Server, click on the Add button. A new record is inserted, which will allow you to enter the Server IP address. Click on the Next button to view the zone and total Master server's information (see Figure 4.20). Click on the Finish button to return to the main screen, and then click on the SAVE button to save the changes.



Figure 4.18 - Select Slave Zone option

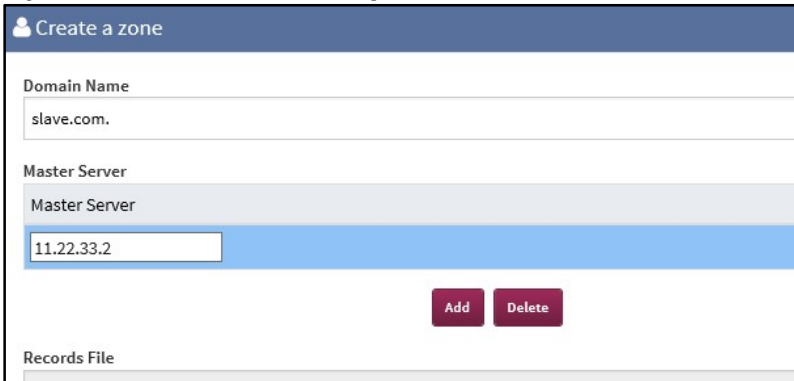


Figure 4.19 - Add a Slave Zone



Figure 4.20 - Summary of the Slave Zone configured
Click Zone Info tab to view the Zone information (see Figure 4.21).

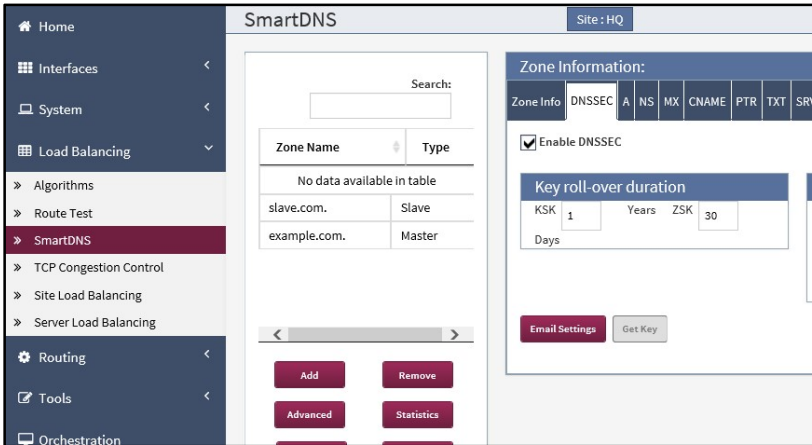


Figure 4.21 - SmartDNS with Zone information for slave zone
To remove a zone, select it from the list and click on the Remove button. The zone will be removed from the list. Click on the SAVE button to make the changes permanent.

Advanced Settings

Click the Advanced button to configure zone transfer for slave servers and interface to network mappings.

Zone Transfers

If you have slave servers that will initiate zone transfers, then enable “Allow Zone Transfers.” If you want to allow zone transfers from any IP in the internet, choose “Any IP.”

Note: It is considered as a security risk to allow zone transfers from any IP in the Internet.

Click on the SAVE button to make the changes permanent (see Figure 4.22).

The screenshot shows the SmartDNS configuration interface. On the left is a sidebar with navigation options: Home, Interfaces, System, Load Balancing (expanded), Algorithms, Route Test, SmartDNS (selected), TCP Congestion Control, Site Load Balancing, Server Load Balancing, Routing, Tools, and Orchestration. The main content area is titled 'SmartDNS' and has a 'Site: HQ' dropdown. It features a search bar, a table for Zone Name and Type (currently empty with the message 'No data available in table'), and buttons for Add, Remove, Advanced, and Statistics. On the right, the 'Advanced' settings for 'Zone Transfer' are shown. The 'Allow Zone Transfers' checkbox is checked. The 'FROM:' section has two radio buttons: 'Any IP' (selected) and 'Specify IPs'. Below this is an 'Interface-To-Network Mapping' table with columns for Site Name, Interface, and Network Address/Mask. The table contains one row: 'Choose' (Site Name), 'WAN1' (Interface), and '11.1.5.0/24' (Network Address/Mask). At the bottom right of this section are buttons for Add, Edit, and Delete.

Figure 4.22 - Zone Transfers

If you want to allow zone transfers from a particular set of IPs, choose “Specify IPs.” Click on the Add button, which will show a popup window where you can enter a valid IP address/mask. If you want to add multiple addresses, click the Add button again, click on the SAVE button to make the changes permanent (see Figure 4.23).

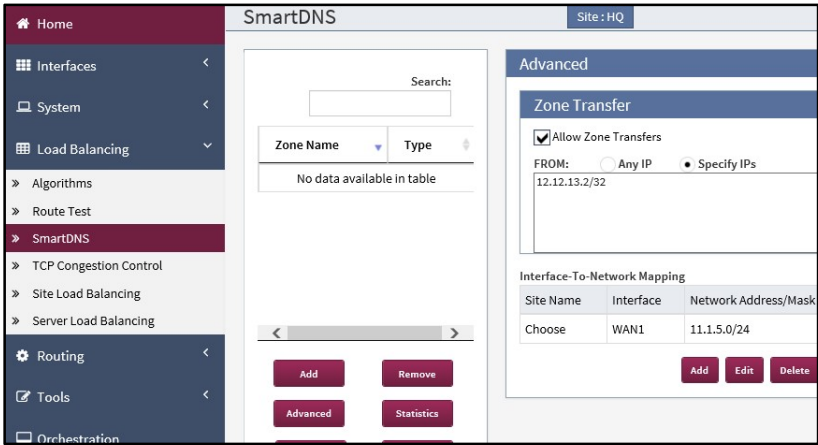


Figure 4.23 - Zone Transfers

To edit an IP Address/mask, select it from the list and click on the Edit button. Click on the SAVE button to make the changes permanent.

To delete an IP Address/mask, select it from the list and click on the Delete button. Click on the SAVE button to make the changes permanent.

Interface-To-Network Mappings

Interface-To-Network Mappings are necessary for SmartDNS to function properly (see Figure 4.24). The mappings are used to specify the network(s) that belong to each interface. This will tell SmartDNS which IPs belongs to which interface when answering DNS queries. The mappings are also used with Site Load Balancing (see Chapter 8) to specify which networks belong to each interface of each site.

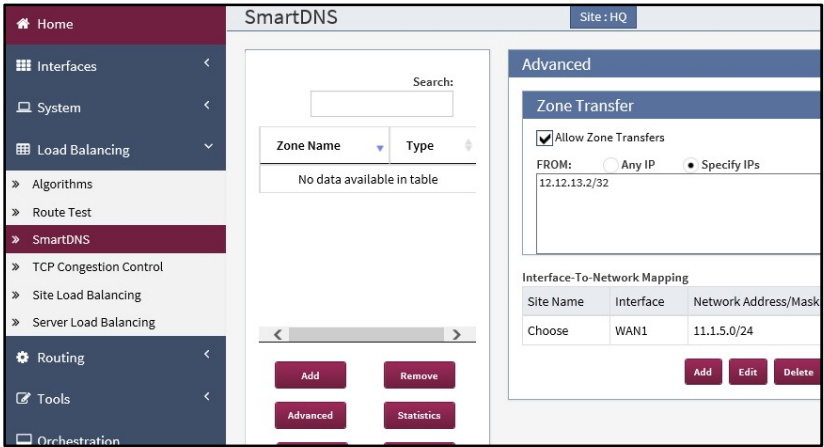


Figure 4.24 - Interface-To-Network Mappings

Click on the Add button to add the new mapping information (see Figure 4.25).

Choose the WAN interface from the Interface dropdown menu. If you are setting up Site Load Balancing you will also select a site from the Site Name dropdown menu. Choose a Role for this Interface from the Select a Role dropdown menu. "Primary" specifies that IPs associated with this WAN Interface will be handed out in a DNS request as long as the link is up. "Backup" specifies that IPs associated with this link will be handed out in a DNS request only if all primary links are down (unavailable).

"Weight" affects how often IPs from this particular mapping is handed out in DNS requests. The number entered is the number of times an IP will be handed out before using the next mapping. If all mappings have a weight of one, then they are all treated equal and IPs are handed out in a round-robin fashion. Click on the Add button to add the Network IP Address/Mask that is associated with the selected interface. Then click on the OK button to return to the SmartDNS page. Click on SAVE button to save the changes.

Site Name Choose ▼	Interface WAN2
Role Primary ▼	weight 1
Network IP Address/Masks 11.1.5.0/24	Add Edit

Figure 4.25 - Add Interface-To-Network Mappings

To edit mapping information, select it from the list and click the Edit button, click on SAVE button to make the changes permanent (see Figure 4.26).

Site Name : Choose	Current Interface :
Role Primary ▼	weight 1
Network IP Address/Masks 11.1.5.0/24	Add Edit

Figure 4.26 - Edit Interface - To - Network Mappings

To delete mapping information, select it from the list and click on the Delete button, click on SAVE button to make the changes permanent.

Statistics

View Statistics

To view a record of SmartDNS statistics for all the zones, click on the Statistics button. Click on the VIEW button to tabulate the DNS responses based on the IP Addresses (see Figure 4.27, Figure 4.28).

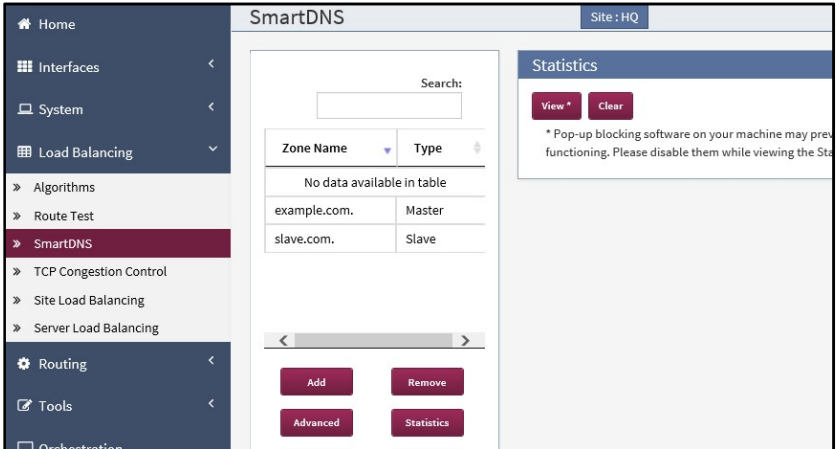


Figure 4.27 - View Statistics

DOMAIN NAME	IP ADDRESS	RESOLVED TIMES
ns1.first.com	207.140.140.2	299
ns2.first.com	207.150.150.2	299
ns3.first.com	207.160.160.2	299
www.first.com	207.140.140.100	100
	207.150.150.100	99
	207.160.160.100	100

Figure 4.28 - DNS Statistics

Clear Statistics

To clear all the SmartDNS statistics stored, click on the CLEAR button.

Setup Steps for Moving DNS to MPVPN

Register a new domain with a registrar. If you have an existing domain, get all domain information from your DNS provider (the group managing your DNS, typically one of your ISPs).

Register new name server names with the registrar using your domain name (e.g., ns1.yourdomain.com and ns2.yourdomain.com).

Setup the DNS Zone (domain information) on FatPipe MPVPN. Initiate a transfer of your domain name with the registrar and point it to your newly registered name

server names (e.g., ns1.yourdomain.com and ns2.yourdomain.com).

Step 1: Register a New Domain Name

You must contact a domain registrar to register a domain name. You can get a full list of ICANN-accredited registrars from InterNIC.com. Directnic.com and Networksolutions.com are two of the competing ICANN-accredited registrars you can use. In the course of registering the new domain, you may be required to provide two name servers that will handle your domain name. If the registrar provides default name servers, you can use them. Otherwise, just specify any existing name servers. (E.g., just put in ns.yahoo.com and ns1.yahoo.com and their corresponding IP addresses). You will transfer these domains to your name server names in a future step.

Step 2: Register Name Servers

Contact your registrar to initiate the creation of your new name servers using your domain name. (E.g., ns1.yourdomain.com and ns2.yourdomain.com). Each name server name will map to its own WAN port IP address on MPVPN. As far as the registrar knows, your domain name is handled on multiple physical name servers, but in reality you are simply mapping a different name server name to each of the WAN port IP addresses.

Step 3: Set Up DNS Zone (Domain Information)

To achieve inbound redundancy, each domain name record, (e.g., www), will have multiple IP addresses assigned to it -- one from each WAN IP block.

SmartDNS will hand out these IP addresses based on the interface-to-network mappings.

Step 4: Initiate Zone Transfer

The last step is to change the name servers for your domains at your registrar's website. This is commonly referred to as "initiating a zone transfer." You will change the name servers for your domains to the name servers you registered in Step 2. The transfer will take from 24 to 48 hours. Once the transfer is complete and the root name servers are updated with the new name server information, SmartDNS will be live.

Note: There may be name servers out in the world that have information cached for a week or more though, so make sure you do not take down your pre-existing name you keep servers. We recommend those stay in place for at

least two weeks or even a month, if you want to be extra careful. Eventually, nobody will be using your pre-existing name servers and it will be safe to remove your domains from those servers.

Basic SmartDNS Example

```
1st WAN IP Block    7.0.0.0 - 7.0.0.255
2nd WAN IP Block    8.0.0.0 - 8.0.0.255
3rd WAN IP Block    9.0.0.0 - 9.0.0.255
```

IP Addresses on FatPipe WAN Ports

```
WAN1                7.0.0.2
WAN2                8.0.0.2
WAN3                9.0.0.2
```

Registered Name Servers

```
ns1.yourdomain.com. 7.0.0.2
ns2.yourdomain.com. 8.0.0.2
ns3.yourdomain.com. 9.0.0.2
```

SmartDNSName Server Entries (NS records)

Name	Name Server
@	ns1.yourdomain.com.
@	ns2.yourdomain.com.
@	ns3.yourdomain.com.

SmartDNS Host Name Entries (A records)

Name	IP Address
@	7.0.0.5
@	8.0.0.9
@	9.0.0.44
www	7.0.0.5
www	8.0.0.9
www	9.0.0.44
ftp	7.0.0.7
ftp	8.0.0.35
ftp	9.0.0.19

Time to Live (TTL)

SmartDNS uses a short TTL to ensure the information about the IP addresses for the hosts it serves are accurate and up-to-date. This means that the machines on the Internet will always connect to the host using a route that is available instead of trying to access the host using an IP address that is not accessible due to a line failure.

The TTL value informs all DNS servers on the Internet how long they should store information about your domain. For example, a name server caches your domain information following a request for a website that uses your domain. Until the TTL value is exceeded, that name server will continue using the information supplied by the first request each time your domain is requested. When your domain is requested after the TTL period, the name server

will conduct a new query for updated information about your domain. The TTL value is measured in seconds.

MPVPN ensures that DNS information is up-to-date. You can change the TTL to your own preferences, along with Refresh, Expire, and Retry entry settings.

Import

DNS zone files can be imported from the host system into FatPipe using the Import button. It will ask you for the location where the zone files are saved in the system. You can also import multiple zone files by selecting all the required files.

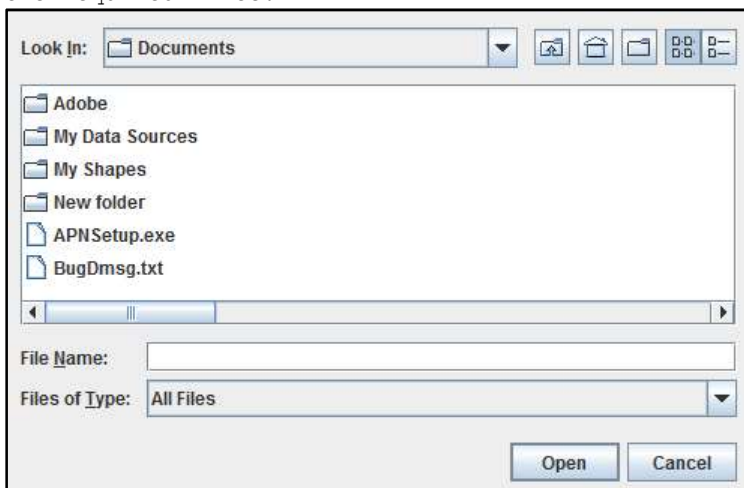


Figure 4.29 – Import Smart DNS

Export

DNS zone files can be exported to the host system from FatPipe using the Export button. Select a zone and click the export button, it will ask you to specify a location where the zone file will be saved.

Reverse DNS (PTR Records)

SmartDNS supports Reverse DNS (PTR Records). To set this up, you must know the exact name of the zone that your ISP will use to delegate the Reverse DNS. The zone name will always end in "in-addr.arpa." The only valid record types in a Reverse DNS zone are NS and PTR.

There are several different zone naming conventions used to delegate Reverse DNS, so you must contact your ISP to find out what zone name to enter under SmartDNS. Here are some examples showing common zone naming conventions:

"Class C" delegation using 1.2.3.0/24 subnet:

3.2.1.in-addr.arpa - notice that it begins with the first three octets backwards

"Less than Class C" delegation using 1.2.3.0/25 subnet:

0.3.2.1.in-addr.arpa - "first octet" convention

0/25.3.2.1.in-addr.arpa - "first octet slash mask bits" convention

0-25.3.2.1.in-addr.arpa - "first octet dash mask bits" convention
0-127.3.2.1.in-addr.arpa - "first octet dash last octet" convention

TCP Congestion Control

This feature is available as an optional add-on feature. See Chapter 12 for details.

Site Load Balancing

This feature is available as an optional add-on feature. See Chapter 8 for details.

Server Load Balancing

This feature is available as an optional add-on feature. See Chapter 9 for details.

Chapter 5: Routing

MPVPN supports the hosting of internal servers including web, e-mail, firewall, and load balancing servers. It features Inbound Policy to control inbound sessions and Outbound Policy for outbound load balancing.

Application Profile

Application Profile feature on FatPipe can be used to define several line conditions with separate template name and can be used where ever required in Policy Routing Rules. Click on Routing button and select Application Profile tab to add templates and select applications (see Figure 5.1).

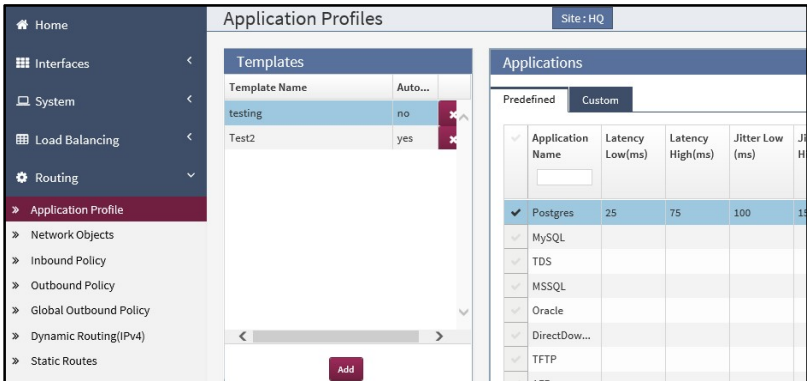


Figure 5.1 - Application Profile

Click on Add to get a Template name. The default name is Template-1 and the number keeps on increasing by adding templates. It can be edited by clicking on the name or deleted by clicking on the cross mark. Select yes or no for Auto Deploy.

Predefined

Predefined tab in Application Profile has a list of Applications that are already represented to select. Select the applications needed for each Template and fill Latency, Jitter, Packet loss, Bandwidth and Priority parameters. Search by typing application name in the search box under Application Name from the list of applications.

Custom

Custom tab under Application Profile is for creating an application by clicking on Add and give application name, port range, host name, if TCP protocol or not, latency, jitter, packet Loss, Bandwidth and Priority. The default Application Name is App-1 but can be changed according to usage. To delete an application, check mark it and click on delete button (see Figure 5.2).

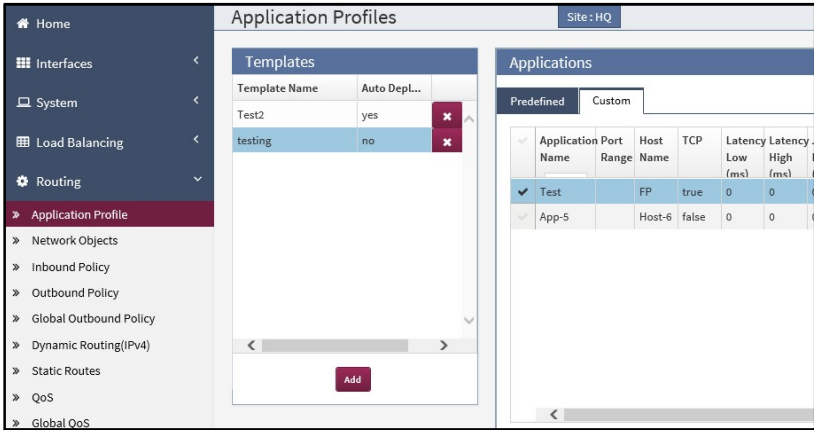


Figure 5.2 - Custom Applications

Network Objects

Network Objects helps to differentiate IP addresses into various types of network entities. These network entities are used to represent Inbound and Outbound PRR. Click on Routing button and select Network Objects tab to get in (see Figure 5.3)

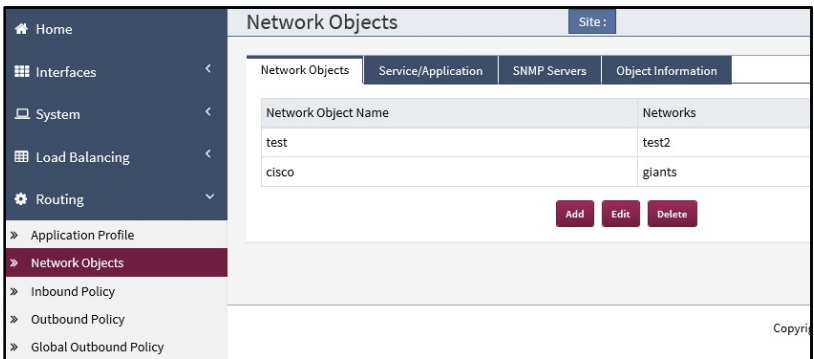


Figure 5.3 - Network Objects

Click on Add to insert a new network object to the list. Enter object name, click on the New button to add a new network select an existing network and click on edit button. Select a Network Name and Address type if it is Static or Dynamic and enter the static IP range. Once after entering the details click on Finish and save the changes (see Figure 5.4)

Add/Edit Network Objects

Network Object Name: ☐ Template

Network Name	Exclude Network	Address
test2	false	Static IPv4 : 192.168.1.0-192.168.1.254

New Edit Delete

Network

Network Name: ☐ Exclude this Network

Network

Address Type: Static IPv4 Range:
(Comma separated ranges in network prefix or using single)

Figure 5.4 - Add/Edit Network Objects

Service/Application

Service or Application in Network Objects is used to identify using port number used or protocol. Select Service/Application tab from Network Objects (see Figure 5.5).

Network Objects Site:

Network Objects Service/Application SNMP Servers Object Information

Service Name	Service/Application(s)
GL	2Wire CSS,2Wire RPC

Add Edit Delete

Copyright

Figure 5.5 -Service/Application

Click on Add button to create a new service and give it a name and select the name of service from a predefined list of services. Click on add or remove after selecting a service to move it to selected. Customized tab is used to add service name manually, select a protocol and enter port number. Search option can be used to look for a specific service from the list (see Figure 5.6)

Add/Edit Service or Application

Name: ☐ Template

Predefined Customized

Search:

Name	Port	Protocol
1Ci Server Management	3091	
1Ci Server Management	3091	
2Wire CSS	3428	
2Wire CSS	3428	
2Wire RPC	3479	
2Wire RPC	3479	

Add -> <- Remove

Name	Port
2Wire CSS	3428
2Wire RPC	3479

Figure 5.6 -Add/Edit Service or Application

SNMP Servers

Used to gather information about routes from other SNMP devices in the network. All the SNMP servers can be added here. Maximum SNMP servers limit is 1.Click on SNMP Servers tab under Network Objects to add an SNMP Server. (see Figure 5.7)

Network Objects Site:

Network Objects Service/Application **SNMP Servers** Object Information

Server Name	Server IP	Port	Version
group	172.16.52.24	8989	2

Add Edit Delete

Application Profile
Network Objects
 Inbound Policy
 Outbound Policy
 Global Outbound Policy

Figure 5.7 - SNMP Servers

To add an SNMP server, click on Add button and give server name, IP Address, port number, version, Community it belongs to and password (see Figure 5.8). Select the server name and click on Edit button to make any changes

or click on Delete to remove the details and click on Save to record the changes

Add SNMP Server	
Name	IP Address
group	172.16.52.24
Port	Version
8989	2
Community	Password
FP	••••

Figure 5.8 - Add SNMP Server

Inbound Policy

Inbound Policies are configured to allow access from the WAN to servers or machines on the LAN. By default, the FatPipe unit denies all inbound connections. To override this default action, Inbound Policies can be created. Inbound Policy, short for Inbound Policy Routing, applies to any traffic that is initiated on the outside (WAN side) of MPVPN coming in. Any traffic matched by these inbound traffic rules (also called inbound policy route rules) will be handled based on the settings of the rule. If you have the QoS add-on, you can apply QoS rules to your inbound policy route rules.

If you have used a prior version of our software, please note that we have now combined the functionality of Pass-Through and Reverse Mapping into one page called Inbound Policy. This change was necessary to facilitate the use of QoS with inbound policy route rules. Each rule can be configured to forward traffic inbound with or without doing Reverse Mapping (NAT).

To configure Inbound Policy route rules, click on the Routing button in the main menu and click on the Inbound Policy tab (see Figure 5.9).

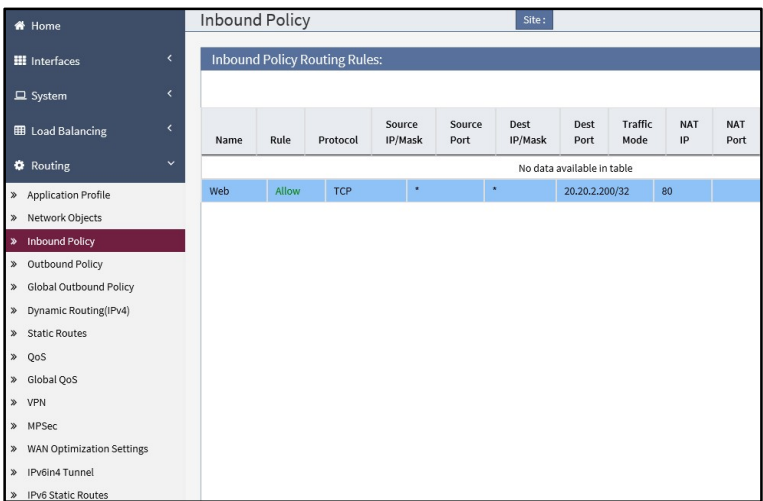


Figure 5.9 - Inbound Policy Routing

To add a new inbound policy routing rule, click on the Add button (see Figure 5.10), click on SAVE button to make the changes permanent.

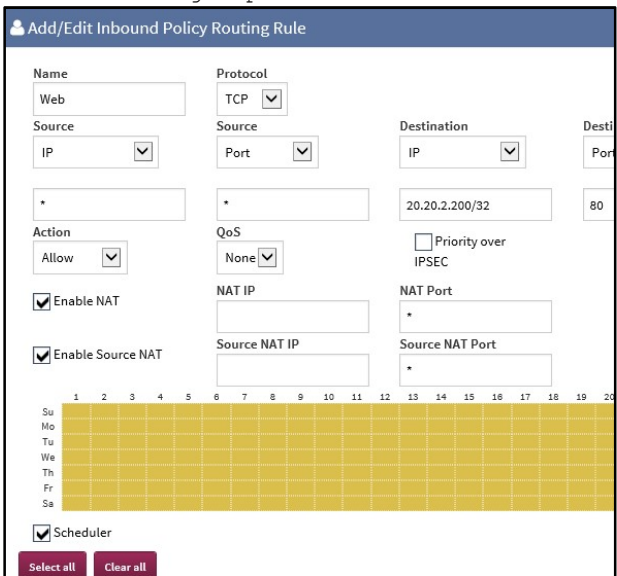


Figure 5.10 - Add Inbound Policy Routing Rule

Name

You can give each rule a unique name. Use this to identify the purpose of the rule.

Protocol

Choose an IP protocol from the list. ALL will match all protocols. Also note that port numbers only apply when using TCP or UDP.

Source IP/Mask

Specify a source IP and mask (using CIDR notation). If you want to match a single IP, use a /32 mask (e.g., 1.2.3.4/32). If you want to match an entire subnet, use the network number with the network mask (e.g., 1.2.3.0/24). If you want to match any IP, use an asterisk (*). MPVPN will display asterisk (*) as 0.0.0.0/0 meaning all IP's.

Source Port

Specify a single port number or a port range separated by a hyphen (e.g., 1-1023). If you want to match any port number, use an asterisk (*).

Note: The Source Port will be enabled only for "TCP/UDP" protocols. All other protocols will be grayed out.

Destination IP/Mask

Specify a destination IP and mask (using bit notation). If you want to match a single IP, use a /32 mask (e.g., 1.2.3.4/32). If you want to match an entire subnet, use the network number with the network mask (e.g., 1.2.3.0/24). If you want to match any IP, use an asterisk (*). MPVPN will display asterisk (*) as 0.0.0.0/0 meaning all IP's.

Destination Port

Specify a single port number or a port range separated by a hyphen (e.g., 1-1023). If you want to match any port number, use an asterisk (*).

Note: The Destination Port will be enabled only for "TCP/UDP" protocols. All other protocols will be grayed out.

Action

Choose "Allow" to allow traffic that matches the rule. Choose "Deny" to deny traffic that matches the rule.

WAN-WAN action

WAN-WAN action serves as an outbound policy for the traffic matching the inbound rule. The rule has to be configured in Outbound Policy Routing fashion, but the source subnet should be that of remote network, whose traffic would reach FatPipe through a WAN interface (especially private) or through a VPN tunnel. This traffic is routed to the Internet or a specific subnet through another WAN interface or through the same WAN interface (in case of IPSec). The destination subnet will be * incase the traffic is routed to the Internet (see Figure 5.11) or a network (see Figure 5.12) depending on user requirement.

Add/Edit Inbound Policy Routing Rule

Name: toInternet

Protocol: TCP

Source: IP

Destination: IP

Action: WAN-WAN

QoS: None

☐ Priority over IPSEC

Traffic Mode: ☐ Interface Priority ☐ Interface Specific

Interface	NAT	Port NAT	NAT IP/Mask
WAN1	Yes	Yes	
WAN2	Yes	Yes	
WAN3	Yes	Yes	

Add Edit Delete Up Down

Su Mo Tu We Th Fr Sa

☒ Scheduler

Figure 5.11 - WAN-WAN routed to Internet

Add/Edit Inbound Policy Routing Rule

Name: toNetwork

Protocol: All

Source: IP

Destination: IP

Action: WAN-WAN

QoS: None

☐ Priority over IPSEC

Traffic Mode: ☐ Interface Priority ☐ Interface Specific

Interface	NAT	Port NAT	NAT IP/Mask
WAN3	Yes	No	0.0.0.0-0.0.0.0

Add Edit Delete Up Down

Su Mo Tu We Th Fr Sa

☐ Scheduler

Figure 5.12 - WAN-WAN routed to Network

Quality of Service

Choose a pre-defined QoS rule that will apply to the traffic matched by this policy route rule. (QoS is a feature add-on). The default is "None."

Enable NAT

Check this box if you want to NAT traffic that matches this rule.

NAT IP

Specify the IP and subnet mask (using bit notation) that the traffic will be mapped to. If you want to map the traffic to a single IP, use a /32 mask (e.g., 1.1.1.1/32). If you want to map the traffic one-to-one, use a full subnet mask (e.g., 1.1.1.0/24).

NAT Port

Here is where you specify the port number the traffic will be mapped to. If you want to map all ports, use an asterisk (*).

Please note that if you do not select NAT, then the rule will default to Pass-Through, which means that MPVPN simply forwards traffic matching the rule.

This requires that you use a smaller subnet -- typically a /30 (255.255.255.252) subnet -- on the corresponding WAN interface of MPVPN. The router, firewall, and any other device with a public IP will be assigned the full subnet mask. The LAN interface of MPVPN will also be assigned the full subnet mask. MPVPN will use Proxy ARP to receive the traffic and route it back to the LAN for any IPs that are part of the Destination IP/Mask.

Scheduler

All policy routing rules will be followed at all times unless you specify times and days for specific policies to run using the Scheduler. If you want a particular rule to be followed, including QoS rules, during a specific period, then it can be scheduled using the scheduler.

To provide a schedule, enable the Scheduler by checking the Scheduler checkbox. The scheduler allows for configuring a schedule on a weekly basis. Every cell represents an hour of the day. You can select a cell by clicking on it wherein it will turn green. Clicking on a selected cell will de-select it. You can choose multiple cells by click-dragging the mouse over the cells. To select all the cells, click on the Select all button. To clear all the cells click on the Clear all button. Click on the OK button to return to the Inbound Policy page. Click on SAVE button to make the changes permanent.

Note: Rule matching is done from the top to the bottom of the rule list. If a packet matches more than one rule, then the routing will be decided based on the first rule that matches it from the top. So the order of the rules is very important. You can change the order of a rule by choosing the rule and moving it to the top, or one level up or down, or to the bottom by clicking the appropriate buttons on the left of the rules list.

To edit an inbound policy rule, select it from the list and click on the Edit button, click on SAVE button to make the changes permanent (see Figure 5.13).

Add/Edit Inbound Policy Routing Rule

Name

Web

Source

IP

*

Action

WAN-WAN

Protocol

TCP

Source

Port

*

QoS

None

Destination

IP

20.20.2.200/32

☐ Priority over IPSEC

Destin

Port

80

Traffic Mode:

☐ Interface Priority

☒ Interface Specific

Interface	NAT	Port NAT	NAT IP/Mask	NA
WAN1	Yes	Yes		
WAN2	Yes	Yes		
WAN3	Yes	Yes		
WAN4	Yes	Yes		
WAN5	Yes	Yes		

Add

Edit

Delete

Up

Down

Su

Mo

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Figure 5.13 – Edit Inbound Policy Routing Rule

To delete an inbound policy rule, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

Clear Session

You can clear all sessions that match the inbound policy routing rule you have selected.

Session Info

You can view all sessions that match the inbound policy routing rule you have selected.

Below are screenshots of example Inbound Policies that pass traffic through to the LAN side:

Inbound Policies - Directs Inbound sessions							
Pass-Through Equivalent Inbound Rules							
Proto col	Source IP	Sour ce Port	Destination IP	Destinat ion Port	Enabl e NAT	NAT IP	NAT Por t
All	*	*	11.1.5.50./32	*	No	-	-
TCP	25.25.25.25/32	*	11.1.5.16./32	*	No	-	-

TCP	25.25.25.25/32	*	11.1.5.16/32	80	No	-	-
UDP	25.25.25.0/24	*	11.1.5.0/24	*	No	-	-
ICMP	*	-	11.1.5.25/32	-	No	-	-

The policy shown below ensures that all traffic originating from Internet destined to host 11.1.5.50 will be allowed inside (see Figure 5.14).

Add/Edit Inbound Policy Routing Rule

Name

RemoteHost

Source

IP

Action

Allow

☐ Enable NAT

☐ Enable Source NAT

Protocol

All

Source

Port

QoS

None

NAT IP

Source NAT IP

Destination

IP

11.1.5.50/24

X

☐ Priority over IPSEC

NAT Port

Source NAT Port

Su

Mo

Tu

We

Th

Fr

Sa

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

☐ Scheduler

Select all

Clear all

Figure 5.14 - Internet Traffic

The below rule ensures that all TCP traffic originating from a remote IP of 25.25.25.25 destined to the LAN IP of 11.1.5.16 will be allowed inside (see Figure 5.15).

Add/Edit Inbound Policy Routing Rule

Name

RemoteHost

Source

IP

25.25.25.25/32

Action

Allow

☐ Enable NAT

☐ Enable Source NAT

Protocol

TCP

Source

Port

*

QoS

None

NAT IP

Source NAT IP

Destination

IP

11.1.5.16/32

☐ Priority over IPSEC

NAT Port

Source NAT Port

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Su

Mo

Tu

We

Th

Fr

Sa

☐ Scheduler

Select all

Clear all

Figure 5.15 - TCP traffic

The rule ensures that all TCP traffic originating from a remote IP of 25.25.25.25 destined to port 80 of the LAN IP 11.1.5.16 will be allowed inside (see Figure 5.16).

Add/Edit Inbound Policy Routing Rule

Name

RemoteHost

Source

IP

25.25.25.25/32

Action

Allow

☐ Enable NAT

☐ Enable Source NAT

Protocol

TCP

Source

Port

*

QoS

None

NAT IP

Source NAT IP

Destination

IP

11.1.5.16/32

☐ Priority over IPSEC

NAT Port

Source NAT Port

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Su

Mo

Tu

We

Th

Fr

Sa

☐ Scheduler

Select all

Clear all

Figure 5.16 - TCP traffic

The rule ensures that all UDP traffic originating from a remote 25.25.25.0 network destined to a local 11.1.5.0 network will be allowed inside (see Figure 5.17).

The screenshot shows the 'Add/Edit Inbound Policy Routing Rule' window. The 'Name' field is 'RemoteHost'. The 'Protocol' is 'UDP'. The 'Source' is 'IP' with a dropdown menu. The 'Destination' is 'IP' with a dropdown menu. The 'Source' field contains '25.25.25.0/24'. The 'Destination' field contains '11.1.5.0/24'. The 'Action' is 'Allow'. The 'QoS' is 'None'. The 'Priority over IPSEC' checkbox is unchecked. The 'Enable NAT' checkbox is unchecked. The 'Enable Source NAT' checkbox is unchecked. The 'NAT IP' field is empty. The 'Source NAT IP' field is empty. The 'NAT Port' field is empty. The 'Source NAT Port' field is empty. Below the fields is a grid for scheduling, with columns numbered 1 to 24 and rows labeled Su, Mo, Tu, We, Th, Fr, Sa. The 'Scheduler' checkbox is unchecked. At the bottom are 'Select all' and 'Clear all' buttons.

Figure 5.17 - UDP traffic

The rule ensures that all ICMP traffic originating from the Internet destined to a LAN host 11.1.5.25 will be allowed inside (see Figure 5.18).

The screenshot shows the 'Add/Edit Inbound Policy Routing Rule' window. The 'Name' field is 'RemoteHost'. The 'Protocol' is 'ICMP'. The 'Source' is 'IP' with a dropdown menu. The 'Destination' is 'IP' with a dropdown menu. The 'Source' field contains '*'. The 'Destination' field contains '11.1.5.25/32'. The 'Action' is 'Allow'. The 'QoS' is 'None'. The 'Priority over IPSEC' checkbox is unchecked. The 'Enable NAT' checkbox is unchecked. The 'Enable Source NAT' checkbox is unchecked. The 'NAT IP' field is empty. The 'Source NAT IP' field is empty. The 'NAT Port' field is empty. The 'Source NAT Port' field is empty. Below the fields is a grid for scheduling, with columns numbered 1 to 24 and rows labeled Su, Mo, Tu, We, Th, Fr, Sa. The 'Scheduler' checkbox is unchecked. At the bottom are 'Select all' and 'Clear all' buttons.

Figure 5.18 - ICMP traffic

Below are screenshots of some of the example Inbound Policies that reverse map (NAT) specific hosts from the WAN side to the LAN side:

Inbound Policies - Directs Inbound sessions							
Reverse Mapping Equivalent Inbound Rules							
Protocol	Source IP	Source Port	Destination IP	Destination Port	Enable NAT	NAT IP	NAT Port
All	*	*	11.1.5.50/32	*	Yes	10.2.0.10/32	*
TCP	25.25.25.25/32	*	11.1.5.16/32	*	Yes	10.2.0.25/32	*
TCP	25.25.25.25/32	*	11.1.5.17/32	80	Yes	10.2.0.17/32	80
TCP	25.25.25.25/32	*	11.1.5.17/32	2222	Yes	10.2.0.18/32	80
TCP	*	*	11.1.5.25/32	20-21	Yes	10.2.0.18/32	20-21
UDP	25.25.25.0/24	*	11.1.5.0/24	*	Yes	10.2.0.0/24	*
ICMP	*	-	11.1.5.5/32	-	Yes	10.2.0.25/32	-

The below rule ensures that all traffic originating from Internet that are destined to 11.1.5.50 host gets NATed to 10.2.0.10 and is allowed inside (see Figure 5.19).

Add/Edit Inbound Policy Routing Rule

Name RemoteHost		Protocol All	
Source IP	Source Port	Destination IP	Destination Port
*	*	11.1.5.50/32	*
Action Allow	QoS None	☐ Priority over IPSEC	
☒ Enable NAT	NAT IP 10.2.0.10/32	NAT Port	
☐ Enable Source NAT	Source NAT IP	Source NAT Port	

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Su																				
Mo																				
Tu																				
We																				
Th																				
Fr																				
Sa																				

☐ Scheduler

Select all Clear all

Figure 5.19 – NAT Internet traffic

The below rule ensures that all TCP traffic originating from 25.25.25.25 with any port that are destined to 11.1.5.16 host with any port gets NATed to 10.2.0.25 and is allowed inside (see Figure 5.20).

Add/Edit Inbound Policy Routing Rule

Name RemoteHost		Protocol TCP	
Source IP	Source Port	Destination IP	Destination Port
25.25.25.25/32	*	11.1.5.16/32	*
Action Allow	QoS None	☐ Priority over IPSEC	
☒ Enable NAT	NAT IP 10.2.0.25/32	NAT Port	
☐ Enable Source NAT	Source NAT IP	Source NAT Port	

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Su																				
Mo																				
Tu																				
We																				
Th																				
Fr																				
Sa																				

☐ Scheduler

Select all Clear all

Figure 5.20 – NAT TCP traffic

The below rule ensures that all TCP traffic originating from host 25.25.25.25 with any port that are destined to 11.1.5.17 host with port 80 gets NATed to 10.2.0.17 with port 80 and is allowed inside (see Figure 5.21).

Add/Edit Inbound Policy Routing Rule

Name

RemoteHost

Source

IP

25.25.25.25/32

Action

Allow

☒ Enable NAT

☐ Enable Source NAT

Protocol

TCP

Source

Port

*

QoS

None

NAT IP

10.2.0.17/32

Source NAT IP

Destination

IP

11.1.5.17/32

☐ Priority over IPSEC

NAT Port

80

Source NAT Port

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Su

Mo

Tu

We

Th

Fr

Sa

☐ Scheduler

Select all

Clear all

Figure 5.21 – NAT TCP traffic

The below rule ensures that all TCP traffic originating from host 25.25.25.25 with any port that are destined to 11.1.5.17 host with port 2222 gets NATed to 10.2.0.18 with port 80 and is allowed inside (see Figure 5.22).

Add/Edit Inbound Policy Routing Rule

Name

RemoteHost

Source

IP

25.25.25.25/32

Action

Allow

☒ Enable NAT

☐ Enable Source NAT

Protocol

TCP

Source

Port

*

QoS

None

NAT IP

10.2.0.18/32

Source NAT IP

Destination

IP

11.1.5.17/32

☐ Priority over IPSEC

NAT Port

80

Source NAT Port

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Su

Mo

Tu

We

Th

Fr

Sa

☐ Scheduler

Select all

Clear all

Figure 5.22 – Nat TCP traffic

The below rule ensures that all TCP traffic originating from Internet that are destined to 11.1.5.25 host with port range 20-21 gets NATed to 10.2.0.18 with port range 20-21 and is allowed inside (see Figure 5.23).

The screenshot shows the 'Add/Edit Inbound Policy Routing Rule' window. The 'Name' field is 'RemoteHost'. The 'Protocol' is 'TCP'. The 'Source' is 'IP' with a dropdown menu. The 'Destination' is 'IP' with a dropdown menu. The 'Source' field contains '25.25.25.25/32'. The 'Destination' field contains '11.1.5.25/32'. The 'Action' is 'Allow'. The 'QoS' is 'None'. The 'NAT IP' is '10.2.0.18/32'. The 'NAT Port' is '20-21'. The 'Source NAT IP' is empty. The 'Source NAT Port' is empty. The 'Enable NAT' checkbox is checked. The 'Enable Source NAT' checkbox is unchecked. The 'Scheduler' checkbox is unchecked. The 'Select all' and 'Clear all' buttons are at the bottom.

Figure 5.23 - NAT TCP traffic

The below rule ensures that all UDP traffic originating from 25.25.25.0 network with any port that are destined to 11.1.5.0 network with any port gets NATed to 10.2.0.0 network with any port and is allowed inside (see Figure 5.24).

The screenshot shows the 'Add/Edit Inbound Policy Routing Rule' window. The 'Name' field is 'RemoteHost'. The 'Protocol' is 'UDP'. The 'Source' is 'IP' with a dropdown menu. The 'Destination' is 'IP' with a dropdown menu. The 'Source' field contains '25.25.25.0/24'. The 'Destination' field contains '11.1.5.0/24'. The 'Action' is 'Allow'. The 'QoS' is 'None'. The 'NAT IP' is '10.2.0.0/24'. The 'NAT Port' is '1'. The 'Source NAT IP' is empty. The 'Source NAT Port' is empty. The 'Enable NAT' checkbox is checked. The 'Enable Source NAT' checkbox is unchecked. The 'Scheduler' checkbox is unchecked. The 'Select all' and 'Clear all' buttons are at the bottom.

Figure 5.24 - NAT UDP traffic

The below rule ensures that all ICMP traffic originating from Internet that are destined to 11.1.5.5 host gets NATed to 10.2.0.25 and is allowed inside (see Figure 5.25).

Add/Edit Inbound Policy Routing Rule

Name: RemoteHost

Protocol: ICMP

Source: IP

Destination: IP

Action: Allow

☒ Enable NAT

☐ Enable Source NAT

QoS: None

NAT IP: 10.2.0.25/32

Source NAT IP: 10.2.0.25/32

NAT Port: *

Source NAT Port: *

☐ Scheduler

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Su																				
Mo																				
Tu																				
We																				
Th																				
Fr																				
Sa																				

Select all Clear all

Figure 5.25 - NAT ICMP traffic

Outbound Policy

Outbound Policy, short for Outbound Policy Routing, applies to any traffic that is initiated on the inside (LAN side) of MPVPN going out. Any traffic matched by these outbound policy route rules will be treated differently than the default load balanced and NATed traffic. If you have the QoS add-on, you can apply QoS rules to your outbound policy route rules. To configure Outbound Policy route rules, click on the Routing button in the main menu and click on the Outbound Policy tab (see Figure 5.26).

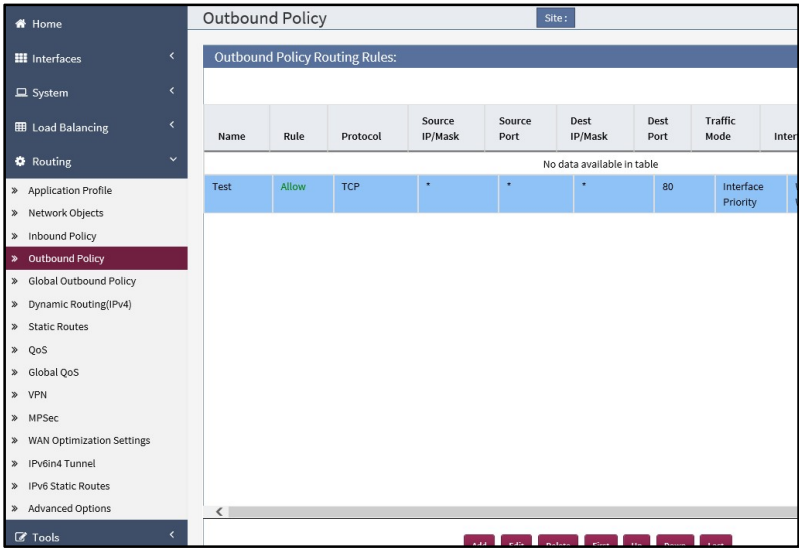


Figure 5.26 – Outbound Policy Routing

To add a new outbound policy routing rule, click on the Add button (see Figure 5.27), click on SAVE button to make the changes permanent.

Add Outbound Policy Routing Rule

Name:

Protocol:

DSCP:

Source:

Source Port:

Destination:

Action:

QoS:

Application Profiles:

☐ Source NAT

☐ Follow System Route ☐ Equal Bandwidth Distribution ☐ Maintain Session Persist

Traffic Mode:

☒ Interface Priority ☐ Interface Specific ☐ Mixed Priority

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	Packet
WAN1	Yes	Yes			No		No			
WAN2	Yes	Yes			No		No			
WAN3	Yes	Yes			No		No			

Figure 5.27 – Add Outbound Policy Routing Rules

Name

You can give each rule a unique name. Use this to identify the purpose of the rule.

Protocol

Choose an IP protocol from the list. ALL will match all protocols. Also note that port numbers only apply when using TCP or UDP.

DSCP (Differentiated Services Code Point)

FatPipe helps perform traffic shaping based solely on Differentiated Services Code Points (DSCP) on outgoing packets. Specify the DSCP value that you want to match to the outbound session. FatPipe will check the DSCP value in the outgoing packets with the DSCP value that is configured in the outbound policy routing rule. If it matches, then it will follow the actions specified in the policy routing rule. The default value is 0.

Note: FatPipe will not set the DSCP value in the packets. It will only check for this value in the outgoing packets. To set DSCP values in the outgoing packet see the WAN Interface List section.

Source IP/Mask

Specify a source IP and mask (using bit notation). If you want to match an IP, use a /32 mask (e.g., 1.2.3.4/32). If you want to match an entire subnet, use the network number with the network mask (e.g., 1.2.3.0/24). If you want to match any IP, use an asterisk (*).MPVPN will display asterisk (*) as 0.0.0.0/0 meaning all IP's.

Source Port

Specify a single port number or a port range separated by a hyphen (e.g., 1-1023). If you want to match any port number, use an asterisk (*).

Note: The Source Port will be enabled only for "TCP/UDP" protocols. All other protocols will be grayed out.

Destination IP/Mask or Domain Name

Specify a destination IP and mask (using bit notation). If you want to match an IP, use a /32 mask (e.g., 1.2.3.4/32). If you want to match an entire subnet, use the network number with the network mask (e.g., 1.2.3.0/24). If you want to match any IP, use an asterisk (*).

Specify Domain Name of the destination. E.g. www.FatPipeinc.com, www.salesforce.com, www.linux.org and so on. At present wild cards are not supported with the domain names.

Destination Port

Specify a single port number or a port range separated by a hyphen (e.g., 1-1023). If you want to match any port number, use an asterisk (*).

Note: The Destination Port will be enabled only for "TCP/UDP" protocols. All other protocols will be grayed out.

Application Rules

Select the Application QoS (rule) and Action from the drop down menus (see Figure 5.28).

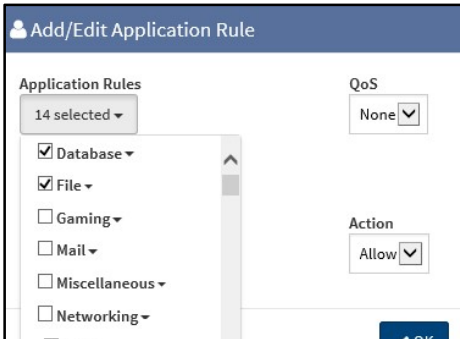


Figure 5.28 - Add/Edit Application Rule

Action

Choose "Allow" to allow traffic that matches the rule.
Choose "Deny" to deny traffic that matches the rule.

Quality of Service

Choose a pre-defined QoS rule that will apply to the traffic matched by this policy route rule. Default is "None."

Equal Bandwidth Distribution

This option ensures that all matching sessions will be provided with equal bandwidth within the selected QoS committed bandwidth. Every time a new session is created, the bandwidth of the existing sessions is re-adjusted to accommodate the new session. Every time a session expires, the bandwidth of the existing sessions is re-adjusted.

Note: This feature depends on QoS and will be available only when you choose a QoS rule from the dropdown.

Traffic Mode

Interface Priority directs traffic out the first live line, using the WAN interface order you specify.

Interface Specific Mode load balances the traffic based on the Load Balancing Algorithm between the line(s) chosen in the WAN list. For example, if you want to send traffic out WAN2 and WAN3 only, then you need to choose only these two interfaces in the WAN list and remove all other interfaces.

Mixed Priority allows the user to derive the benefits of both Interface Priority and Interface Specific Modes in a single rule. You can assign priorities to the WAN interfaces and traffic will be sent out of the high priority lines as long as those lines are up. If 2 or more lines have the same priority, then traffic will be load balanced between those lines based on the load balancing algorithm you selected. For example, assume you want to send a particular type of traffic out of WAN1 as long as it is up and you want the traffic to go out either WAN2 or WAN3 if WAN1 goes down. You can set the priority of WAN1 to 1 and the priority of WAN2 and WAN3 as 2. This way traffic will go out of WAN1 as long as WAN1 is up and it will get load balanced between WAN2 and WAN3 if WAN1 goes down.

WAN Interface List

You can enter a list of WAN interfaces that you want this policy route rule to use. For each interface you can specify whether or not you want to do NAT. If you do use

NAT, you can specify whether you want to NAT to a specific IP and port or if you want to have the system automatically assign an IP and port. The IP will be the IP of the WAN interface the traffic goes out. Click Add to choose the WAN interface you want to use (see Figure 5.29). Then select whether or not to use NAT and/or Port NAT. If the Auto options are chosen, the system will handle the IP and port assignments for you dynamically. This is recommended in most scenarios.

Enable DSCP Tagging

This will enable tagging of packets based on the DSCP tag value provided.

Enable Threshold Based Session Failover

This option provides a higher level of failover based on the thresholds defined for Latency, jitter and packet loss. If any of the parameters crosses the defined threshold, the sessions will fail over to the next link even if the link on which the sessions are established is UP.

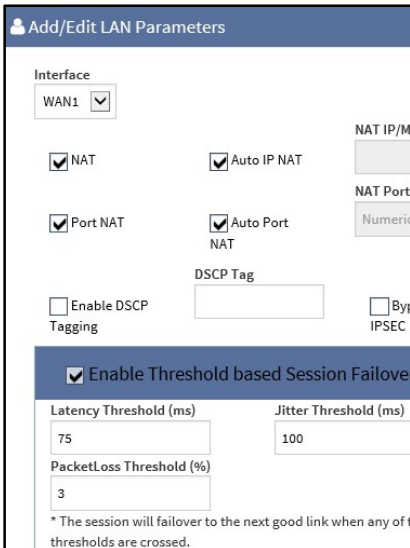


Figure 5.29 - Add/Edit WAN Parameters

Follow System Route

Enable Follow System Route check box to ensure matching Static Route entry takes precedence over default Load balancing behavior for the configured Destination subnet or host.

Scheduler

Policy Routing rules are in effect at all times. However, you can schedule different Policy Routing rules and QoS rules (QoS is an add-on feature) to run at different times and on different days by using FatPipe MPVPN's Scheduler.

To setup a schedule, enable the Scheduler by clicking on the checkbox. The Scheduler allows for configuring a schedule on a weekly basis. Every cell represents an hour of the day. You can select a cell by clicking on it. It will turn green. Clicking on a selected cell will de-select it. You can choose multiple cells by click-dragging the mouse over the cells. To select all the cells, click on the Select all button and to clear all the cells click on Clear all button. Click on Ok button to return to the Outbound Policy page. Click on SAVE button to make the changes permanent.

UDP Aggregation

This feature aggregates smaller UDP packets into a bigger UDP packets thereby reducing bandwidth consumption. See Chapter 12 for details.

HTTPs Acceleration

Selecting this feature ensures optimization of SSL based traffic matching this rule. See Chapter 12 for details.

WAN Optimization

Selecting this feature ensures optimization of traffic matching this rule. See Chapter 12 for details.

Note: The order of the rules is important. Rule matching is done from top to bottom of the rule list. If a packet matches more than 1 rule, then the routing will be decided based on the first rule that matches it from the top. You can change the order of a rule by choosing the rule and moving it to the top or one level up or down, or to the bottom by clicking the appropriate buttons on the right of the rules list.

Note: Global Outbound Policy Top rule takes precedence over any Outbound Policy Routing Rule (please refer to Chapter 14: Central Manager for further details)

To edit an Outbound Policy Rule, select it from the list and click on the Edit button. Click on the SAVE button to make the changes permanent (see Figure 5.30).

Add Outbound Policy Routing Rule

Name

SSL

Protocol

TCP

DSCP

5

UDP Pa

Enable

Enable

Enable

Source

IP

Source

Port

Destination

IP

Destina

Port

*

*

*

*

Action

Allow

QoS

None

Application Profiles

None

Applica

Add

Edit

Delete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Sess

Traffic Mode:

☒ Interface Priority

☐ Interface Specific

☐ Mixed Priority

Interface	NAT	Port	NAT	DSCP	Enable	Latency	Jitter	Pac		
Interface	NAT	NAT	IP/Mask	Port	Tagging	Value	DynLoadBalOpt	Threshold	Threshold	Thre
WAN1	Yes	Yes			No		No			

Figure 5.30 – Edit Outbound Policy Routing Rule

To delete an Outbound Policy Rule, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

Clear Session

You can clear all sessions that match the outbound policy routing rule you have selected.

View Session

You can view all sessions that match the outbound policy routing rule you have selected.

Below are five Outbound Policies that should be added to the FatPipe: (Assuming WAN 1 shares the LAN IP block)

Default Outbound Policies – Directs Outbound Sessions									
Protocol	Source IP	Srce Port	Dest IP	Dest Port	Trfc Mode	Interface	NAT		
UDP	*	*	*	500	Specific	1	No		
ESP	*	*	*	*	Specific	1	No		
TCP	*	*	*	443	Priority	1,2,3	All		
TCP	*	*	*	1723	Specific	1	No		
GRE	*	*	*	*	Specific	1	No		

One additional Outbound Policy that may need to be created is for outbound Mail traffic (port 25). If a PTR record exists for the public IP of the Mail server, then that traffic must be directed out the WAN interface of the FatPipe that shares the same IP block as the LAN. See the example below:

Outbound Mail (port 25) Outbound Policy							
Protocol	Source IP	Srce Port	Dest IP	Dest Port	Trfc Mode	Interface	NAT
TCP	*	*	*	25	Specific	1	No

This policy above would send outbound Mail (port 25) traffic out WAN 1 as its original source IP -- the public IP that was given by the Firewall if private IPs exist on the LAN -- ensuring PTR reverse lookup compatibility. To provide outbound load balancing and redundancy for your outbound mail (port 25), it is suggested to have a PTR record created for each of the additional WAN IPs of the FatPipe that point back to the original name belonging to the actual public IP of the mail server. See the example below to see what the policy would look like for load balancing.

Outbound Mail (port 25) Outbound Policy							
Protocol	Source IP	Srce Port	Dest IP	Dest Port	Trfc Mode	Interface	NAT
TCP	*	*	*	25	Specific	1,2	NAT WAN 2 only

Below are screenshots of some of the default and other miscellaneous Outbound Policies:
HTTPS/SSL Outbound Policy (WAN 2 set as priority to failover to WAN 1 in case of failure and then to WAN 3) (see Figure 5.31).

Add Outbound Policy Routing Rule

Name

SSL

Protocol

TCP

DSCP

5

UDP Port

Enable

Source

IP

Source

Port

Destination

IP

Destination

Port

*

*

*

*

Action

Allow

QoS

None

Application Profiles

None

Application

Add

Edit

Delete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☒ Interface Priority

☐ Interface Specific

☐ Mixed Priority

Interface	NAT	Port	NAT	NAT	DSCP	Value	Enable	Latency	Jitter	Pack
			IP/Mask	Port	Tagging		DynLoadBalOpt	Threshold	Threshold	Thre
WAN2	Yes	Yes			No		No			

Figure 5.31 - HTTPS/SSL Outbound Policy

VPN UDP 500 Outbound Policy (for VPN sessions initiated behind the FatPipe utilizing UDP port 500) (see Figure 5.32).

The screenshot shows the 'Add Outbound Policy Routing Rule' configuration page. The rule is named 'VPN_UDP_500'. The protocol is set to 'UDP'. The DSCP value is '5'. The source is configured with 'IP' and a wildcard '*' address. The destination is configured with 'IP' and a wildcard '*' address. The action is set to 'Allow'. The QoS is set to 'None'. The application profiles are set to 'None'. There are buttons for 'Add', 'Edit', and 'Delete'. Below these, there are checkboxes for 'Source NAT', 'Follow System Route' (checked), 'Equal Bandwidth Distribution', and 'Maintain Session'. The 'Traffic Mode' section has three radio buttons: 'Interface Priority' (selected), 'Interface Specific', and 'Mixed Priority'. At the bottom, there is a table with columns: Port, NAT, DSCP, Enable, Latency, Jitter, and Priority.

Figure 5.32 - VPN UDP Outbound Policy

Corresponding VPN ESP Protocol Outbound Policy (see Figure 5.33).

The screenshot shows the 'Add Outbound Policy Routing Rule' configuration page. The rule is named 'VPN_ESP'. The protocol is set to 'ESP'. The DSCP value is '0'. The source is configured with 'IP' and a wildcard '*' address. The destination is configured with 'IP' and a wildcard '*' address. The action is set to 'Allow'. The QoS is set to 'None'. The application profiles are set to 'None'. There are buttons for 'Add', 'Edit', and 'Delete'. Below these, there are checkboxes for 'Source NAT', 'Follow System Route' (checked), 'Equal Bandwidth Distribution', and 'Maintain Session'. The 'Traffic Mode' section has three radio buttons: 'Interface Priority', 'Interface Specific' (selected), and 'Mixed Priority'. At the bottom, there is a table with columns: Port, NAT, DSCP, Enable, Latency, Jitter, and Priority.

Figure 5.33 - Corresponding VPN ESP protocol Outbound Policy

AIM port 5190 Outbound Policy (WAN 1 set to the Primary interface for traffic to failover to WAN 2) (see Figure 5.34).

Add Outbound Policy Routing Rule

Name	Protocol	DSCP	UDP Port
AIM	TCP	5	Enable
Source	Source	Destination	Destination
IP	Port	IP	Port
*	*	*	5190
Action	QoS	Application Profiles	Application
Allow	None	None	

Add **Edit** **Delete**

☐ Source NAT

☒ Follow System Route ☐ Equal Bandwidth Distribution ☐ Maintain Session

Traffic Mode:

☒ Interface Priority ☐ Interface Specific ☐ Mixed Priority

Interface	NAT	Port	NAT	DSCP	Enable	Latency	Jitter	Pa		
Interface	NAT	NAT	NAT IP/Mask	Port	Tagging	Value	DynLoadBalOpt	Threshold	Threshold	Th
WAN1	Yes	Yes	0.0.0.0/0.0.0.0	0	No	0	No	0	0	0

Figure 5.34 - AIM Port 5190 Outbound Policy

Outbound Policy - SMTP/Port 25 (non-load balancing - only leave WAN 1 with no NAT applied) (see Figure 5.35).

Add Outbound Policy Routing Rule

Name	Protocol	DSCP	UDP Port
SMTP_LB	TCP	0	Enable
Source	Source	Destination	Destination
IP	Port	IP	Port
*	*	*	25
Action	QoS	Application Profiles	Application
Allow	None	None	

Add **Edit** **Delete**

☐ Source NAT

☒ Follow System Route ☐ Equal Bandwidth Distribution ☐ Maintain Session

Traffic Mode:

☒ Interface Priority ☐ Interface Specific ☐ Mixed Priority

Interface	NAT	Port	NAT	DSCP	Enable	Latency	Jitter	Pa		
Interface	NAT	NAT	NAT IP/Mask	Port	Tagging	Value	DynLoadBalOpt	Threshold	Threshold	Th
WAN1	Yes	Yes	0.0.0.0/0.0.0.0	0	No	0	No	0	0	0

Figure 5.35 - SMTP/Port 25 Outbound Policy, non-load balancing

Outbound Policy - SMTP/Port 25 (load balancing - created once corresponding PTR record for WAN 2's IP address is

created with the ISP. PTR will be created for WAN 2's IP address to correspond to the actual Mail name) (see Figure 5.36).

Add Outbound Policy Routing Rule

NameSMTP_LB

ProtocolTCP

DSCP0

UDP Port Enable Enable Enable

SourceIP

SourcePort

DestinationIP

DestinationPort25

ActionAllow

QoSNone

Application ProfilesNone

Application

AddEditDelete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☒ Interface Priority☐ Interface Specific☐ Mixed Priority

InterfaceNATNATNAT IP/MaskPortDSCP TaggingValueEnableDynLoadBalOptLatency ThresholdJitter ThresholdPa

Figure 5.36 – SMTP/Port 25 Outbound Policy, load balancing

Outbound Policy – PPTP traffic utilizing TCP port 1723 (non-load balancing – Only leave WAN 2 with no NAT applied) (see Figure 5.37). Corresponding VPN PPTP GRE Outbound Policy (see Figure 5.38).

Add Outbound Policy Routing Rule

NamePPTP

ProtocolTCP

DSCP5

UDP Port Enable Enable Enable

SourceIP

SourcePort

DestinationIP

DestinationPort1723

ActionAllow

QoSNone

Application ProfilesNone

Application

AddEditDelete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☐ Interface Priority☒ Interface Specific☐ Mixed Priority

PortNATDSCPEnableLatencyJitterPa

Figure 5.37 – PPTP traffic utilizing TCP port 1723 Outbound Policy

Add Outbound Policy Routing Rule

Name

PPTP

Protocol

GRE

DSCP

5

UDP Port

Enable

Source

IP

Source

Port

Destination

IP

Destination

Port

*

*

Action

Allow

QoS

None

Application Profiles

None

Application

Add

Edit

Delete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☐ Interface Priority

☒ Interface Specific

☐ Mixed Priority

	Port	NAT	DSCP	Enable	Latency	Jitter	Pa

Figure 5.38 Corresponding VPN PPTP GRE Outbound Policy

Outbound Policy - Sending traffic out one WAN Interface to a specific remote host (non-load balancing, but will provide redundancy should the WAN interface listed at the top of the list fail. No NAT applied out WAN 1) (see Figure 5.39).

Add Outbound Policy Routing Rule

Name

RemoteHost

Protocol

All

DSCP

0

UDP Port

Enable

Source

IP

Source

Port

Destination

IP

Destination

Port

*

*

12.11.10.9/32

*

Action

Allow

QoS

None

Application Profiles

None

Application

Add

Edit

Delete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☒ Interface Priority

☐ Interface Specific

☐ Mixed Priority

Interface	Port	NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	Pa Th
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0

Figure 5.39 - Outbound Policy to specific host, non-load balancing

Outbound Policy - Sending traffic out all WAN Interfaces to a specific remote host (with load balancing and redundancy should one of the WAN interfaces listed fail - NATing all interfaces - Source IP will be FatPipe WAN) (see Figure 5.40).

Add Outbound Policy Routing Rule

Name

RemoteHost

Protocol

All

DSCP

0

UDP Pe

Enable

Enable

Enable

Source

IP

Source

Port

Destination

IP

Destina

Port

*

*

12.11.10.9/32

*

Action

Allow

QoS

None

Application Profiles

None

Applica

Add

Edit

Delete

☐ Source NAT

☒ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Sess

Traffic Mode:

☐ Interface Priority

☒ Interface Specific

☐ Mixed Priority

Interface	NAT	Port	NAT	NAT IP/Mask	Port	DSCP	Tagging	Value	Enable	DynLoadBalOpt	Latency	Jitter	Pa
											Threshold	Threshold	Th
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	0	No	0	No	0	No	0	0	0

Figure 5.40 - Outbound Policy to specific host, load balancing

Global Outbound Policy

This tab is similar to the Outbound Policy tab except the rules are created and maintained from the Central Manager Console(please refer to Chapter 14: Central Manager for further details)

Static Routes

Static Routes are used to route additional subnets that are not locally connected; they are not part of one of the Interface subnets. This section describes how to configure static routes in MPVPN.

To configure static routes, click on the Routing button in the main menu and click on the Static Routes tab (see Figure 5.41).

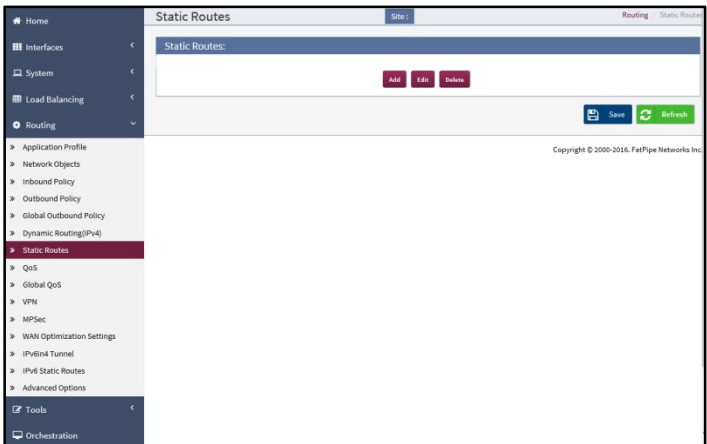


Figure 5.41 – Static Routes

To add a static route, click on the Add button (see Figure 5.42), click on the SAVE button to make the changes permanent.

Enter the Destination IP, Subnet Mask, Gateway and Metric. The Gateway should belong to one of the local subnets and should be reachable. Metric specifies the number of hops to the gateway, and is at least 2 with the way MPVPN routes.

Type: ☒ IP ☐ Network Objects	
Destination Network	Gateway
10.10.10.0	172.17.127.100
Subnet Mask	Metric
255.255.255.0	2

Figure 5.42 – Add Static Routes

To edit a static route, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 5.43).

Type: ☒ IP ☐ Network Objects	
Destination Network	Gateway
10.2.0.0	10.2.5.1
Subnet Mask	Metric
255.255.255.0	2

Figure 5.43 – Edit Static Routes

To delete a static route, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

Quality of Service (QoS)

This feature is available as an optional add-on feature. See Chapter 7 for details.

Global Quality of Service (QoS)

This tab is similar to the QoS tab except the rules are created and maintained from the Central Manager Console. This feature is available only if QoS add-on is enabled. See Chapter 14 for more details.

VPN

This feature is available as an optional add-on feature. See Chapter 10 for details.

MPSec

MPVPN features MPSec™ (Multi-Path Security) technology, which provides significantly more security of data transmission over data connections. MPSec creates multiple, independent data pathways between two or more locations. This enhances the security of the data transmission, and achieves full bandwidth utilization between locations. MPSec will direct traffic out multiple lines and if a line fails, communication will continue uninterrupted. You must have an MPVPN unit at each location to take advantage of this feature.

To configure FatPipeMPSec, click on Routing button in the main menu and click the MPsec tab (see Figure 5.44).

The screenshot displays the MPsec configuration page in a web interface. On the left is a navigation menu with options like Home, Interfaces, System, Load Balancing, and Routing (which is expanded to show sub-options like Application Profile, Network Objects, Inbound Policy, Outbound Policy, Global Outbound Policy, Dynamic Routing(IPv4), and Static Routes). The main content area is titled 'MPsec' and includes a 'Site:' dropdown. Below this, there are input fields for 'Local VPN Name' (containing 'Chicago'), 'Local VPN IP' (containing '11.11.11.20'), and 'Polling Interval (ms)' (containing '1000'). A section titled 'Remote Location' contains a table with the following data:

Index	Remote VPN Name	Remote VPN IP	Load Balancing Option
0	Denver	14.14.14.30	Packet

Below the table are buttons for 'Add', 'Edit', 'Delete', 'Up', and 'Down'. At the bottom, there is a 'Select Site Name' dropdown menu currently set to 'None'.

Figure 5.44 – FatPipeMPSec

Enter the Local VPN Name and the Local VPN IP. The Local VPN Name is user-defined and can be any descriptive name. The Local VPN IP Address is the external IP address of the local VPN device.

The Polling Interval specifies how often the FatPipe MPVPN unit checks the connections of all the entries in the table. The default is 5 seconds, which is an acceptable amount of time for stable lines. If your lines are not stable and tend to go up and down periodically, then you may want to set this polling interval to a smaller value, such as 3 seconds.

To add a Remote Location entry, click on the Add button (see Figure 5.45).

The screenshot shows a web-based configuration interface titled "Add Entry". It contains several sections for configuring a Remote VPN entry. At the top, there are input fields for "Remote VPN name" (containing "Denver") and "Remote VPN IP" (containing "14.14.14.30"). A "Load Balancing" section has two radio buttons: "Session" (selected) and "Packet". Below this is a "Dynamic Mpsec Load Balancing" section with a checked checkbox "Enable Bandwidth Detection (Kbps)" and a "Detect Bandwidth Every" input field set to "1". Further down, there are four sub-sections, each with a checked checkbox: "Use Available Bandwidth" (with a "Weight Reduce Factor" of 1), "Use Latency" (with a "Threshold (ms)" of 5000 and "Weight Red" of 1), "Use Packet Loss" (with a "Threshold (%)" of 50 and "Weight Reduce Factor" of 1), and "Use Jitter" (with a "Threshold (ms)" of 1000 and "Weight Red" of 1).

Figure 5.45 – Add Remote VPN Entry

The Remote VPN Name is user-defined and can be any descriptive name. The Remote VPN IP Address is the IP address of the remote VPN device. Your Load Balancing method can be either Session based or Packet based.

Enable Bandwidth Detection

Checking this checkbox will enable bandwidth detection between the two units on all the configured paths. The detected bandwidth will be used to manage QoS on the MPsec paths.

Dynamic MPsec Load Balancing

The Dynamic MPsec Load Balancing will load balance MPsec traffic based on dynamic weight value. Dynamic weight

value is calculated periodically based on the parameters defined and their weightage to the overall path weight. Use available Bandwidth - checking this check box will ensure the Available bandwidth on the path is taken into account for calculating the path weight. The more the upload bandwidth on the path, the greater the weights. Available bandwidth of MPsec path is the lesser of the remaining upload bandwidth in one site and remaining download bandwidth of remote side. The download bandwidth is sent through the MPsec ping packets to remote side.

Weight Reduce Factor - This factor helps to determine the weightage to be given to parameter when multiple parameters are selected. The greater the value of the Weight Reduce Factor, the lesser the weightage to the parameter in the overall path weight.

UsePacket Loss - Checking this check box will ensure that the packet loss on the path is taken into account for calculating the path weight.

You can define a threshold for this parameter. If the value of the path crosses the threshold at any time, the path is marked as bad and traffic is not sent on this path until it recovers.

Weight Reduce Factor - This factor helps to determine the weightage to be given to parameter when multiple parameters are selected. The greater the value of the Weight Reduce Factor, the lesser the weightage to the parameter in the overall path weight.

UseLatency - Checking this check box will ensure that the Latency on the path is taken into account for calculating the path weight.

You can define a threshold for this parameter. If the value of the path crosses the threshold at any time, the path is marked as bad and traffic is not sent on this path until it recovers.

Weight Reduce Factor - This factor helps to determine the weightage to be given to parameter when multiple parameters are selected. The greater the value of the Weight Reduce Factor, the lesser the weightage to the parameter in the overall path weight.

Jitter- Checking this check box will ensure that the jitter on the path is taken into account for calculating the path weight.

You can define a threshold for this parameter. If the value of the path crosses the threshold at any time, the path is marked as bad and traffic is not sent on this path until it recovers.

Weight Reduce Factor - This factor helps to determine the weightage to be given to parameter when multiple parameters are selected. The greater the value of the Weight Reduce Factor, the lesser the weightage to the parameter in the overall path weight.

Click the Ok button to return to the MPSec tab. Click on Save button to make the changes permanent.

To edit an MPSec entry, select it from the list and click the Edit button, click on SAVE button to make the changes permanent (see Figure 5.46).

Weight reduce Factor

Weight reducing factor reduces the effect of a parameter that has on the overall weight of a path. For example if the weight reducing factor is set to '1' for all parameters, then each parameter has equal importance in calculating the overall weight of a path. If we increase the weight reducing factor of any of the parameters to "n", then that parameter's importance on calculating the overall weight of the path reduces to '1/n'. This helps to negate the influence of a parameter thereby increasing the influence of the other parameters.

☐ Template
Remote VPN name
Denver
Remote VPN IP
14.14.14.30

Load Balancing
☒ Session
☐ Packet

☒ **Dynamic Mpssec Load Balancing**
☒ **Enable Bandwidth Detction (Kbps)** Detect Bandwidth Every 1 Min

☒ **Use Available Bandwidth**
Weight Reduce Factor 1

☒ **Use Latency**
Threshold (ms) 5000 Weight Red 1

☒ **Use Packet Loss**
Threshold (%) 50 Weight Reduce Factor 1

☒ **Jitter**
Threshold (ms) 1000 Weight Red 1

Figure 5.46 - Edit Remote VPN Entry

After creating and saving a new entry in the Remote Location list, you must assign the MPsec paths to that entry. To add MPsec paths to an entry, select the Remote Location from the Select Site Name dropdown menu, and then click on the Configure button (see Figure 5.47).

Add Paths

Remote VPN Name: NAEAST_Rockville
 Remote VPN IP: 42.2.1.2

Load Balancing Options
 TCP Protocol
 • Session based
 Other Protocol
 • Session based
 Packet based

Load Balancing Type: • Static

Remote FatPipe IP: 42.2.1.2
 Remote WAN Interface No: 1

Connect using	Compression	Weight	Usage	Encryption Type	LatencyThreshold	JitterThreshold	PacketLossThreshold
☒ Connect using WAN1	☐ Compression	1	Primary	IPSEC	0	0	0
☐ Connect using WAN2	☐ Compression	0	Primary	IPSEC	0	0	0
☐ Connect using WAN3	☐ Compression	0	Primary	IPSEC	0	0	0
☐ Connect using WAN4	☐ Compression	0	Primary	IPSEC	0	0	0

Remote FatPipe IP: Remote WAN Interface No: Add Edit Delete

More Paths

Cancel

Figure 5.47 – Configure MPsec Paths

The Remote FatPipe IP is the IP address of a remote MPVPN WAN IP address to which the local MPVPN will connect. The load balancing can be either session based or packet based.

Remote WAN Interface No.

To create MPsec paths for the remote FatPipe IP select the WAN interfaces that you wish to use. The number of MPsec paths created for the remote FatPipe should correspond to the number of local WAN interfaces selected.

MPsec Paths with Static IPs on WANs

When the Remote FatPipe IP is a statically assigned IP, select 'None' from the Remote WAN Interface drop down menu. By selecting "None" from the dropdown box, any changes that occur to the WAN IP of the peer site will not be updated.

MPsec Paths with Dynamic IPs on WANs

When the Remote FatPipe IP is a dynamically assigned IP, select the WAN interface from the Remote WAN interface drop down menu that corresponds to the IP address assigned to the remote WAN interface. This will ensure that any changes that occur to the remote WAN IP will be updated accordingly. FatPipe automatically and seamlessly recreates the MPsec paths.

MPsec Paths with a combination of Dynamic IPs and Static IPs on WANs

When two sites have a combination of dynamically assigned IPs and statically assigned IPs to their WAN links, always select the Remote WAN interface from the drop down menu that corresponds with the Remote WAN IP. A site should configure all its MPsec paths as per Dynamic IPs configuration even if it has only one link with Dynamic IPs. This is a requirement for the feature to function effectively.

Weights are assigned to each MPsec path for load balancing. The default weight is 1. This is described in the Load Balancing Type section below.

Load Balancing Type

MPsec Load balancing is based on the weights assigned to each MPsec path. Static Load Balancing allows you to define the weights that will be used. By default the weight is 1. The Upload and Download Bandwidths are defined for each WAN tab in the interfaces menu. Click the OK button to return to the MPsec page, click on SAVE button to make the changes permanent. Select the particular site name from the Select Site Name to configure the path for the particular remote location and click on Configure (see Figure 5.48).

Usage

MPsec paths can be configured either as Primary or as Backup. To configure any MPsec path as Backup you will need at least one Primary path available.

Remote VPN Name: NAEST_Backville
Remote VPN IP: 42.2.1.2

Load Balancing Options
TCP Protocol
• Session based
Other Protocol
• Session based
• Packet based

Load Balancing Type
• Static

Connect using WAN	Compression	Weight	Usage	Encryption Type	LatencyThreshold	JitterThreshold	PacketLossThreshold
☒ Connect using WAN1	☐ Compression	1	Primary	IPSEC	0	0	0
☐ Connect using WAN2	☐ Compression	0	Primary	IPSEC	0	0	0
☐ Connect using WAN3	☐ Compression	0	Primary	IPSEC	0	0	0
☐ Connect using WAN4	☐ Compression	0	Primary	IPSEC	0	0	0

Buttons: Add, Edit, Delete

Figure 5.48 - Configure MPsec Paths

To delete an MPsec entry, select it from the list and click the Delete button, click on SAVE button to make the changes permanent.

To view the MPSec pathway status for any of your single Remote Locations, select the particular site name from the Select Site Name dropdown box and click the Status button (see Figure 5.49).

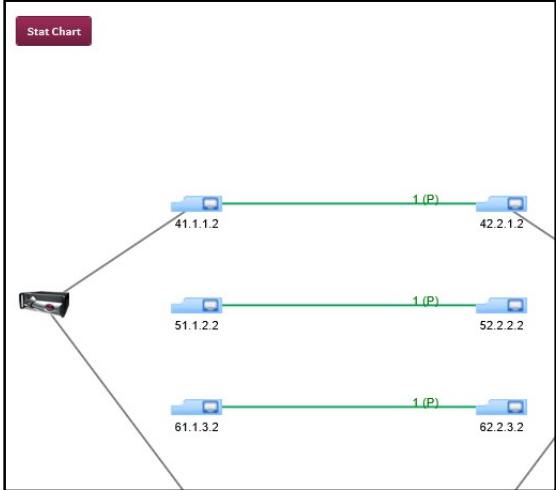


Figure 5.49 - Connection status for a particular remote site

To view the MPSec pathway status for ALL Remote Locations, select "All Sites" from the Select Site Name dropdown box and click the Status button (see Figure 5.50).

The status of each port will read ON, OFF, or (--). ON indicates the connection is established. OFF indicates the connection is not established. The symbol (--) indicates the connection is not enabled.

Connection Status							
Real-Time Path Status							
Index	Remote VPN Name	WAN1	WAN2	WAN3	WAN4	Remote VPN IP	R
1	NAEAST_Rockville	ON	--	--	--	42.2.1.2	4
2	NAEAST_Rockville	ON	ON	--	--	42.2.1.2	5

Figure 5.50 - Connection status for all remote sites

Real-Time Path Status

Real-Time Path status refer to latency, bandwidth, packet loss and jitter of aMPSec path.

To view real time MPsec pathway status for a particular site, click on the “Real-Time Path Status” button (see Figure 5.51).

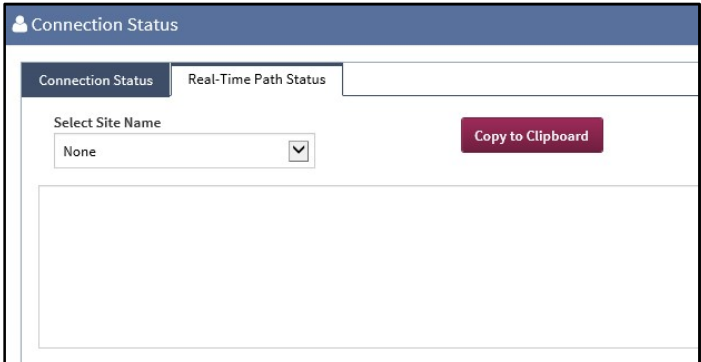


Figure 5.51 - MPsec Path Status

Select the remote site using the drop down menu. FatPipe MPVPN will show the status. You can click the refresh button to refresh the real time status (see Figure 5.52).

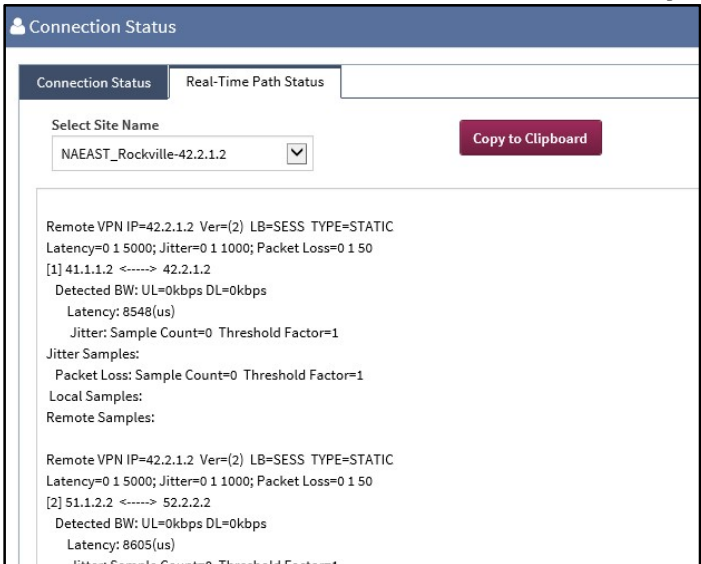


Figure 5.52 - MPsec Path Status

Compression

Compression is an add-on feature that does packet-based data compression and increases network capacity on an average of three times by compressing packet payload and eliminating redundant data traffic over WAN through the MPsec channels. Compression significantly increases bandwidth without having to upgrade line connections (e.g.: from DSL to a T1 or a T1 to a DS3), bringing significant cost savings immediately. Please refer to the contact information in the back of the manual or contact your local FatPipe representative for purchasing information.

You can enable Compression on your FatPipe MPVPN by clicking on the Configure button on the MPsec page. Under Remote FatPipe IP, select Edit, and check the box marked for Use Compression for each WAN interface (see Figure 5.39).

IPv6in4 tunnel

This feature is to encapsulate IPv6 packets within IPv4. This allows the IPv6 packet to be carried across IPv4 routing infrastructures.

To configure IPv6in4 tunnels, click on the Routing button in the main menu and click on the IPv6in4 tunnel tab (see Figure 5.53).

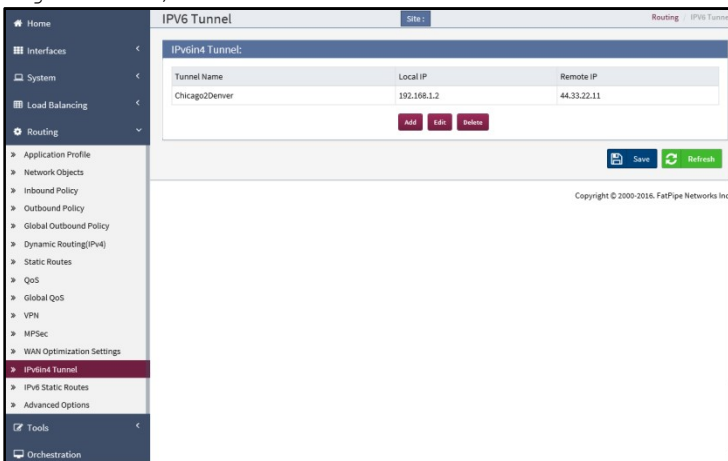


Figure 5.53 - IPv6in4 tunnel

To add a new IPv6in4 tunnel, click on the Add button (see Figure 5.52).



Add IP	
Tunnel Name	Chicago2Denver
Local IP	192.168.1.2
Remote IP	44.33.22.11

Figure 5.54 - Add IPv6in4 tunnel

Tunnel Name

Enter the tunnel name, the name must be unique and no space between characters is allowed.

Local IP

Select the local IP from the dropdown menu. The list includes all the IPs of the LAN and WAN Interfaces.

Remote IP

Enter the remote IP.

Click on the SAVE button to make the changes permanent.

To edit a tunnel after it has been configured, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 5.47).



Edit IP	
Tunnel Name	Chicago2Denver
Local IP	192.168.1.2
Remote IP	44.33.22.11

Figure 5.55 - Edit IPv6in4 tunnel

To delete a tunnel, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

IPv6 Static Routes

Static routes in IPv6 are similar to configuring static routes for IPv4.

To establish the IPv6 static route, click on the Routing button in the main menu and click on the IPv6 Static Routes tab (see Figure 5.56).

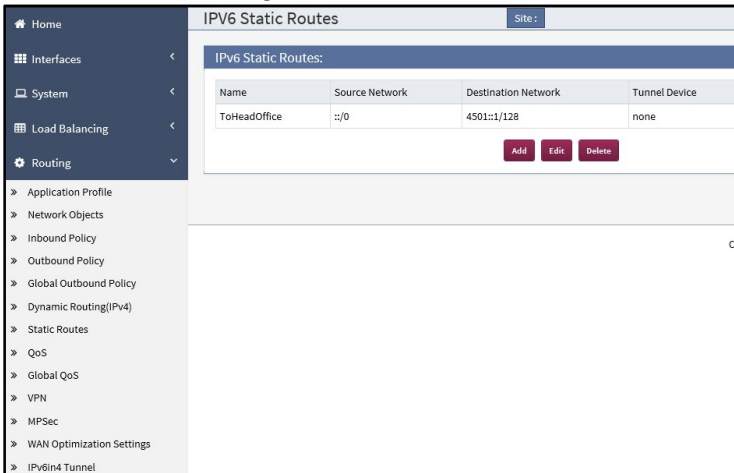


Figure 5.56 - IPv6 Static route

To add a static route, click on the Add button (see Figure 5.57).

Name

Enter a valid tunnel name.

Source Network

The source of the IPv6 traffic - the source can be a host address, subnet address, or network address.

Destination Network

A destination for the IPv6 traffic - the destination can be a host address, subnet address, or network address.

Tunnel Device

Select the IPv6 tunnel name from the dropdown menu if the traffic is to be routed using the IPv4 node.

If the end point is an IPv6 node, then select "None" from the dropdown.

Gateway

The Gateway is enabled only when a tunnel device is not selected ("None"). The Gateway should belong to one of the local subnets and should be reachable.

Metrics

Specifies the number of hops to the gateway. It is always at least 2 hops when using MPVPN. Click on the SAVE button to make the changes permanent.

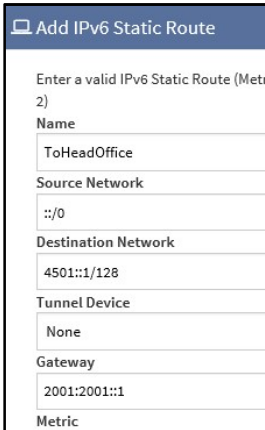
A screenshot of a web-based configuration interface titled "Add IPv6 Static Route". The form contains several input fields: "Name" with the value "ToHeadOffice", "Source Network" with the value "::/0", "Destination Network" with the value "4501::1/128", "Tunnel Device" with the value "None", "Gateway" with the value "2001:2001::1", and "Metric" which is currently empty. The form is styled with a blue header bar and white input fields with grey borders.

Figure 5.57 - Add Static Routes

To edit a static route, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 5.58).

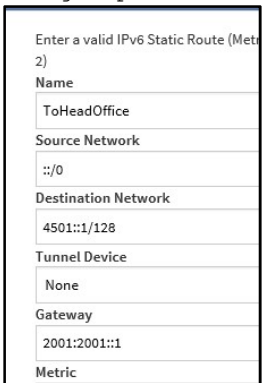
A screenshot of a web-based configuration interface titled "Edit IPv6 Static Route". The form contains several input fields: "Name" with the value "ToHeadOffice", "Source Network" with the value "::/0", "Destination Network" with the value "4501::1/128", "Tunnel Device" with the value "None", "Gateway" with the value "2001:2001::1", and "Metric" which is currently empty. The form is styled with a blue header bar and white input fields with grey borders.

Figure 5.58 - Edit Static Routes

To delete a static route, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

Advanced Options

EIGRP Multicast Forwarding - Enable EIGRP Multicast Forwarding when you want the FatPipe transparent between 2 EIGRP end-points.

Enable LAN Redirect - Enable LAN Redirect when you want a LAN client to access a server in your LAN segment using its public IP, rather than its private LAN IP. For example, if you have a web server in your LAN with a Public IP and it is accessible from the Internet using the domain name, and one or more LAN clients also want to access the web server using the domain name, then you would enable the LAN Redirect option.

Direct Route LAN

Enable Direct Route LAN to route packets directly from LAN to WAN.

Direct Route WAN - Enable Direct Route WAN to route packets directly from WAN to LAN.

Send ESP as GRE

Enable Send ESP as GRE to send ESP packets as GRE.

Hop-based MPSEC Load Balancing - Enable Hop-based MPSEC Load balancing to load balance MPSEC traffic based on hops.

GRE Inspect - Enable GRE Inspect to do deep packet inspection of GRE Packets to identify the sessions to which the actual packets belong to and decide the routing accordingly

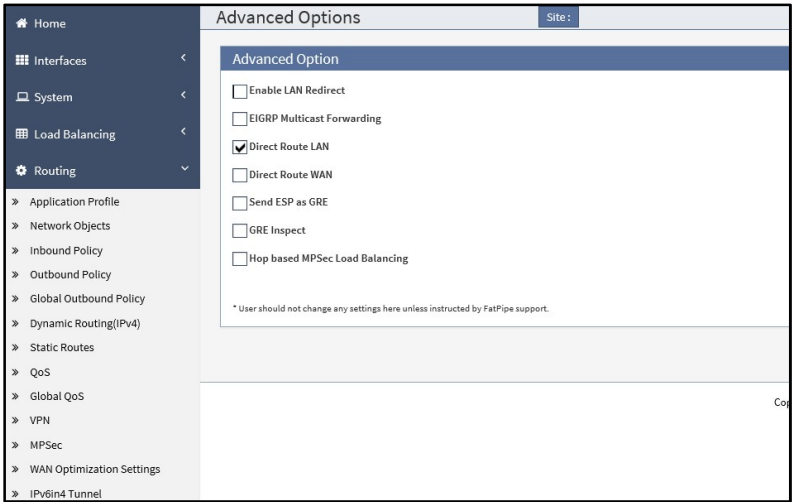


Figure 5.59 - Advanced

Chapter 6: Tools

FatPipeMPVPN branch in a box provides graphical monitoring tools to aid you in monitoring the speed and performance of your Internet connections. This chapter describes the methods to view the Speed Chart. If you have the QoS add-on, then you will also see a QoS Statistics page, that page is covered in Chapter 7.

Speed Chart

Monitor the upload and download or combined speeds of each of the WAN lines independently or in combination by viewing the Speed Chart.

To view the speed chart, click on the Tools button in the main menu and click on the Speed Chart tab (see Figure 6.1).

There are five views to choose from:

- WAN1 - Displays Total Speed, Upload Speed, and Download Speed for WAN1
- WAN2 - Displays Total Speed, Upload Speed, and Download Speed for WAN2
- WAN3 - Displays Total Speed, Upload Speed, and Download Speed for WAN3
- ALL INTERFACES TOGETHER - Displays Total Speed, Total Upload Speed, and Total Download Speed of all WAN ports combined
- ALL INTERFACES - Displays Total Speed for each of the WAN ports on the same graph

The Speed Chart is a dynamic, real-time chart that updates every second. The scale dynamically changes based on the current bandwidth usage.

The speed chart shows information according to the option selected from the dropdown menu.



Figure 6.1 - Speed Chart with All Interfaces Together selected

QoS Statistics

See Chapter 7 for details.

MPsecQoS Statistics

See Chapter 7 for details.

Diagnostics

FatPipe MPVPN can test both physical and Internet service connections for availability. Select the Diagnostics page to run various tests.

To view diagnostics, click on the Tools button in the main menu, and click on the Diagnostics tab.

To ping a host or trace route to a host to test connectivity, enter the IP address or domain name of the host. Select the interface to run these tests from the "Interface" dropdown menu and click the Ping It button or the Trace It button (see Figure 6.2).

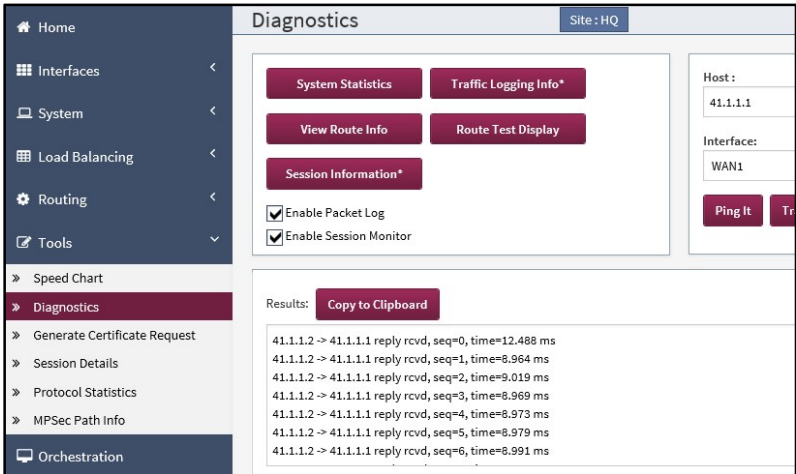


Figure 6.2 - Ping host to test connectivity

System Statistics

Display information about MPVPN including system uptime and interface statistics (e.g., packets received, packets transmitted, and any packet errors (see Figure 6.3)).

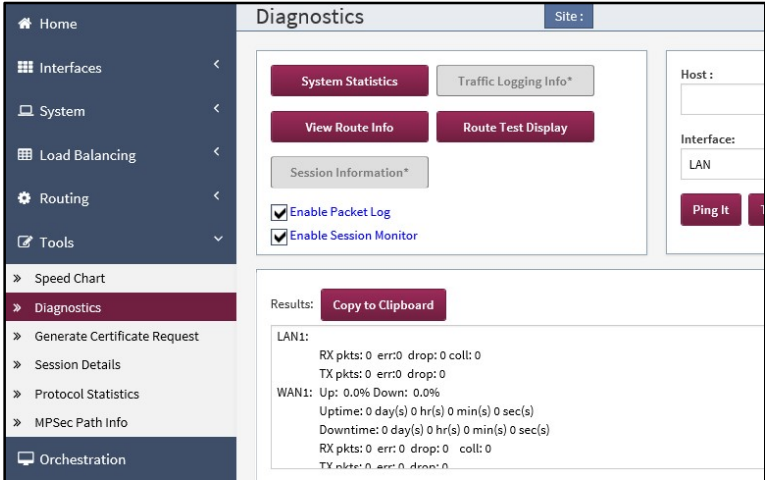


Figure 6.3 - System Statistics

Route Test Display

Displays a pictorial representation of the current line status (see Figure 6.4).

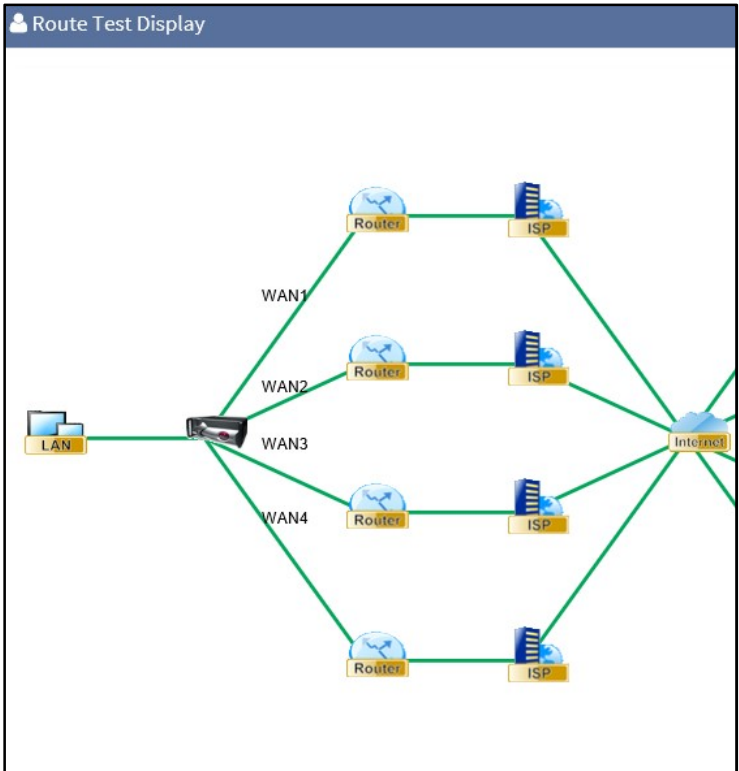


Figure 6.4 - Route Test Display

Session Information

To view the session information, check the "Enable Session Monitor" checkbox and click on the SAVE button to make the changes permanent. Now the Session Information button is enabled. Click the button to view all the sessions currently running on the unit. The Session Information button will be grayed out by default (see Figure 6.5, Figure 6.6).

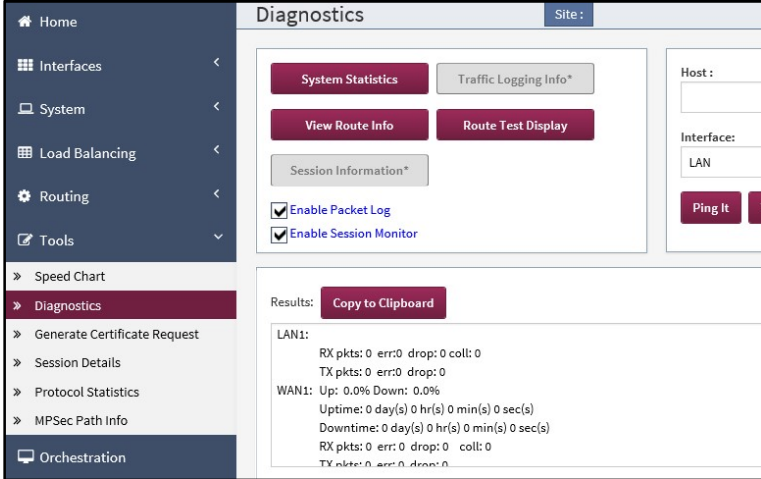


Figure 6.5 - Enabling Session monitor Information

Device	Port	Protocol	Source IP	Destination IP	Port	Protocol	Device	Type
10.2.0.182	40967	10.2.0.182	40967	172.18.2.2	8080	17	eth2	OUTBOUND
10.2.0.182	57673	10.2.0.182	57673	172.18.2.2	8080	17	eth2	OUTBOUND
10.2.0.182	41069	10.2.0.182	41069	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	58626	10.2.0.182	58626	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	58672	10.2.0.182	58672	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	60470	10.2.0.182	60470	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	60459	10.2.0.182	60459	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	44481	10.2.0.182	44481	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	44572	10.2.0.182	44572	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	46378	10.2.0.182	46378	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	46712	10.2.0.182	46712	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	47005	10.2.0.182	47005	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	47273	10.2.0.182	47273	172.18.2.2	8080	17	eth1	OUTBOUND
10.2.0.182	47583	10.2.0.182	47583	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	47608	10.2.0.182	47608	172.18.2.2	8080	17	eth3	OUTBOUND
10.2.0.182	33014	10.2.0.182	33014	172.18.2.2	8080	17	eth3	OUTBOUND

Figure 6.6 - Session Table

Traffic Logging Info

To view the Traffic Logging information, check the "Enable Packet log" checkbox and click on the SAVE button to make the changes permanent. Now the Traffic Logging Info button is enabled. Click the button to display a page where you can monitor the inbound and outbound traffic for individual hosts on your network. Sort by Host IP to view a history graph for that host (see Figure 6.7, Figure 6.8, and Figure 6.9).

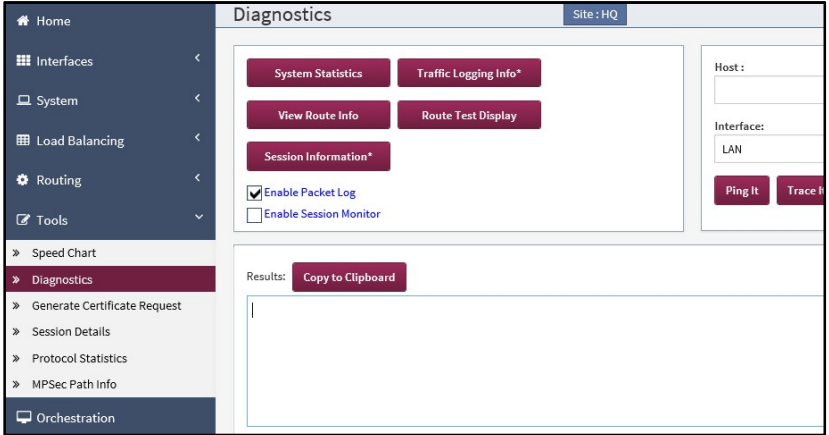


Figure 6.7 - Enabling Packet log Information

Packet Log													
Device	All	Initiate	All	Direction	All	Sort by	Total TCP	Refresh	4	Close Window			
Host IP	Host Name	All* (KB)	TCP* (KB)	UDP* (KB)	ICMP* (KB)	IPSEC* (KB)	GRE* (KB)	IPV6* (KB)	Other* (KB)	Device (KB)	Bandwidth	Session Total	Session Rate
10.33.0.7	Steven Paul Jobs	3696779	3696779	0	0	0	0	0	0	3696779	0	0	0
10.33.0.16	Akshil Fadnis	600407	600407	0	0	0	0	0	0	600407	0	0	0
10.33.0.16	Vladimir Levin	412449	412449	0	0	0	0	0	0	412449	0	0	0
10.33.0.14	Gary McKinnon	420969	420969	0	0	0	0	0	0	420969	0	1	0
10.33.0.11	Jonathan James	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6.8 - Packet Log information

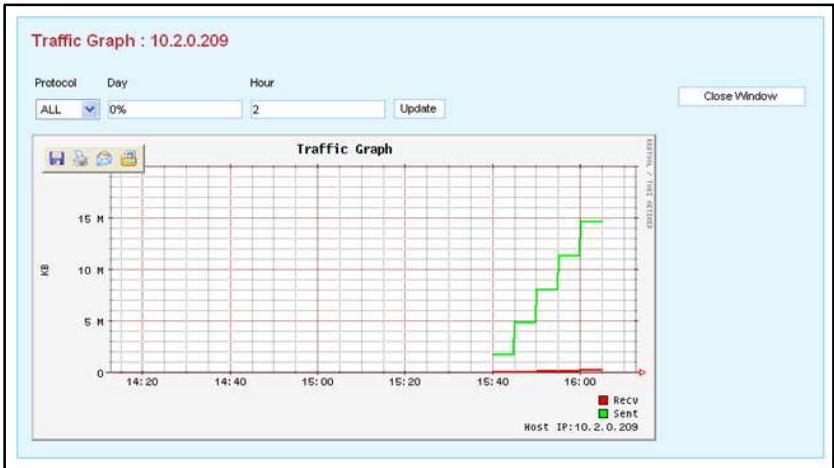


Figure 6.9 - Traffic Log Graph

Server Statistics

See chapter 9 for details.

SatBooster

See chapter 13 for details

Session Details

Session Details provides a comprehensive report on the sessions that flow through the device on a per flow basis. You can view all the sessions that flowed through the FatPipe at a given period of time or live sessions. This screen is further divided into a Sessions tab that shows the details of the sessions and a Reports tab that shows a bar chart. Click on the Session Tab to show the options to choose the sessions.

This screen is further divided into a Sessions tab that shows the details of the sessions and a Reports tab that shows a bar chart. Click on the Session Tab to show the options to choose the sessions. (see Figure 6.10)

Preset

Choose a given preset from the dropdown. Chose custom if you want to see (See Figure 6.11)the report for a given period other than the ones defined in the preset.

Note - If a preset other than custom is chosen, from Date and time and to date and time fields will be disabled.

From Date

Enter from Date in the mm/dd/yyyy format. Click on the date picker tool to choose a date (See Figure 6.12).

From Time

Enter the time in the hh:mm format followed by AM or PM. Click on the timepicker tool to choose a time. (See Figure 6.13)

To Date

Enter the To Date in mm/dd/yyyy format. Click on the date picker tool to choose a date.

To Time

Enter the time in the hh:mm format followed by AM or PM. Click on the timepicker tool to choose a time.

Note - Make sure you give a valid period.

Interface

This dropdown lists all the WAN interfaces. You can chose either ALL to show sessions that flowed through all the WAN interfaces or choose a particular interface to show the sessions that flowed that interface only.

The screenshot displays the 'Session Details' page of a network management system. On the left is a dark blue sidebar with navigation links: Home, Interfaces, System, Load Balancing, Routing, Tools (expanded), Speed Chart, Diagnostics, Generate Certificate Request, Session Details (highlighted), Protocol Statistics, MPsec Path Info, Orchestration, and EnterpriseView. The main content area has a title 'Session Details' and a 'Site:' dropdown. Below this is a section for 'Enable Session Logging' with a checked checkbox and a 'Prepare Database' button. A 'Last Updated Time' field shows '2017:01:04 15:06:10'. There are two tabs, 'Sessions' and 'Reports', with 'Sessions' selected. The 'Sessions' tab contains search filters: 'Preset' (set to 'Custom'), 'From' (date '1/4/2017' and time '12:01 AM'), 'To' (date '1/4/2017' and time '11:59 PM'), 'Interface' (set to 'All'), 'Source IP', 'Destination IP', and 'Protocol'. At the bottom are buttons for '< Prev', 'SHOW', 'Next >', and 'Export'.

Figure 6.10 Session Details

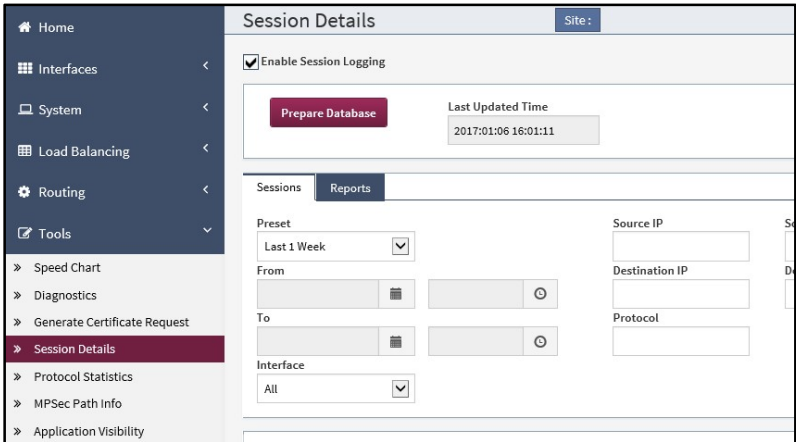


Figure 6.11 Selecting Preset



Figure 6.12 Selecting Custom DateFigure 6.13 Selecting Custom Time

Source IP

Enter a valid IP for viewing sessions for that particular IP. Leave it blank to view sessions for all IPs. (See Figure 6.14)

Destination IP

Enter a valid IP for viewing sessions for that particular IP. Leave it blank to view sessions for all IPs.

Source Port

Enter a valid port for viewing sessions for that particular port. Leave it blank to view sessions for all ports.

Destination Port

Enter a valid port for viewing sessions for that particular port. Leave it blank to view sessions for all ports.

Protocol

Enter a valid Layer 4 protocol to view all sessions for that protocol For example, TCP, UDP etc. Leave it blank to view sessions for all protocols.

Active Sessions

Check this option to see details of sessions that are active or hasn't been closed.

Note:When you check this option, you will get to see a dropdown at the top right corner named Refresh that allows you to choose the refresh period for the GUI. The UI will refresh at the chosen period.

Local Sessions

Check this option to see all the sessions generated by the FatPipe which includes Route test sessions.

You can enter any combination of the above fields to get a more granular report.

Click on the SHOW button to show the sessions matching the given criteria. The report is paginated. Click on the NEXT button to show the next set of sessions. Click on the Previous button to show the previous set of sessions.

The screenshot shows a web interface for 'Session Details'. On the left is a dark sidebar with a menu: Home, Interfaces, System, Load Balancing, Routing, Tools (expanded), Speed Chart, Diagnostics, Generate Certificate Request, Session Details (highlighted), Protocol Statistics, MPsec Path Info, and Application Visibility. The main panel has a title 'Session Details' and a 'Site:' dropdown. Below this is a section with 'Enable Session Logging' checked and a 'Prepare Database' button. A 'Last Updated Time' box shows '2017:01:06 16:01:11'. There are two tabs: 'Sessions' and 'Reports'. Under 'Sessions', there are filters: 'Preset' (Custom), 'From' (01/02/2017 to 12:38 PM), 'To' (01/04/2017 to 12:40 PM), and 'Interface' (All). On the right, there are input fields for 'Source IP' (10.10.10.18), 'Destination IP', and 'Protocol'.

Figure 6.14 Filtering Session Details based on source IP Address

The Reports tab shows bar charts for Top 10 Hosts, Protocols, applications or Conversations. (See Figure 6.15)

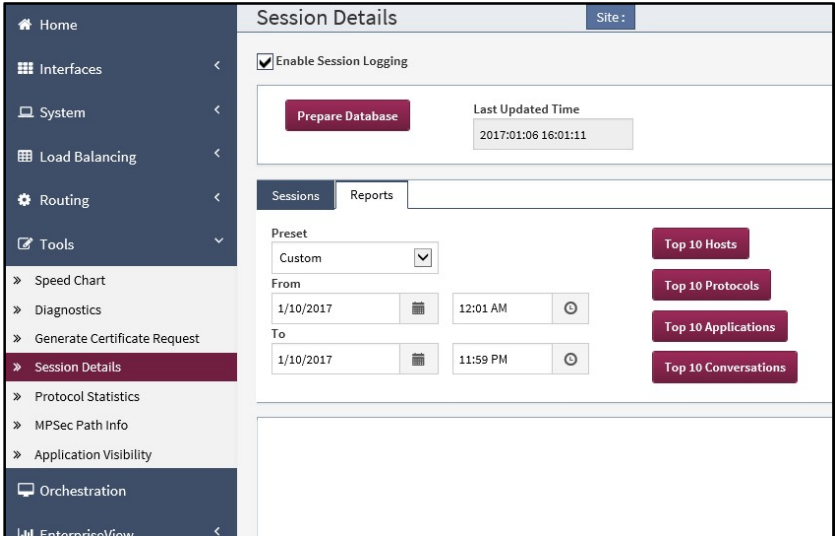


Figure 6.15

Preset

Choose a given preset from the dropdown. Chose custom if you want to see the report for a given period other than the ones defined in the preset.

Note: If a preset other than custom is chosen, the from Date and time and To date and time fields will be disabled.

From Date

Enter From Date in the mm/dd/yyyy format. Click on the datepicker tool to choose a date.

From Time

Enter the time in the hh:mm format followed by AM or PM. Click on the timepicker tool to choose a time.

To Date

Enter the To Date in the mm/dd/yyyy format. Click on the datepicker tool to choose a date.

To Time

Enter the time in the hh:mm format followed by AM or PM. Click on the timepicker tool to choose a time.

Note: Make sure you give a valid period.

Select the required report from the list of Session Reports. Click on SHOW to display the chart.

Chapter 7: Quality of Service (QoS)

Introduction

QoS is an add-on feature from FatPipe. When enabled, it allows you to prioritize your WAN traffic. This is especially useful for ensuring that real-time traffic including -- voice and video -- gets priority over other types of traffic.

The primary purpose of QoS is assurance that packets are transported from a source to a destination with certain characteristics corresponding to the requirements of the service that the packet flow supports. This becomes a challenge in a situation where multiple streams compete for limited available resources. One of these resources is link transmission capacity, which gets divided into throughputs of individual streams. Another important resource is buffer memory, which affects packet loss.

Outgoing network traffic is managed by assigning a priority to each type of traffic. This priority determines the treatment of that traffic type in terms of how many packets are preserved and how urgently they are transmitted, relative to one another.

'0' is the highest priority and Best Effort (7) traffic is the lowest classification. The Best Effort classification does not guarantee any particular level of service; it simply represents the unused capacity of the link at any moment. In addition to QoS priority, a certain amount of bandwidth is also assigned by the user to each type of traffic, and it is defined by committed rate (CR) and Burst rate.

Committed rate of a traffic type defines the amount of bandwidth that is guaranteed to be available for that type of traffic at any time the associated link is up. The amount of traffic forwarded under these conditions is called the primary rate.

Burst rate is required for QoS and defines the upper limit for bandwidth that can be made available to the traffic type. The Burst rate can be set up to the maximum available bandwidth of the associated interface. The Burst rate can be equal to or greater than the commit rate. The amount of bandwidth between CR and Burst rate

is made available only if it is not in use by other quality groups. Traffic above CR is downgraded to Best Effort, without guarantees on packet loss and delay.

FatPipeQoS also provides some degree of control over incoming network traffic by letting the user limit the rate at which the LAN receives traffic from each of the WAN links. While this does not help conserve bandwidth, it can help reduce the occurrence of unwanted connection-oriented traffic. The Inbound Policed Rate defines the limit above which all-incoming traffic that it applies to will be dropped.

Configuration

In order to define QoS characteristics for a traffic type, you must first create a QoS Rule.

To configure QoS settings, click on the Routing button in the main menu and click on the QoS tab (see Figure 7.1).

Home

Interfaces

System

Load Balancing

Routing

» Application Profile

» Network Objects

» Inbound Policy

» Outbound Policy

» Global Outbound Policy

» Dynamic Routing(IPv4)

» Static Routes

» QoS

» Global QoS

» VPN

» MPSec

» WAN Optimization Settings

» IPv6in4 Tunnel

QoS

Site:

QoS Rules:

Name	Qos for MPsec	WAN1 (50000/50000 kbps)			WAN2 (50000/50000 kbps)			WAN3 (50000/50000 kbps)			WAN4 (50000/50000 kbps)
		Priority	OUT		Priority	OUT		Priority	OUT		
QoSRuleA	No	0	256	512	1	240	500	2	360	450	4
Total			256	512		240	500		360	450	

<

Add

Edit

Delete

Figure 7.1 – Quality of Service page

Enter the name for the rule (only letters and numbers are allowed). For each link (interface) that you want to use for this type of traffic, you can define the Inbound Policed Rate and/or the Committed Rate.

Note:Link Bandwidth has to be defined for each link that you want to apply the QoS rules to (see Chapter 2: Interfaces).

The minimum value for the Committed Rate (CR) is 8 kbps and the maximum value is 90% of the link's bandwidth. The actual amount available to a particular quality group depends on the amount of bandwidth that has already been committed. The sum of all CRs on a particular link cannot be greater than 90% of the Link Bandwidth. The remaining 10% is always reserved for Best Effort traffic.

The Burst rate cannot be changed, and it will default to the link bandwidth. The QoS Rules table provides a convenient view of Inbound Policed Rates, Committed Rates, and Priorities, as well as Link Bandwidths and total bandwidth already reserved by CRs for each link. You can select from 10 different priority levels for each type of traffic.

To add a new QoS rule, click on the Add button (see Figure 7.2).

Enter the QoS Rule name, WAN1 and WAN2 values (in Kbps), and click on the OK button to return to the QoS page, click on the SAVE button to make the changes permanent.

Add / Edit Quality of Service Rule

Name

QosRuleA

☐ Apply Qos rule for mpsec paths

Interface (Upload/Download)	Policed Rate	Committed Rate	Burst Rate
WAN1 (45000 / 45000)	256	512	1000
WAN2 (45000 / 45000)	240	500	900
WAN3 (45000 / 45000)	360	450	750
WAN4 (45000 / 45000)	300	420	800
WAN5 (45000 / 45000)	330	480	850

* Upload/Download - Available bandwidth in kbps

Figure 7.2 - Add Quality of Service Rule

Apply QoS Rule to MPsec Paths

FatPipe allows you to create separate QoS Rules that can be applied to MPsec paths. Enabling this checkbox will ensure that the QoS rule is applied to every MPsec path.

To edit a QoS rule, select it from the list and click on the Edit button, click on the SAVE button to make the changes permanent (see Figure 7.3).

Name

☐ Apply Qos rule for mpsec paths

Interface (Upload/Download)	Policed Rate	Committed Rate	Burst Rate
WAN1 (45000 / 45000)	256	512	1000
WAN2 (45000 / 45000)	240	500	900
WAN3 (45000 / 45000)	360	450	750
WAN4 (45000 / 45000)	300	420	800
WAN5 (45000 / 45000)	330	480	850

* Upload/Download - Available bandwidth in kbps

Figure 7.3 - Edit Quality of Service Rule

To delete a QoS rule, select it from the list and click on the Delete button, click on the SAVE button to make the changes permanent.

A QoS rule by itself does nothing without an association with a particular kind of traffic. In order to create this association, go to either the Outbound Policy or Inbound Policy page. If you edit an existing Policy Routing rule or create a new one, you can select a QoS rule, which will be applied to the traffic, defined by the Policy Routing rule.

Note: A QoS rule created with "Apply QoS Rule to MPsec Paths" option enabled should only be attached to a Policy Routing Rule that matches inter-site traffic. This QoS rule should not be applied to normal traffic

QoS Statistics

This page displays information about QoS traffic going through MPVPN (see Figure 7.5). Information is displayed on two real-time charts. The chart at the top displays the rate at which traffic is being forwarded. The chart at the bottom displays the percentage of packets that are being lost. You can filter the view by selecting a QoS Rule, one or more interfaces, and the direction of traffic (either inbound or outbound). Click on Tools in the main menu and select QoS Statistics to view traffic and packet loss.

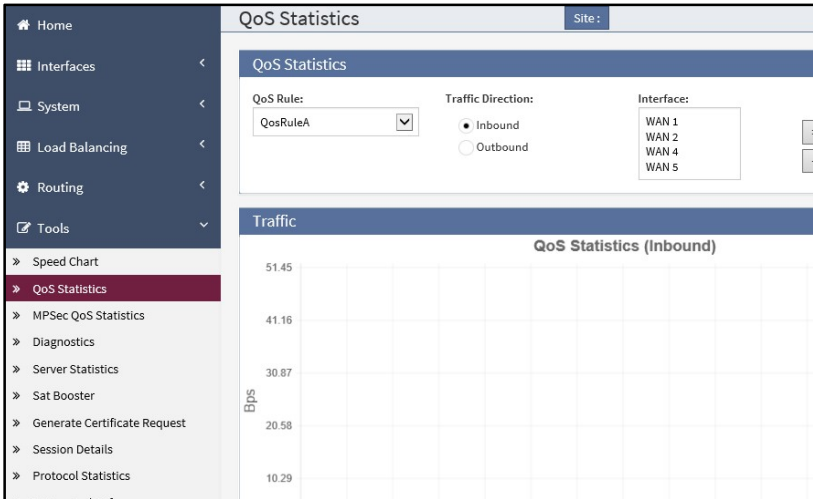


Figure7.5 - QoS Statistics

To view information for a particular QoS Rule, select a QoS Rule from the QoS Rule list. When a QoS Rule is selected, interfaces that the rule applies to will appear in the list box to the right. Select one or more interfaces for which you want to monitor traffic in the interface list and move them to the selected list by using the arrows. Select either Inbound or Outbound from the Traffic Direction dropdown menu depending on which direction you want to monitor. The charts will begin displaying information after at least one interface is selected, provided that traffic matching that QoS rule is passing through the MPVPN.

The charts refresh every five seconds. The Traffic Chart displays the aggregate rate of the traffic that belongs to the selected QoS Rule in stacking area format. Traffic that falls within the Committed Rates (CRs) on respective interfaces will be shown at the bottom in green. Traffic that exceeds the CR but is within limits defined by the Burst Rate and is being forwarded as Best Effort traffic will be displayed above in yellow. Discarded traffic is shown in red on top of the chart. The Packet Loss Chart shows the percentage of packets that are lost.

Note: Due to the variable size of packets, this chart does not represent the actual amount of lost data for each packet.

MPSecQoS Statistics

This page displays the QoS Statistics for a given MPsec path. To view the statistics, choose the QoS Rule from the dropdown, choose the Remote site and the MPsec Path comprising of the Local WAN interface and the Remote WAN Interface. Once the choice is complete, the device will show the statistics. See figure 7.6.

QoS Rule

Select the QoS rule for which you want to see the statistics.

Site Name

Select the Remote site.

Local WAN

Select the WAN interface of the local unit.

Remote WAN

Select the WAN interface of the Remote Site.

Note: These dropdowns are populated in the order of selection - QoS Rule followed by Site Name followed by Local WAN followed by Remote WAN.

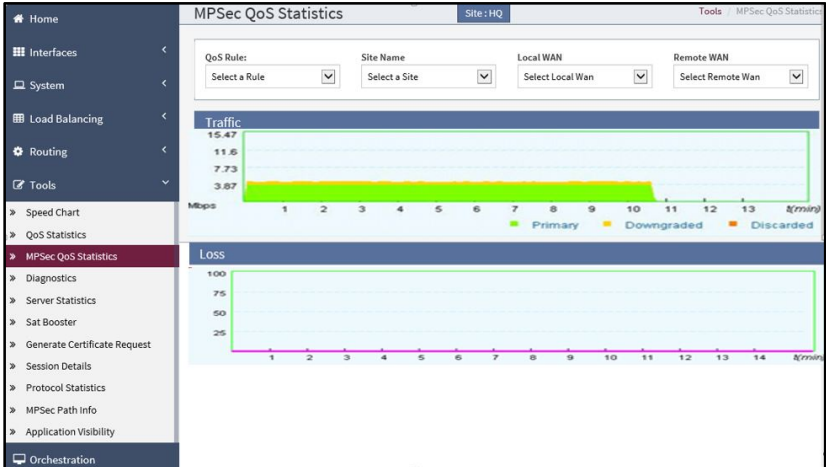


Figure 7.6 - MPsecQoS Statistics

Fine Tuning QoS Rules

The traffic chart and the packet loss chart can be used for fine-tuning of QoS parameters. For example, a consistent, high amount of discarded packets for a

particular type of traffic is an indicator that there is a much higher demand for bandwidth for that traffic than the one that is assigned to it. If all available links (interfaces) are fully utilized, an increase in CR for that type of traffic should be considered.

If increasing the CR does not help, further improvement can be achieved by reassigning a higher priority to that type of traffic.

Traffic rate significantly below the assigned CR may be an indicator that the need for bandwidth was overestimated and a smaller CR should be considered in order to make more bandwidth available for other applications.

If none of this helps, then there are likely too many other QoS Rules with high demand, which compete for the service.

Note: Remember that priorities work in relative terms and assigning the highest priority to all applications does not improve performance for any of them. Tuning QoS is an iterative process and desired results are rarely achieved in the first attempt.

Layer 7 QoS – Application Level QoS

Layer 7 QoS allows you to classify packets based on the application they belong to.

Some applications use well-known port numbers for communication, which simplifies detection of their packets but there are a number of applications that do not have a reserved port. Also, applications may be configured to use non-standard ports, for security, for example. In these cases, the applications can be identified by inspecting the payload of the packets; the Layer 7 (L7) data. An L7 classifier is used to do this inspection and thus must have specific knowledge of a given application.

Application Rules supply the patterns used by the Layer 7 classifier as an extension of Outbound Policy Routing rules. It allows the user to classify traffic based on application-specific information, regardless of port numbers used by transport protocols.

FatPipeQoS supports 180+ predefined Applications. The Application rules are classified into several categories. Each category has a list of applications to select from.

Configuration

When creating a new Outbound Policy Routing rule, you have the option of creating one or more Application Rules.

Click on the Routing button from the main menu and then click on the Outbound Policy tab to edit the existing outbound policy rule (see Figure 7.7).

Name

SSL

Source

IP

Action

Allow

Protocol

TCP

Source

Port

QoS

QoSRuleA

DSCP

5

Destination

IP

Application Profiles

None

Add

Edit

Delete

☐ Source NAT

☐ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session Persistence

Traffic Mode:

Interface Priority

Interface Specific

Mixed Priority

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	Packets
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0
WAN2	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0
WAN3	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0

Add

Edit

Delete

Up

Down

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

Sa

Mo

Tu

We

Th

Fr

Sa

Figure 7.7 - Edit Outbound Policy Routing Rule

Click on the Add button to select an Application Rule. Every Application Rule defines an additional action that is applied to traffic matching both the Outbound Policy Routing rule and the chosen application. Select one or more application from the list. You can add a maximum of 17 applications from one or more categories for a single Outbound rule.

The Outbound Policy Routing rule determines how the traffic is routed while the Application Rules allow you to block matching traffic or assigns a Quality of Service (QoS) rule to matching traffic (see Figure 7.8).

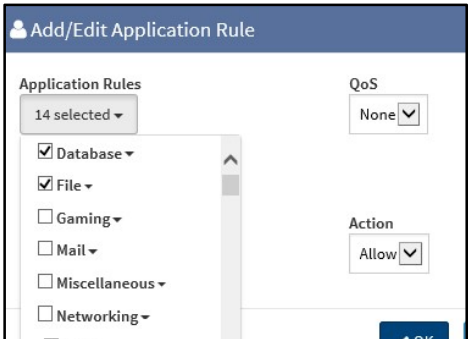


Figure 7.8 - Add/Edit Application Rule

Chapter 8: Site Load Balancing

MPVPN can be configured to provide site load balancing where inbound connectivity to Internet accessible servers is critical. Site Load Balancing also allows for Site Failover between servers located in geographically separate locations that have identical or similar information in both locations. Site Load Balancing is an optional feature available upon request. The Main site is referred to as "Primary," and alternate sites are referred to as "Backup."

Site Load Balancing can share weighted traffic between two sites utilizing all lines available at each site. Please refer to the back of the manual for general contact information or contact your local FatPipe representative for purchasing information.

To implement Site Load Balancing, two or more sites should be configured and ready to accept incoming requests for servers hosted locally. Prior to being configured, the MPVPN at the primary site should contain all the DNS records in SmartDNS for all zones that will be used with Site Load Balancing. SmartDNS on all other sites is configured automatically once Site Load Balancing is established. From that point on, any DNS change made to one site will be propagated to all other sites.

Site Load Balancing determines the status of each site dynamically. Each line at each site is given a priority of Primary or Backup. Only IP addresses belonging to Primary lines will be handed out in DNS requests. If all Primary lines are down, Site Load Balancing will detect the primary failure and switch the DNS service to the backup lines. This means MPVPN will mask IP addresses for hosts that belong to the Primary lines, and IP addresses for hosts belonging to Backup lines will be handed out in DNS requests.

Initial setup

Setup for Site Load Balancing involves these steps:

Enable the Site Load Balancing and SmartDNS options on each interface that you want to use for Site Load Balancing (see the WAN section of Chapter 2).

Setup the primary site unit by clicking on Load Balancing in the main menu and click on the Site Load Balancing

Tab. Select Enable Site Load Balancing to enable the site load balancing feature, and enter a unique name in the Site Name field. Click on the Advanced button to create a secret key in the Key field, and click Ok. This key is the secret key used for securing the communication between peers. Click on the Add button and enter the IPs of the second site (Peer Site), click on the SAVE button to confirm the changes. You may now setup this section in the secondary site unit.

Setup the second site (Peer Site) for site load balancing by clicking on the Load Balancing button in the main menu and click on the Site Load Balancing Tab. Select Enable Site Load Balancing to enable the site load balancing feature, and enter a unique name in the Site Name field. Click on the Advanced button to create a secret key in the Key field, and click Ok. This key is the secret key used for securing the communication between peers. Click on the Add button and enter the IPs of the second site (Peer Site), click on the SAVE button to confirm the changes. You may now setup the Interface-to-Network Mappings in SmartDNS section of the primary site unit.

Configure the Interface-to-Network Mappings for all sites under SmartDNS in the primary site unit (see Chapter 4).

Once you save the Interface-to-Network Mappings on the primary site unit, the DNS records and Interface-to-Network Mappings will be written to the backup unit(s).

Confirm that the backup unit has copied the DNS records and Interface-to-Network Mappings from the primary site unit. If you need to add mappings or DNS records, add it to the primary site unit first, and click SAVE to write the changes to the backup site unit(s).

At this point, any changes made to DNS records on any unit, will be written to all the units associated with Site Load Balancing.

Once the site units are able to communicate with each other, you will see a table showing the status of the lines at each location in the Site Load Balancing section (see Figure 8.1).

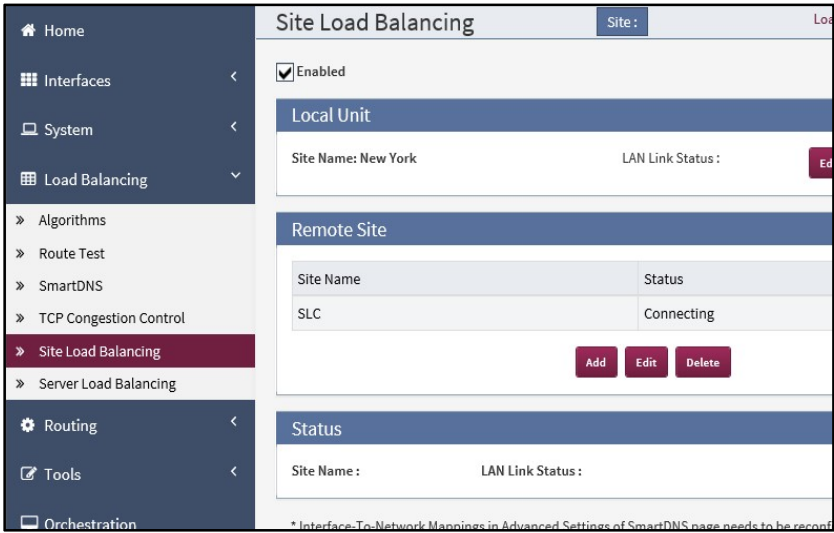


Figure 8.1 - Site Load Balancing

Local Unit

Shows you the site name of the current unit.

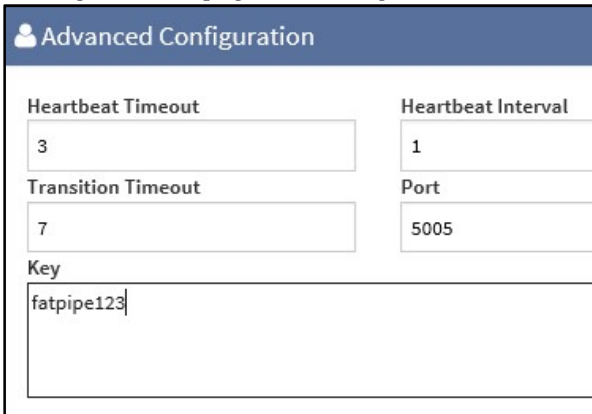
Peer Info

Gives you a list of available peers.

To add a peer, click on the Add button.

To delete a peer, click on the Delete button.

The Advanced Configuration window (Figure 8.2) can be accessed by clicking on the Advanced button on the main configuration page (see Figure 8.1).

The image shows a web-based configuration window titled "Advanced Configuration" with a user icon. It contains several input fields for configuring site load balancing. The fields are arranged in two columns. The left column has "Heartbeat Timeout" (value 3), "Transition Timeout" (value 7), and "Key" (value fatpipe123). The right column has "Heartbeat Interval" (value 1) and "Port" (value 5005).

Advanced Configuration	
Heartbeat Timeout	Heartbeat Interval
3	1
Transition Timeout	Port
7	5005
Key	
fatpipe123	

Figure 8.2 - Advanced Configuration for Site Load Balancing

Heartbeat Timeout

Specifies the time to wait for a heartbeat a peer before determining that the connection to the peer is lost. The default is 3.0 seconds.

Heartbeat Interval

Specifies time interval between two heartbeats sent from this unit to other peers. The default is 1.0 second.

The heartbeat is a small network packet sent periodically between peers. It keeps each peer updated with the status of other peers. The absence of the heartbeat from any peer within Heartbeat Timeout will signal hardware failure and all lines belonging to the remote peer will be considered down.

Note: Heartbeats use UDP protocol that does not guarantee delivery. Therefore, it is important to have Heartbeat Timeout at least several times longer than Heartbeat Interval. The timeout should be bigger than any possible network delay to avoid false positives.

When setting a timeout, it is also important to consider a balance between the network load and the speed of failover. Faster failover means that more heartbeats per second have to be sent.

Transition Timeout

Specifies a time interval after a line has failed during which connectivity problems will be ignored. This could be necessary should MAC and IP addresses change as a result of transition, (e.g., if Unit Failover and Site Load Balancing are both enabled), and routers/switches need some time to relearn routes. During this timeout all site units will ignore lack of heartbeats from other site units. The default is 7.0 seconds.

Port

It is the port number used for communication between peers.

Key

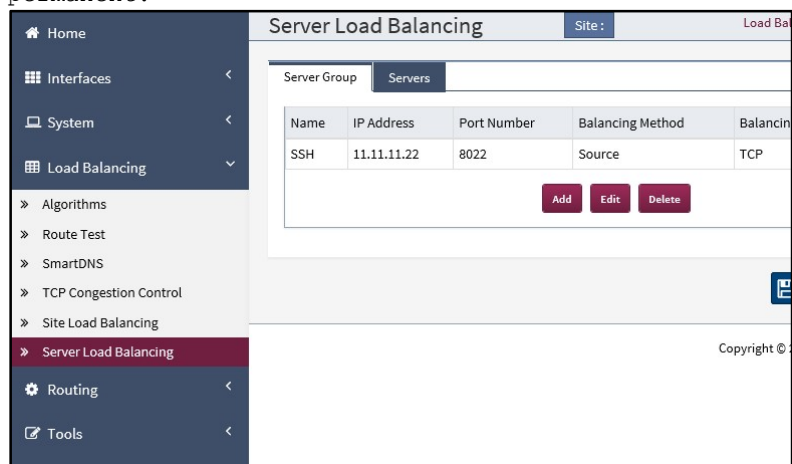
Secret Key used for securing the communication between peers. It is recommended that this be a long random mix of characters, numbers, and symbols.

Chapter 9: Server Load Balancing

The server load balance feature is an add-on feature for inbound connection load balancing. It load balances and failovers servers based on the availability of the servers, load of the servers and number of server connection counts. This branch in a box feature eliminated the need for a separate server load balancer. The server load balance feature is an add-on feature for inbound connection load balancing. It load balances and failovers servers based on the availability of the servers, load of the servers and number of server connection counts.

Server Group

To add a new Server Group, click on the Add button (See Figure 9.2), click on the SAVE button to make the changes permanent.



The screenshot displays the 'Server Load Balancing' configuration page. On the left, a dark blue sidebar contains a menu with options: Home, Interfaces, System, Load Balancing (expanded), Algorithms, Route Test, SmartDNS, TCP Congestion Control, Site Load Balancing, Server Load Balancing (selected), Routing, and Tools. The main content area has a header 'Server Load Balancing' with a 'Site:' dropdown and a 'Load Balancing' button. Below this, there are two tabs: 'Server Group' and 'Servers'. The 'Servers' tab is active, showing a table with the following data:

Name	IP Address	Port Number	Balancing Method	Balancing
SSH	11.11.11.22	8022	Source	TCP

Below the table, there are three buttons: 'Add', 'Edit', and 'Delete'. At the bottom right of the main panel, there is a 'Copyright ©' notice.

Figure 9.1 – Server Group

Figure 9.2 - Add Server Group

Name

A unique name for Server group needs to be configured.

IP Address and Port Number

The IP Address for the server group can be configured. Normally IP Address and port number configured here will be the FatPipe WAN IP and any non-standard port. SLB will make FatPipe listen on that port.

Balance Method

The Balance Method defines the load balancing algorithm to be used in the backend for the server group.

Round Robin

In this algorithm each server is used in rotating order. This algorithm ensures smooth and fair distribution of server's processing time as the sessions are distributed equally.

Least Connection

In this algorithm the server with the least number of connections or sessions receives the connection. This algorithm is suggested where the sessions are expected to

be longer as in LDAP, SQL and so on. Least connection algorithm does not work well with for shorter sessions like HTTP.

Source

The source IP address is hashed and divided by the total weight of the running servers to designate which server will receive the request. This ensures that the same client IP address will always reach the same server as long as no server goes down or up. This algorithm is generally used in TCP mode where no cookie is inserted.

Balance Mode

The Balance Mode option will set the running mode or protocol of the instance.

TCP

The instance will work in pure TCP mode. A full-duplex connection will be established between clients and servers, and no layer 7 examination will be performed. This is the default mode. It should be used for SSL, SSH, SMTP, etc.

HTTP

The instance will work in HTTP mode. The client request will be analyzed in depth before connecting to any server. Any request which is not RFC-compliant will be rejected. Layer 7 filtering, processing and switching will be possible. This is the mode which brings SLB most of its value.

HTTP Options

The HTTP Options will be enabled for HTTP balance mode.

Force HTTP Close

Enabling this flag ensures that the HTTP connection is closed after each response. This ensures that the connection is closed even if the client fails to close the session causing a buildup of number of inactive session.

Add X-Forwarded-For

Enabling this flag adds the 'X-Forwarded-For' header to the requests sent to servers which might be required by

the servers for logging and other purposes to identify the actual client initiating the session.

Check Cache

This flag enables SLB to analyze all server responses and block requests with cacheable cookies. When a session cookie is returned on a cacheable object, there is a high risk of session crossing or stealing. In some situations, it is better to block the response. The option 'Check Cache' enables deep inspection of all server responses for strict compliance with HTTP specification in terms of cache ability.

HTTP Redistribution

Redispatch -This flag ensures session redistribution in case of connection failure. In HTTP mode, if a server designated by a cookie is down, clients may definitely stick to it because they cannot flush the cookie, so they will not be able to access the service anymore. Selecting 'Redispatch' option allows the proxy to break their persistence and redistribute them to an available server.

Persist

This flag ensures forced persistence on servers that are down. When an HTTP request reaches a backend with a cookie that refers to an unavailable server, it is dispatched to another server. This option enables the request to be sent to the unavailable server first if absolutely needed. A common use case is when servers are under extreme load and spend their time flapping. This option is used in conjunction with 'Redispatch' so that in the event when the server is completely down, the client request be redirected to an available server.

Cookie Options

Rewrite

This keyword indicates that SLB has to modify the cookie's value to set the server's identifier in it. In this mode all responses needs to be monitored so it works only with HTTP close mode. This option is not recommended for new deployment unless it is absolutely required.

Insert

This options ensures that SLB inserts the persistence cookie to the responses. This option can be used to

upgrade existing configurations running in the 'Rewrite' mode.

Prefix

This option enables SLB to prefix a value to the already existing value. This option is helpful in certain environments where multiple cookies are not supported. This option can function only in HTTP close mode as the requests and responses require modifications.

Indirect

This check box option when enabled in 'Insert' mode, will add cookies when the server is accessed for the first time after being selected by a load-balancing algorithm. There is no need to insert the cookie again for the further requests as the client has all the required information.

No cache

This check box option works in conjunction with 'Insert' mode when there is cache between SLB and the client. It ensures that a cacheable response is tagged non-cacheable if a cookie needs to be inserted. This option is significant to avoid one server getting over loaded due to persistence cookies that are added on a cache-able home page.

Post only

This check box option enables SLB to insert cookie only on responses to POST requests. This option can be looked as an alternative to 'Nocache' option as POST responses are non-cacheable. To optimize caching this option is very efficient as most sites do not require persistence before the first POST which is generally a login request.

Protocol Tests

SSL Check

This option enables SSL based health checks for testing the server's availability. SLB relays SSL-based protocols in TCP mode to test and see if the server correctly talks SSL. This option ensures that the client Hello health check packet is sent to the server and the server responds with a SSL server Hello message.

SMTP Check

This option enables SMTP based health checks for server testing. When 'SMTP Check' is enabled, the health check connection is followed by a SMTP command. The server's reply code is analyzed and looked for valid response. Any invalid response or lack of response will mark the server unavailable.

HTTP Check

This option enables HTTP protocol check on the server's health. In contrast to the default server health checks where only a TCP connection is tried to establish, HTTP Check ensures that a complete HTTP request is sent once the TCP connection is established. Any invalid response or lack of response will mark the server as dead. The port to connect and the time interval are specified in the server configuration.

HTTP URL

This string is the URL referenced in the HTTP requests. It defaults to "/" which is accessible by default on almost any server, but may be changed to any other URL.

Disable on 404

This option enables maintenance mode upon HTTP/404 response to health-checks. When this option is set, a server which returns an HTTP code 404 will be excluded from further load-balancing, but will still receive persistent connections. This provides a very convenient method for Web administrators to perform a graceful shutdown of their servers.

Ignore on 404

This option will still consider the server for load balancing even if 404 errors are returned during health check.

Group Timeouts

The timeout values for a session established through FatPipe SLB will be controlled using the following parameters.

Connection - This timeout sets the maximum time to wait for a successful connection to a server. If the server is located on the same LAN as SLB, the connection should be immediate.

Queue

This timeout sets the maximum time to wait in the queue for a connection slot to be free. When a server's maximum connections are reached, connections are left pending in a queue; a timeout is applied to requests pending in the queue. If the timeout is reached, it is considered that the request will almost never be served, so it is dropped.

Client

This timeout sets the maximum inactivity time on the client side. The inactivity timeout applies when the client is expected to acknowledge or send data. It is a good practice to set the client timeout equal to the server timeout.

Server - This timeout sets the maximum inactivity time on the server side. The inactivity timeout applies when the server is expected to acknowledge or send data.

HTTP Request

This timeout sets the maximum allowed time to wait for a complete HTTP request. In order to offer DoS protection, it may be required to lower the maximum accepted time to receive a complete HTTP request without affecting the client timeout. Note that this timeout only applies to the header part of the request, and not to any data. As soon as the empty line is received, this timeout is not used anymore.

Tarpit

This timeout sets the duration for which tarpitted connections will be maintained. When a connection is tarpitted, it is maintained open with no activity for a certain amount of time, and then closed. Tarpit timeout defines how long it will be maintained open.

Standard Options

All Backups

This flag enables the use of all backup servers at a time during primary failure. By default, the first operational backup server gets all traffic when normal servers are all down. When option 'All Backups' is enabled, the load balancing will be performed among all backup servers when all normal ones are unavailable. There will not be any priority order between the backup servers.

Servers

This tab will allow Adding/Deleting/Editing servers to a server group. Select the server from the drop-down menu.

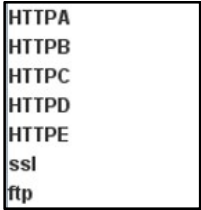


Figure 9.3 - Add Server Name

Multiple servers can be configured in a server group. Servers added in TCP balance mode should not have cookie value in its configuration. Cookie value can only be configured on servers to be added under HTTP Balance mode.

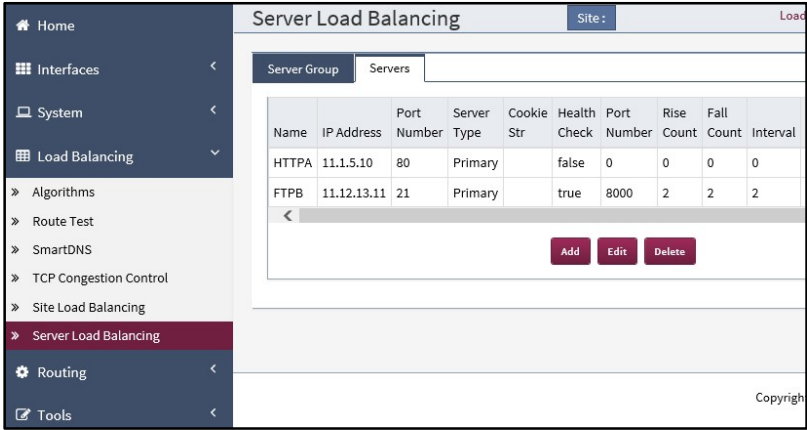


Figure 9.4 - Servers

To add a new server, click on the Add button (See Figure 9.5). Click on the SAVE button to make changes permanent.

Add/Edit Server

Name: IP Address:

Port Number:

Performance Options

Maximum Connections: Maximum Queue:

Weight:

Standard Options

Server Type: ☒ Primary ☐ Backup

Cookie Value:

Health Checks

☒ Health Checking

Port Number:

Rise Count: Fall Count:

Interval (Seconds):

Figure 9.5 - Add Servers

Name

The unique name of the server is configured.

IP Address

IP address of the server is configured.

Port Number

Specify the port number on which the server is listening.

Performance Options

Maximum Connections

The 'Maximum Connections' parameter specifies the maximal number of concurrent connections that will be sent to this server. If the number of incoming concurrent requests goes higher than this value, they will be queued.

Maximum Queue

The 'Maximum Queue' parameter specifies the maximal number of connections which will wait in the queue for this server. If this limit is reached, next requests will be re-dispatched to other servers instead of indefinitely waiting to be served.

Weight

The 'Weight' parameter is used to adjust the server's weight relative to other servers. All servers will receive a load proportional to their weight relative to

the sum of all weights, so the higher the weight, the higher the load. The default weight is 1, and the maximal value is 256. A value of 0 means the server will not participate in load-balancing but will still accept persistent connections.

Standard Options

Server Type - Primary or Backup - The server's role or type can be marked as either Primary or Backup. Only servers marked as Primary will participate in load balancing. Servers marked as Backup will participate only when all Primary marked Servers are down. Requests coming with a persistence cookie referencing the server will always be served though.

Health Checks

This option enables health checks on the server. If 'Health Checking' check box is selected, the server will receive periodic health checks to ensure that it is really able to serve requests. A different port number can be configured for health checks.

Server Statistics

Server Statistics page displays the usage details and error details for all the server groups in real time (See Figure 9.6).

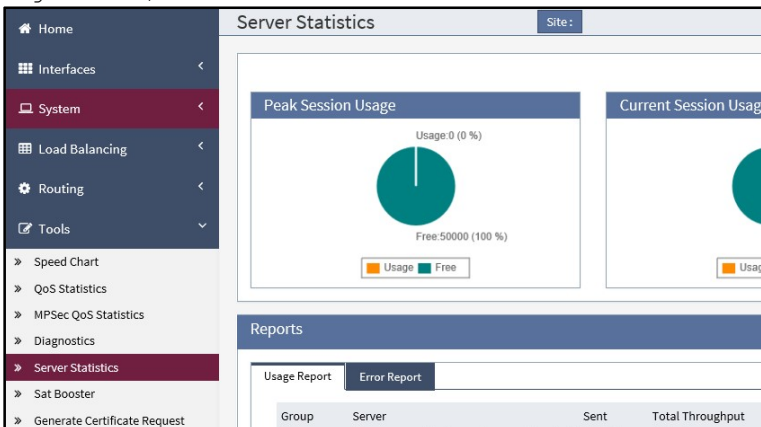


Figure 9.6 - Server Statistics

This page gives an idea into the current load and the session handling capacity of each server. These statistics will help to fine tune servers' configuration to achieve optimal performance. "Current Session Usage" and "Peak Session Usage" pie charts give a bird's eye

view of the sessions count handled by the appliance. The user can set the Refresh time, so that the chart will refresh automatically.

Chapter 10: VPN

FatPipe VPN allows you to create and configure VPN tunnels between

- a) Two or more remote networks (site-to-site VPNs) and
- b) With remote users using mobile VPN clients (sometimes referred to as Road Warriors)

FatPipe VPN is an add-on feature. You need to configure just one tunnel per LAN subnet per interface for Remote users. All remote users can connect to the VPN interface using this tunnel.

Remote End VPN

The Remote End VPN user feature provides connectivity for mobile VPN clients. It allows individual users to connect to hosts on the LAN behind FatPipe by using a VPN client.

Remote End VPN User (mobile user also known as Road Warrior) failover is possible if the Remote End VPN User client allows the user to specify a gateway as a host name, rather than with an IP address. The user can create a SmartDNS entry for the gateway with multiple WAN IP addresses. Then, the user creates one Remote End VPN User tunnel for each of these IP addresses.

When a remote VPN client connects for the first time, it will use whichever IP address the gateway host name was resolved to. When that line goes down, the client will try to reconnect and re-resolve the gateway host name.

SmartDNS will send a "live" IP and the VPN connection will be established on another WAN interface.

For each remote VPN policy, as for each net-to-net IPSEC policy, there is one row in the IPSEC table. Unlike net-to-net policies, policy information about the status and IP addresses are not displayed for remote clients. Instead, we show the number of established remote end VPN user connections in the "Status" column, and string "0.0.0.0" in the Remote network and the Remote External columns.

Currently supported clients are:

- Greenbow VPN Client
- Windows XP VPN Client
- Windows Vista (Business & Enterprise) VPN Client

- Windows 2000 pro VPN client
- Shrew Soft VPN Client

Go to the respective company websites for more information on the different VPN clients listed above. To configure FatPipe VPN settings, click on the Routing button in the main menu and click the VPN tab (see Figure 10.1)

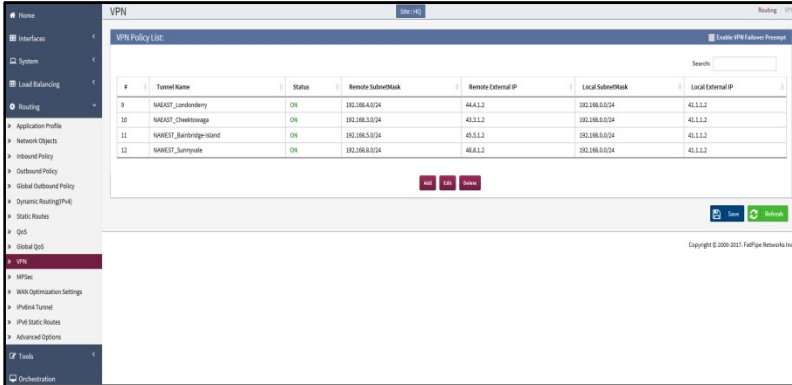


Figure 10.1 - FatPipe VPN

To add a new VPN entry, click on the Add button (see Figure 10.2).

The Add VPN Policy window has the following elements:

Tunnel Name

Specify a unique name for the policy.

Encryption

Select the Encryption type (either "AES" or "3DES" encryption) to be used for the policy. The encryption must match the encryption used on the VPN peer. AES is the strongest encryption, 3DES is next strongest. The default is "AES" encryption.

Authentication

Select the Authentication method (either "SHA1" or "MD5" authentication) to be used for the policy. The authentication method must match the authentication method used on the VPN peer. The default is "SHA1" authentication method.

Remote End

Select the Remote End User type (either "User" or "Network") to create VPN tunnels. The Default is "Network."

Choose the "User" option to create VPN tunnels between the local network and the remote hosts that will connect using mobile VPN clients also known as Road Warriors. The check-box indicates that the connection being added will arrive from an unspecified IP address and it will be established by personal VPN client, rather than a gateway. The "Remote Info" text-fields for remote end IP information will be grayed out.

PFS

By enabling PFS, if someone breaks a key, PFS ensures that the attacker is not able to derive any other key. If PFS is not enabled, someone can potentially break the IKE SA secret key.

TCP Maximum Segment size

The MSS value helps set the maximum segment size. The size range is from 576 -1460. The default value is 1372. You will not be required to change the default value in most situations. If you do see performance issues, adjust the settings.

NAT-T (NAT-Traversal)

NAT-T functionality helps create tunnels between VPN devices even if they are behind NAT devices like firewalls. NAT-T can be setup in either "Auto" (automatic) or "Forced" mode. Choosing the Auto mode leaves the VPN devices to negotiate NAT-Traversal. The IKE port and encapsulated UDP port hold the default values 500 and 4500 (see Figure 10.2) for this mode.

☐ Template

Tunnel Name
NAEAST_Rockville

Remote End
☒ Network ☐ User

Encryption
AES128 ☒

Authentication
SHA1

NAT-T
☒ Auto ☐ Forced
☐ Custom Ports
IKE Port: Numeric
Encapsulated UDP Port: Numeric

Other
TCPMSS: 1372
DPD Timeout: 120
DPD Del: 30
☐ PFS

Local Info
☐ Local LAN Networks
Network IP Address/Mask: 192.168.0.0/24
External IP: 41.1.1.2
Add Edit Delete

Remote Info
☐ Remote LAN Networks
Network IP Address/Mask: 192.168.2.0/24

Key Management
☒ Pre-Shared Secret ☐ RSA Signature ☐ RSA Certificates
Pre-Shared Key: 1234
Remote ID: 42.2.1.2
IKE Lifetime: 1 hour
Key Lifetime: 0 minute 1 hour

Figure 10.2 – Add VPN Policy with NAT-T Auto configured.

NAT-T Forced mode will force the VPN devices to encapsulate IPsec packets into UDP frames to solve traversal problems that may occur with intermediate NAT routers.

To configure Forced mode option, select the Forced option, (see Figure 10.3). The normal ports for NAT-Traversal are UDP 500 for Key negotiation ports and UDP 4500 for data exchange. You can change these values by checking the Custom Ports checkbox, which allows you change these values to any valid UDP Port number.

☐ Template

Tunnel Name
NAEAST_Rockville

Remote End
☒ Network ☐ User

Encryption
AES128

Authentication
SHA1

NAT-T
☐ Auto ☒ Forced
☒ Custom Ports
IKE Port: 1097
Encapsulated UDP Port: 4006

Other
TCPMSS: 1372
DPD Timeout: 120
DPD Delay: 30
☐ PFS

Local Info
☐ Local LAN Networks
Network IP Address/Mask: 192.168.0.0/24
External IP: 41.1.1.2
Add Edit Delete

Remote Info
☐ Remote LAN Networks
Network IP Address/Mask: 192.168.2.0/24

Key Management
☒ Pre-Shared Secret ☐ RSA Signature ☐ RSA Certificates
Pre-Shared Key: 1234

Figure 10.3 – Add VPN Policy with NAT-T Forced configured

Configure a Local Network to Remote Network VPN Tunnel by selecting the Network Remote End option.

Local Info

Network - local network IP for the policy
Subnet - local subnet mask for the policy
External IP - local external IP used for the policy (this should be one of the WAN interface IPs)

Remote Info

Network - remote network IP for the policy
Subnet - remote subnet mask for the policy
External IP - remote external IP used for the policy (this should be one of the WAN interface IPs)

Key Management

Select the key management type to use for the policy. The key management type must match the key management type used on the VPN peer. You can use a Pre-Shared Secret, RSA Signature. Each has its own set of sub-options. Below are the steps for using each of the key management types. Here are the details:

Pre-Shared Secret

Enter an alphanumeric pre-shared secret phrase (must be the same on the VPN peer).
Configure the IKE Lifetime and Key Lifetime. Standard lifetime for both is 8 hours (see Figure 10.4).

The screenshot shows the 'Add/Edit VPN Policy Rule' configuration window. The 'Template' checkbox is unchecked. The 'Tunnel Name' is 'NAEAST_Rockville'. The 'Remote End' is set to 'Network'. The 'Encryption' section shows 'AES128' selected. The 'Authentication' section shows 'SHA1' selected. The 'NAT-T' section has 'Auto' selected. The 'Custom Ports' checkbox is unchecked. The 'IKE Port' is 'Numeric'. The 'Encapsulated UDP Port' is 'Numeric'. The 'Local Info' section has 'Local LAN Networks' unchecked. The 'Network IP Address/Mask' is '192.168.0.0/24'. The 'External IP' is '41.1.1.2'. The 'Remote Info' section has 'Remote LAN Networks' unchecked. The 'Network IP Address/Mask' is '192.168.2.0/24'. The 'Key Management' section has 'Pre-Shared Secret' selected. The 'Pre-Shared Key' field is empty.

Figure 10.4 - Pre-Shared Secret key for "Network" configuration

RSA Signature

Enter Left RSA ID as a Fully Qualified Domain Name preceded by an@ sign (e.g., @chicago.example.com.). It must end with a dot. It does not need to be a real domain name; it is simply used as a unique identifier.

Enter Right RSA ID as a Fully Qualified Domain Name preceded by an @ sign (e.g., @denver.example.com.). It must end with a dot. It does not need to be a real domain name; it is simply used as a unique identifier.

Click on the Get Local Key button to generate a public key. This key will be used on the remote VPN peer that will connect to this peer. You will need to generate a public key on the remote VPN peer and paste its public key into the Remote Public Key text field for this

policy. In other words, each VPN peer will have the other's public key specified under Remote Public Key.

Configure IKE Lifetime and Key Lifetime. Standard lifetime for both is 8 hours (see Figure 10.5, Figure 10.6).

Caution:

Do not click on the Re-Create Local Key button unless you want to change your public key. Click Ok. The new VPN policy will display in the VPN tab. Click on the SAVE button to make the changes permanent.

The screenshot shows a VPN configuration window with the following sections:

- Template:** A checkbox labeled "Template".
- Tunnel Name:** A text field containing "NAEAST_Rockville".
- Remote End:** Radio buttons for "Network" (selected) and "User".
- Encryption:** A dropdown menu showing "AES128".
- Authentication:** A text field showing "SHA1".
- NAT-T:** Radio buttons for "Auto" (selected) and "Forced". Below are checkboxes for "Custom Ports", and fields for "IKE Port" and "Encapsulated UDP Port", all labeled "Numeric".
- Other:** Fields for "TCP MSS" (1372), "DPD Timeout" (120), "DPD Delay" (30), and a checkbox for "PFS".
- Local Info:** A checkbox for "Local LAN Networks", a text field for "Network IP Address/Mask" (192.168.0.0/24), and an "External IP" field (41.1.1.2) with "Add", "Edit", and "Delete" buttons.
- Remote Info:** A checkbox for "Remote LAN Networks", a text field for "Network IP Address/Mask" (192.168.2.0/24).
- Key Management:** Radio buttons for "Pre-Shared Secret", "RSA Signature" (selected), and "RSA Certificates". Below are fields for "Local RSA ID" (jrockville.example.com) and "Remote RSA ID" (@SLC.example.com). A "Remote Public Key" field is at the bottom.

Figure 10.5 – RSA Signature key for “Network” configuration

The screenshot shows a text area titled "Local Public key" containing the following RSA public key string:

```
0sAQOG4uLdJehMa40XLDpb5F21ZbQokP9Ja+YPyaheOclXY3
ao0+ttmmP1YQR5NiQjp/Jlu89+QxYvi36QjgmAxGjnlVs07g4
ruwG3xelAWuJai9cdEDYwwBbCl8PjeuZ0kgfYhh7ruCbUJKSF
bMbtZsPeUlenjm9oq6td4Ne+6YcNQVIEB0Wa9pV9i+HCUZQyu
0Nvv/pA4P16t7nGGXWuQJBPP+jinwMjET//NzvmZG1cels+Lsc
lpNwv1T2THzHJZlkxJpsT5pWasMIXOUCEU3CXWB8Zo8FR
URiQZ
```

Figure 10.6 – RSA Local public key

Configuring a Local Network to Remote user VPN Tunnel by selecting the User Remote End option.

Local Info

Network

Local network IP for the policy

Subnet

Local subnet mask for the policy

External IP

Local external IP used for the policy (this should be one of the WAN interface IPs)

Remote Info

Fields for network, subnet and external IP are not required for Mobile user configuration. This information will be disabled.

Key Management

Select the key management type to use for the policy. The key management type must match the key management type used on the VPN peer. You can use a Pre-Shared Secret key or Certificates. Each has its own set of sub-options. Steps for using IPSEC Certificates are provided below.

Pre-Shared Secret

Enter an alphanumeric pre-shared secret phrase, which must be same on the VPN peer.

Configure the IKE Lifetime and the Key Lifetime. The Standard lifetime for both is 8 hours (see Figure 10.7, Figure 10.8).

☐ Template

Tunnel Name
NAEAST_Rockville

Remote End
Network • User

Encryption
AES128

Authentication
SHA1

NAT-T
• Auto
☐ X-Auth

Other
TCPMSS 1372
DPD Timeout 120
DPD De 30
☐ PFS

Local Info
☐ Local LAN Networks
Network IP Address/Mask
192.168.0.0/24

External IP
41.1.1.2

Remote Info
☐ Remote LAN Networks
Network IP Address/Mask

Key Management
• Pre-Shared Secret
☐ Certificates

Pre-Shared Key
Hsecur@451222

Figure 10.7 - Pre-Shared Secret key for "User" with NAT-T Auto configured

☐ Template

Tunnel Name
NAEAST_Rockville

Remote End
Network • User

Encryption
AES128

Authentication
SHA1

NAT-T
☐ Auto • Forced
☐ X-Auth

Other
TCPMSS 1372
DPD Timeout 120
DPD De 30
☐ PFS

Local Info
☐ Local LAN Networks
Network IP Address/Mask
192.168.0.0/24

External IP
41.1.1.2

Remote Info
☐ Remote LAN Networks
Network IP Address/Mask

Key Management
• Pre-Shared Secret
☐ Certificates

Pre-Shared Key
Hsecur@451222

Figure 10.8 - Pre-Shared Secret key for "User" with NAT-T Forced configured

IPSEC Certificates

IPSEC Certificate is an authentication method for remote, mobile user IPSEC tunnels. Remote mobile users are sometimes referred to as Road Warriors.

IPSEC Certificates give you better control over key management for different users (see Figure 10.9).

Without the IPSEC Certificate, IPSEC uses a Pre-shared key as the only authentication and key-management method. One of its main drawbacks is that the same key is shared for all users of a particular tunnel. Disabling access for one user thus requires changing the key for all others. IPSEC certificate allows you (the Administrator) to create one certificate per user. Each certificate can be generated or revoked separately. When you (the Administrator) generate a certificate, it will be exported and handed over to the user. The user then stores it on his/her computer and imports it into his/her client.

The screenshot displays a web-based configuration interface for IPSEC. At the top, there's a 'Template' checkbox and a 'Tunnel Name' field set to 'NAEAST_Rockville'. To the right, a 'Remote End' dropdown is set to 'User'. The interface is divided into several sections: 'Encryption' with 'AES128' selected; 'Authentication' with 'SHA1' selected; 'NAT-T' with 'Auto' selected and 'X-Auth' disabled; 'Local Info' with 'Local LAN Networks' disabled and an 'External IP' of '41.1.1.2'; 'Remote Info' with 'Remote LAN Networks' disabled; and 'Key Management' with 'Pre-Shared Secret' disabled and 'Certificates' selected. At the bottom, there are three buttons: 'Generate Remote Certificate', 'Export Remote Certificate', and 'Revoke Remote Certificate'.

Figure 10.9 - Certificates for "User" configuration

Generate Remote Certificate

To generate the Remote Certificate, switch the Key Management radio option to Certificates. The local certificate installed on the FatPipe is created with the

Local ID, the Remote ID, and Remote Certificate password. When the certificate is created, it is signed internally by MPVPN's root CA certificate.

Local ID

Enter the name of the local site in the Local ID field.

Remote ID

Click on the Generate Remote Certificate button. Enter the name of the remote site/user in the Remote ID field and click Ok. You will see a message in the bottom part of the window showing that the Certificate was created successfully. The generated certificates of the tunnel will be displayed in the Remote ID dropdown list. The certificate is now ready for export to the remote user (see Export Remote Certificate below) (see Figure 10.10, Figure 10.11, and Figure 10.12).

The screenshot shows a VPN configuration window with the following sections:

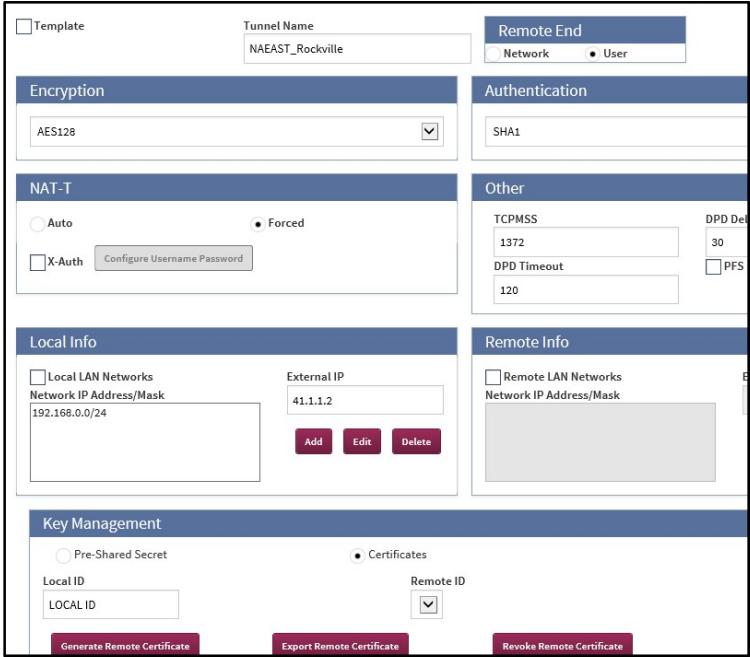
- Template:** A checkbox labeled "Template".
- Tunnel Name:** A text field containing "NAEAST_Rockville".
- Remote End:** A dropdown menu with "Network" and "User" (selected).
- Encryption:** A dropdown menu showing "AES128" with a checkmark.
- Authentication:** A text field containing "SHA1".
- NAT-T:** Radio buttons for "Auto" and "Forced" (selected). Below is a checkbox for "X-Auth" and a button "Configure Username Password".
- Other:** Fields for "TCPMSS" (1372), "DPD Timeout" (120), "DPD Delay" (30), and a checkbox for "PFS".
- Local Info:**
 - Checkbox "Local LAN Networks".
 - Text field "Network IP Address/Mask" containing "192.168.0.0/24".
 - Text field "External IP" containing "41.1.1.2".
 - Buttons "Add", "Edit", and "Delete".
- Remote Info:**
 - Checkbox "Remote LAN Networks".
 - Text field "Network IP Address/Mask".
- Key Management:**
 - Radio buttons for "Pre-Shared Secret" and "Certificates" (selected).
 - Text field "Local ID" containing "LOCAL ID".
 - Dropdown menu "Remote ID" with a checkmark.
 - Buttons "Generate Remote Certificate", "Export Remote Certificate", and "Revoke Remote Certificate".

Figure 10.10 - Generate Remote Certificate



The screenshot shows a window titled "Remote ID" with a blue header bar. Below the header, there is a section labeled "Remote ID" containing a text input field with the word "Remote" inside it.

Figure 10.11 - Entering Remote ID



The screenshot shows a complex VPN configuration window. It has a blue header bar with a "Template" checkbox. Below the header, there are several sections: "Encryption" with a dropdown set to "AES128"; "NAT-T" with radio buttons for "Auto" and "Forced" (selected), and a checkbox for "X-Auth"; "Local Info" with a checkbox for "Local LAN Networks" and a text area for "Network IP Address/Mask" containing "192.168.0.0/24"; "External IP" with a text field containing "41.1.1.2" and buttons for "Add", "Edit", and "Delete"; "Remote End" with a dropdown set to "User"; "Authentication" with a dropdown set to "SHA1"; "Other" with fields for "TCPMSS" (1372), "DPD Timeout" (120), and "DPD Delay" (30); "Remote Info" with a checkbox for "Remote LAN Networks" and a text area for "Network IP Address/Mask"; and "Key Management" with radio buttons for "Pre-Shared Secret" and "Certificates" (selected), and fields for "Local ID" (LOCAL ID) and "Remote ID" (selected from a dropdown). At the bottom, there are three buttons: "Generate Remote Certificate", "Export Remote Certificate", and "Revoke Remote Certificate".

Figure 10.12 - Created Certificate saved to list

Export Remote Certificate

Select the generated certificates of the tunnel from the dropdown list and click on the Export Remote Certificate button (see Figure 10.13). Enter the password to save the certificate (see Figure 10.14, Figure 10.15) The selected certificate will be converted to a- Personal Information Exchange Syntax Standard - PKCS12 - (.p12) file format and exported, which can be saved and distributed to the client(s) who use the Remote End User tunnel.

The screenshot shows a VPN configuration window with the following sections:

- Template:** A checkbox labeled "Template".
- Tunnel Name:** A text field containing "NAEAST_Rockville".
- Remote End:** Radio buttons for "Network" and "User", with "User" selected.
- Encryption:** A dropdown menu showing "AES128" with a checkmark.
- Authentication:** A text field containing "SHA1".
- NAT-T:** Radio buttons for "Auto" and "Forced", with "Forced" selected. Below it is a checkbox for "X-Auth" and a button "Configure Username Password".
- Local Info:** A checkbox for "Local LAN Networks". Below it is a text field "Network IP Address/Mask" containing "192.168.0.0/24". To the right is a text field "External IP" containing "41.1.1.2" and three buttons: "Add", "Edit", and "Delete".
- Remote Info:** A checkbox for "Remote LAN Networks". Below it is a text field "Network IP Address/Mask".
- Key Management:** Radio buttons for "Pre-Shared Secret" and "Certificates", with "Certificates" selected. Below "Pre-Shared Secret" are text fields for "Local ID" and "Remote ID". Below "Certificates" is a dropdown menu for "Remote ID" showing a checkmark. At the bottom are three buttons: "Generate Remote Certificate", "Export Remote Certificate", and "Revoke Remote Certificate".

Figure 10.13 - Export Remote Certificate

The screenshot shows a dialog box titled "Password" with a single text input field for entering a password.

Figure 10.14 - Entering password

The screenshot shows a file save dialog box with the following content:

- Text: "You have chosen to open:"
- File icon and name: "Remote-cert.p12"
- Text: "which is: p12 File (2.4 KB)"
- Text: "from: http://10.2.0.180"
- Text: "What should Firefox do with this file?"
- Radio buttons: "Open with" (with a "Browse..." button) and "Save File" (selected).
- Checkbox: "Do this automatically for files like this from now on." (unchecked).
- Buttons: "OK" and "Cancel".

Figure 10.15 - Saving the certificate

Revoke Remote Certificate

To revoke the generated certificates, Select the Remote ID from the dropdown list and click on the Revoke Remote Certificate button (see Figure 10.16).

The screenshot displays a VPN configuration web interface. At the top, there's a 'Template' checkbox and a 'Tunnel Name' field containing 'NAEAST_Rockville'. To the right, a 'Remote End' section has 'Network' and 'User' radio buttons, with 'User' selected. Below this, the interface is divided into several panels: 'Encryption' with an 'AES128' dropdown; 'Authentication' with a 'SHA1' dropdown; 'NAT-T' with 'Auto' and 'Forced' radio buttons ('Forced' is selected) and an 'X-Auth' checkbox; 'Other' with 'TCPMSS' (1372), 'DPD Timeout' (120), and 'DPD Del' (30) fields, plus a 'PFS' checkbox; 'Local Info' with 'Local LAN Networks' (192.168.0.0/24) and 'External IP' (41.1.1.2) fields, and 'Add', 'Edit', 'Delete' buttons; 'Remote Info' with 'Remote LAN Networks' field; and 'Key Management' with 'Pre-Shared Secret' and 'Certificates' radio buttons ('Certificates' is selected). Under 'Key Management', there are 'Local ID' and 'Remote ID' (None) dropdowns. At the bottom, three buttons are visible: 'Generate Remote Certificate', 'Export Remote Certificate', and 'Revoke Remote Certificate'.

Figure 10.16 - Revoke Remote Certificate

Click Ok. The new VPN policy will display in the VPN tab. Click on the SAVE button to make the changes permanent. To edit a VPN Policy Rule, select it from the list and click on the Edit button. Click on the OK button to return to the VPN page, then click on the SAVE button to make the changes permanent (see Figure 10.17).

☐ Template

Tunnel Name

NAEAST_Rockville

Remote End

☐ Network ☒ User

Encryption

AES128

✓

Authentication

SHA1

NAT-T

☐ Auto ☒ Forced

☐ X-Auth

Configure Username Password

Other

TCPMSS

1372

DPD Timeout

120

DPD Del

30

☐ PFS

Local Info

☐ Local LAN Networks

Network IP Address/Mask

192.168.0.0/24

External IP

41.1.1.2

Add

Edit

Delete

Remote Info

☐ Remote LAN Networks

Network IP Address/Mask

Key Management

☐ Pre-Shared Secret ☒ Certificates

Local ID

LOCAL ID

Remote ID

✓

Generate Remote Certificate

Export Remote Certificate

Revoke Remote Certificate

Figure 10.17 – Edit VPN Policy

To delete a VPN Policy Rule, select it from the list and click on the Delete button. Click on the SAVE button to make the changes permanent.

Chapter 11: VPN Site Failover

MPVPN can be configured to provide failover of VPN tunnels from one site to another. The networks can be located in geographically separate locations, but have identical or similar information at both the main office and the disaster recovery site. This feature is used primarily by companies with disaster recovery sites. This is an optional feature available upon request.

VPN Site Failover can provide failover for VPN traffic between branch office(s), main office and disaster recovery sites, utilizing all lines available at each site. The MPVPNs at the main site and the disaster recovery site need to have identical configurations including VPN tunnels and MPSEC channels. Each branch office should have one tunnel to the main office and one tunnel to the disaster recovery site.

VPN Site Failover determines the status of connectivity between sites dynamically using the MPSEC feature. At the branch office, each VPN tunnel is given a priority and needs to be in the same VPN site Failover Group. The MPVPN at the branch office will establish a VPN tunnel to the site which has the highest priority first. If connectivity to that site fails, then the VPN Site Failover will establish the VPN tunnel to the next Disaster Recovery site that has the next priority in the group.

Initial Setup for VPN Site Failover

VPN Site Failover can be configured only for VPN tunnels between Remote Networks, not Remote Users. To configure VPN Site Failover, click the VPN tab under the Routing main menu (see Figure 11.1).

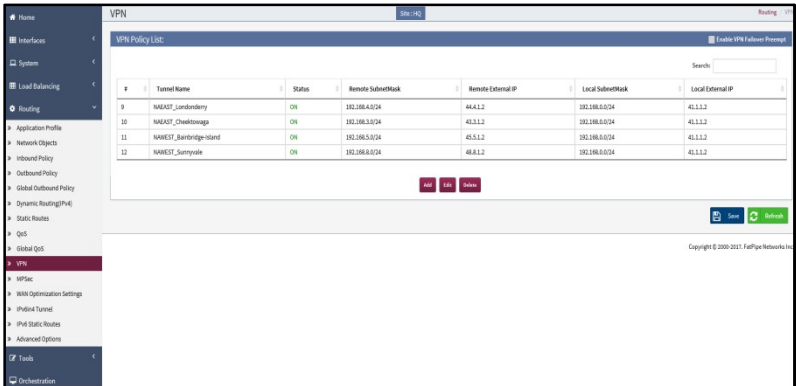


Figure 11.1 - VPN Site Failover

In addition to the configuration of VPN tunnel such as Tunnel Name, Encryption, Authentication, Local Info, Remote Info, Key Management, you will need to configure the VPN Site Failover information such as Group Name, Priority and Failover after [n] failed MPsec polls (see Figure 11.2).

Once the VPN Site Failover is configured, it will show the status of the tunnels in the main VPN Policy List with a status of either ON or OFF. The tunnel with the highest priority will be ON and the tunnel with the lower priority will be OFF (see Figure 11.1).

As part of the VPN Site Failover configuration, tunnels participating in VPN Site Failover need to be configured in the same group. Priority should be set differently for the tunnels at different MPVPN sites.

☐ Template

Tunnel Name: NAEAST_Rockville

Remote End: ☒ Network ☐ User

Encryption
AES128 ☒

Authentication
SHA1

NAT-T
☒ Auto ☐ Forced
☐ Custom Ports
IKE Port: Numeric
Encapsulated UDP Port: Numeric

Other
TCP MSS: 1372
DPD Timeout: 120
DPD Del: 30
☐ PFS

Local Info
☐ Local LAN Networks
Network IP Address/Mask: 192.168.0.0/24
External IP: 41.1.1.2
Add Edit Delete

Remote Info
☐ Remote LAN Networks
Network IP Address/Mask: 192.168.2.0/24

Key Management
☒ Pre-Shared Secret ☐ RSA Signature ☐ RSA Certificates
Pre-Shared Key: 1234
Remote ID: 42.2.1.2
IKE Lifetime: 1 hour
Key Lifetime: 0 minute 1 hour

Figure 11.2 - Add VPN Site Failover

Group Name

Enter a group name for the tunnels to create a Failover group. Only tunnels with the same group name will be participating in VPN Site Failover.

Priority

The Priority Value range is 1-10, with 1 being the highest priority. At each location, you must assign the same priority number to the tunnel going to the main site. Likewise, you must assign a different number to all of the tunnels going to the disaster recovery site.

Each tunnel in a group is given a priority. The lower priority value will have the highest priority. Normally, the tunnel going to the same VPN devices will have same priority value.

Failover after [n] failed MPSec polls

MPSec polls each Remote Location at 1-15 second intervals, as defined in the MPSec tab. This is so the local MPVPN can determine whether the MPVPN at the Remote Location can still be reached or not. The amount of time it takes for the number of failed polls to be reached is a relative value based on the MPSec Polling Interval. The value range is 4-20.

You can add or edit VPN Site Failover Policy Rule entries by using the Add/Edit buttons.

To delete a VPN Site Failover Policy Rule entry from the list, select the particular Tunnel Name and click the Delete button.

MPSec channels needs to be configured for both the tunnels in the VPN Site Failover group (see Figure 11.3).

Figure 11.3 - MPsec channels configured for the tunnels in the VPN Site

Failover group

The status of the MPsec channels should be ON for all the channels configured for these tunnels (see Figure 11.4).

Index	Remote VPN Name	WAN1	WAN2	WAN3	WAN4	Remote VPN IP
1	NAEAST_Rockville	ON	---	---	---	42.2.1.2
2	NAEAST_Rockville	---	ON	---	---	42.2.1.2
3	NAEAST_Rockville	---	---	ON	---	42.2.1.2

Figure 11.4 - Status of MPsec channels for all tunnels

Site Failover Preempt

Site Failover Preempt determines if the traffic remains at the secondary site after the primary is back online. Assigning a high priority and enabling Site Failover Preempt on a site ensures that traffic will stay with that site as long as it is up. If you do not select this option, traffic will continue to be processed by the disaster recovery site, even after the primary site is back online.

Chapter 12: WAN Optimization

FatPipe WAN Optimization is a branch in a box add-on feature to further reduce the need for additional appliances that provide HTTPS acceleration, UDP Aggregation and WAN Optimization for site-to-site traffic. To configure HTTPS acceleration, you can specify specific subnets or leave them as empty with wildcards (*). Please use port number 443 for HTTPS when configuring either of these rules.

Configuring HTTPS Acceleration

To configure HTTPS acceleration, an outbound policy needs to be created. Click on Routing the main menu, and click on the Outbound Policy tab. Select Add and create a new policy route. Select the HTTPS Acceleration check box to accelerate SSL based traffic.

Interface	NAT	Port	NAT IP/Mask	NAT Port	DSCP Tagging	Enable Value	DynLoadBalOpt	Latency Threshold	Jitter Threshold	Packet Threshold
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0

Figure 12.1 – HTTPS Acceleration

Configuring WAN Optimization

To configure WAN optimization, an outbound policy needs to be created. Click on Routing the main menu, and click on the Outbound Policy tab. Select Add and create a new policy route. Select the WAN Optimization check box to optimize site to site WAN traffic.

Name

web

Protocol

All

DSCP

Source

IP

Source

Port

Destination

IP

Action

Allow

QoS

None

Application Profiles

None

Add

Edit

Delete

☐ Source NAT

☐ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☐ Interface Priority

☒ Interface Specific

☐ Mixed Priority

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	Pac Thr
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0

Figure 12.2 – WAN Optimization

Configuring UDP Aggregation

To configure UDP Aggregation, an outbound policy needs to be created. Click on Routing the main menu, and click on the Outbound Policy tab. Select Add and create a new policy route. Select the UDP Aggregation check box to aggregate UDP packets for site to site traffic.

Name

UDP

Protocol

UDP

DSCP

Source

IP

Source

Port

Destination

IP

Action

Allow

QoS

None

Application Profiles

None

Add

Edit

Delete

☐ Source NAT

☐ Follow System Route

☐ Equal Bandwidth Distribution

☐ Maintain Session

Traffic Mode:

☐ Interface Priority

☒ Interface Specific

☐ Mixed Priority

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	Pac Thr
WAN1	Yes	Yes	0.0.0.0-0.0.0.0	0	No	0	No	0	0	0

Figure 12.3 – UDP Aggregation

TCP Congestion Control

TCP is a stream based protocol that ensures reliable data delivery. To manage data delivery and speed of delivery, different algorithms are used to compute the parameters based on network latency, loss, time taken for acknowledgement to receive, etc. These algorithms control the size of the sender's send window size, receiver's receive window size, how long to wait for acknowledgements, reset loss timer or wait timer, etc. These algorithms are collectively called "TCP Congestion Avoidance" algorithms. By utilizing different algorithms for different network line scenarios, the optimal speed or TCP throughput can be achieved.

With FatPipe's TCP Congestion Control feature, located in the Load Balancing main menu, you can select the Congestion Avoidance Algorithm for different network latency ranges. Use the FatPipe defined Congestion Control Algorithms listed. By default, the latency ranges and Congestion Control Algorithms mapped to those is defined by FatPipe. However, you may change the latencies and apply different Congestion Control Algorithms available using drop-down boxes associated with them.

Latency Range

Enter the minimum and maximum latencies for which the chosen algorithm should be applied.

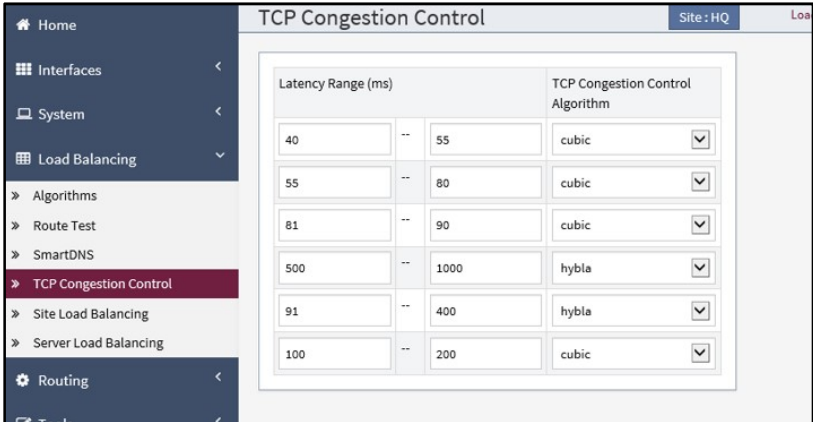


Figure 12.4 - TCP Congestion Control

Application or Protocol Optimization

To configure Application or Protocol Optimization, go to the WAN Optimization Settings page under Routing. Select the Enable WAN Optimization check box to select from a list of Applications or Protocols. (See Figure 12.5)

WAN Optimization Settings Site: HQ

☒ Enable WAN Optimization

☐ Select All

Mail Applications		
	Compress	Cache
POP3	☒	☒
IMAP	☒	☒
SMTP	☒	☒
MS Exchange	☐	☐
Lotus Notes	☐	☐

Database Application	
	Compress
MYSQL	☒
MSSQL	☐
SYBASE	☐
ORACLE	☐

Thin Client		
	Compress	
RDP	☒	
Citrix ICA	☐	
SunRay	☐	

Web		
	Compress	Cache
HTTP	☒	☒
*HTTPS	☐	

Messaging		
	Compress	Cache
ISCSI	☐	☐

Other	
	Compress
Undefined App	☒

File Sharing

- CIFS
- FTP
- NFS

Messaging

- Yahoo Messenger
- MSN

Business Applications

- Share Point
- SAP
- ORACLE ERP

Figure 12.5 – Enabling Application based Optimization.

Click on 'Select All' to select all the listed Applications to optimize using both Compression

andCaching

technology.

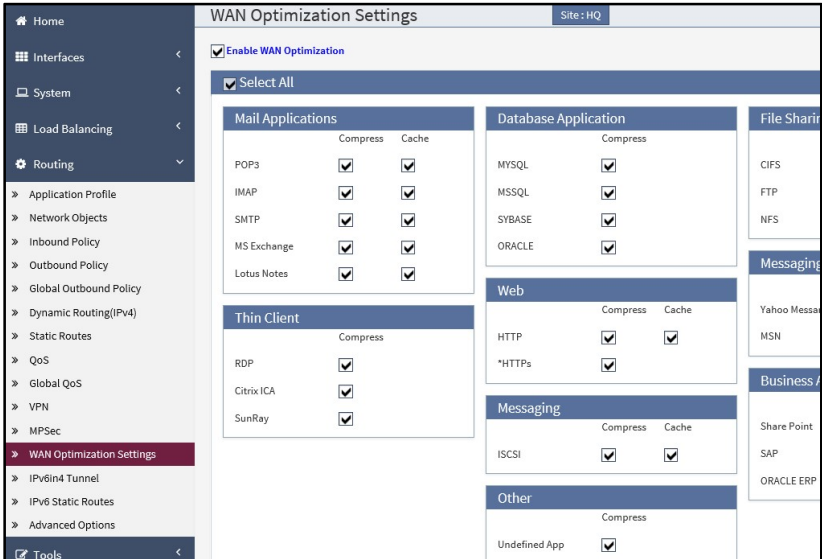


Figure 12.6 – Configuring protocol based optimization

To customize compression and caching for a specific application or protocol, check or uncheck the corresponding option. (See Figure 12.6)

Citrix and RDP Compression

By default, Citrix and RDP traffic are encrypted and compressed. To improve bandwidth usage, you can use FatPipe to perform the encryption and compression of Citrix and RDP. To enable this, you must disable encryption and compression settings in Citrix and RDP. Go to Addendum A to read how to disable these

MPSec Traffic Compression

Select the MPsec site name for which Compression is required and click on configure.

Index	Remote VPN Name	Remote VPN IP	Load Balancing Option
0	Denver	14.14.14.30	Packet

Figure 12.7 - Configuring MPsec Traffic Compression

Connect using WAN	Compression	Weight	Usage	Encryption Type
☒ Connect using WAN1	☐ Compression	1	Primary	IPSEC
☐ Connect using WAN2	☐ Compression	0	Primary	IPSEC
☐ Connect using WAN3	☐ Compression	0	Primary	IPSEC
☐ Connect using WAN4	☐ Compression	0	Primary	IPSEC

Figure 12.8 - Configuring MPsec Traffic Compression

Enable compression by checking the compression check box for different path. For maximum compression check all the paths, if only some paths are selected for compression, traffic that goes on via that specific path would get compressed and otherwise it won't (see Figure 12.8).

MPsec Stats

To view the Stat Chart that show the upload and download of LAN and WAN data, click on the MPsec Tab located in

the Routing menu, and click on Stat Chart (see Figure 12.7).

There, you will see two charts: one for uploads and one for downloads (see Figure 12.9). You will see in the upper left hand corner the site name, which is the remote VPN entry you assigned in MPSec.

The upload chart shows “LAN in” data and “WAN out” data rates calculated. The rate speed is shown in bits per second.

The download section shows the “LAN out” data and “WAN in” data rates calculated. The Total Data Reduction shows the amount of data reduced for uploads and downloads.

Home

Interfaces

System

Load Balancing

Routing

- » Application Profile
- » Network Objects
- » Inbound Policy
- » Outbound Policy
- » Global Outbound Policy
- » Dynamic Routing(IPv4)
- » Static Routes

MPsec

Site :

Local VPN Name: Chicago

Local VPN IP: 11.11.11.20

Polling Interval (ms): 1000

Remote Location

Search

Index	Remote VPN Name	Remote VPN IP	Load Balancing Option
0	Denver	14.14.14.30	Packet

Add Edit Delete Up Down

Select Site Name

None

Figure 12.10 - MPsec Statistics

Clearing Cache

The FatPipe cache repository can be cleared by using this button (see Figure 12.10).

Note: If you clear the cache repository, then the repository may take time to build to obtain optimization and acceleration.

Chapter 13: Web Content Blocking - SatBooster

Web Content Blocking is an add-on feature that allows a user to block advertisements on web pages. It also provides options for blocking a specific file types, including exe, mp3, mp4, swf, wmv (see Figure 13.1). FatPipe references this feature as SatBooster because companies that use Satellite links benefit greatly from blocking these types of traffic, increasing bandwidth capacity by blocking unwanted traffic.

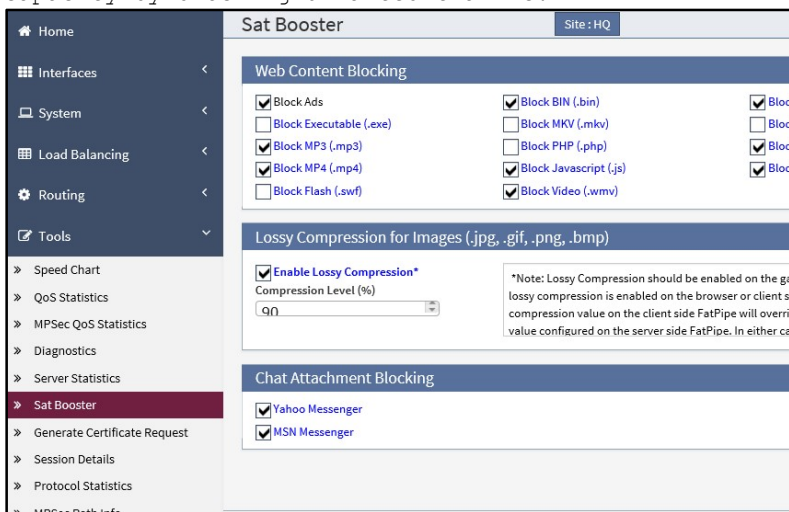


Figure 13.1 - Web Content Blocking

Click on the Tools button in the main menu, and then on the Web Content Blocking tab and select which file types you would like to block.

Lossy Compression

FatPipe WAN Optimization can also provide Lossy compression of images in JPEG, TIFF, GIF, BMP, PNG formats. Applying this feature for web browsing greatly reduces your bandwidth usage and speeds up web browsing, as most web content on the Internet is made up of images.

To enable Lossy Compression, click on the Tools button, and select the SatBooster Tab. There, click on the radio

button next to Enable Lossy Compression for Images and select the level of compression.

Note: It is recommended to use no more than 80% Lossy compression level for most web browsing activity, as images could be seriously distorted and almost illegible if higher compression levels are chosen.

Chapter 14 Central Manager

Central Management Console is an add-on module that allows a user to manage all the FatPipe appliances in their network from a single console without the burden of logging into each unit individually. The units can be organized into groups and configured with a secret key for inter unit communication. This communication is secure with the combined reporting module, Orchestration, provides a bird's eye view of the network with capabilities to drill down to the individual session passing through a FatPipe device at any point of time along with the ability to do a historical search. The system also provides the ability to monitor the FatPipe appliances and raise alerts whenever an event is triggered. You can define any combination of parameters for monitoring and apply them to all the units or to a select group of units.

The management of all you devices are divided into groups. Each group can have multiple devices which can be managed independently. Only one device can be managed at a time.

To configure a group go to Orchestration tab (see figure 14.1)

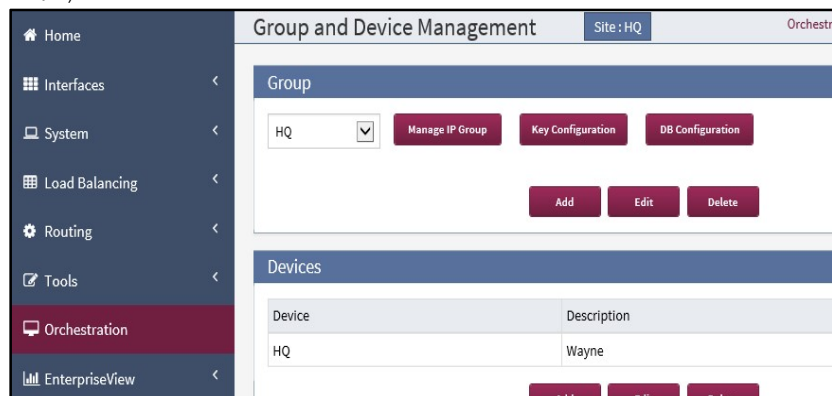


Figure 14.1 Managing Groups and devices

Choose "Default" group as all devices will be added to the default group and you can create logical groups and map devices to them for easy administration. Virtual VPN Network Range will be the Virtual IP subnet used to

assign one IP per brance device. Assign virtual IP if you are using dynamic IP's on any of your WAN interfaces. To add a group click on the add button (see figure 14.2)

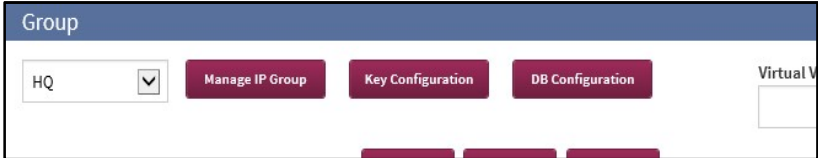


Figure 14.2 Adding a Group in Central Manager

Enter the name and description for the group, then click okay and then click save (see figure 14.3)

Figure 14.3 Adding a Group

Editing a group click the manage button in the CM Header panel, select the group name from the drop down menu and click the edit button (see figure 14.4)

Figure 14.4 Editing a Group.

Devices can be automatically detected and added. If you want to add manually, configure devices by click on add button at the bottom of the window, this will open up a new window for adding a device (see figure 14.5)

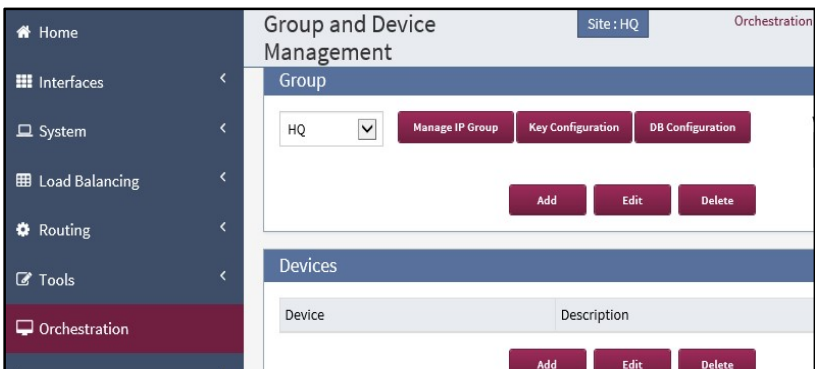


Figure 14.5 Adding a device in a group

Enter a Name for the device, FatPipe Serial Number of the device (Serial Number is case sensitive therefore ensure that the serial number is correct). Enter a relevant description of the device select the group name where it belongs. (See figure 14.6)

The 'Add /Edit User' form is displayed. It includes the following fields and sections:

- Name:** A text input field containing 'NewYork'.
- Description:** A large text area.
- Serial Number:** A text input field containing 'fwrps200110377'.
- Group:** A dropdown menu with 'HQ' selected.
- Virtual VPN IP:** An empty text input field.
- External VPN IP:** An empty text input field.
- IP Addresses:** A section with three text input fields containing '11.1.1.2', '12.1.1.2', and '13.1.1.2'. Below these fields are 'Add' and 'Delete' buttons.
- Device Location:** A section with four text input fields:
 - Address:** Contains 'Sample Addr'.
 - City:** Contains 'NewYork'.
 - State:** Contains 'NY'.
 - Country:** Contains 'USA'.

Figure 14.6 Adding a device in a group.

To access the GUI of the device WAN IP address are needed. Enter the WAN IP address of the remote device which you want to access. To add an IP address click the Add button under the IP address section. Add all the wan IP address one by one. (See figure 14.7), then enter the devicelocation. This needs to be correct in order to map it correctly in the enterprise view. Click ok and then click save (see figure 14.8)

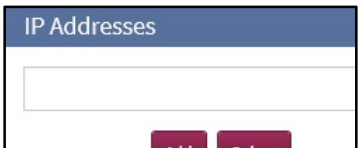
A screenshot of the 'IP Addresses' form. It has a title bar 'IP Addresses' in a blue header. Below the header is a large empty text input field. At the bottom of the form are two red buttons: 'Add' and 'Delete'.

Figure 14.7 Adding IP address

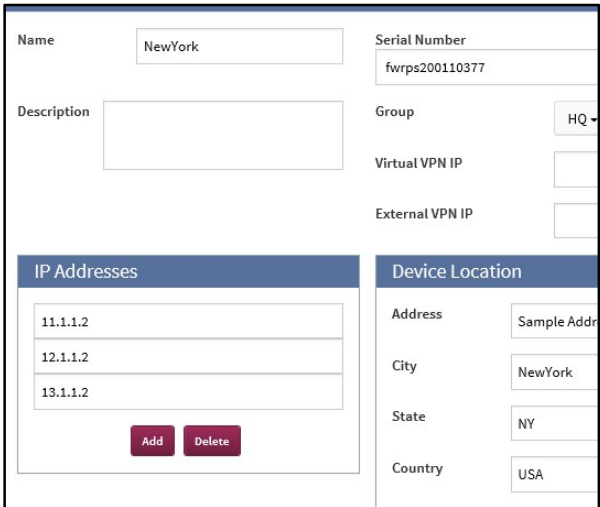
A screenshot of the 'Saving device in a Group' form. The form is divided into several sections. At the top, there are fields for 'Name' (containing 'NewYork'), 'Serial Number' (containing 'fwrps200110377'), 'Description' (empty), 'Group' (a dropdown menu showing 'HQ'), 'Virtual VPN IP' (empty), and 'External VPN IP' (empty). Below these are two side-by-side sections. The left section is titled 'IP Addresses' and contains a list of three IP addresses: '11.1.1.2', '12.1.1.2', and '13.1.1.2'. At the bottom of this list are 'Add' and 'Delete' buttons. The right section is titled 'Device Location' and contains fields for 'Address' (containing 'Sample Addr'), 'City' (containing 'NewYork'), 'State' (containing 'NY'), and 'Country' (containing 'USA').

Figure 14.8 Saving device in a Group

To access the GUI of the branch FatPipeunit's Central manager login must be enabled on the branch unit as well as the HQ unit. They must have the same secret key as well. To enable central manager login click Central manager check box under Users tab under Systems page. (See figure 14.9)

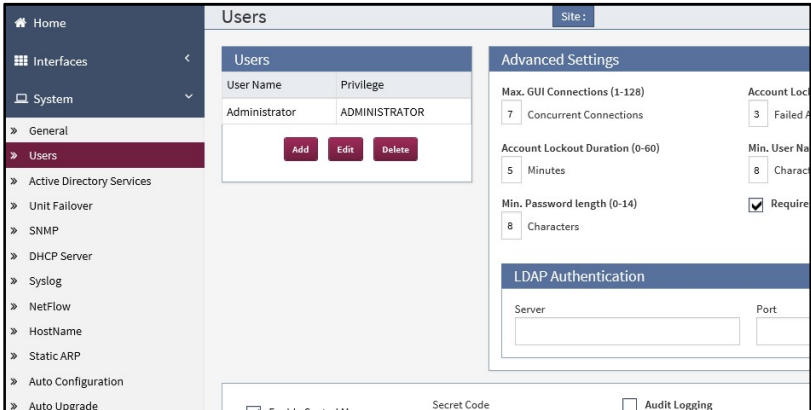
A screenshot of the 'Central Manager Login' form. The form is divided into several sections. On the left is a sidebar with a navigation menu. The main content area is titled 'Users' and contains a table with columns 'User Name' and 'Privilege'. The table has one row with 'Administrator' and 'ADMINISTRATOR'. Below the table are 'Add', 'Edit', and 'Delete' buttons. To the right of the 'Users' section is the 'Advanced Settings' section. It contains several settings: 'Max. GUI Connections (1-128)' (7), 'Concurrent Connections' (3), 'Account Lockout Duration (0-60)' (5 minutes), 'Min. Password length (0-14)' (8 characters), 'LDAP Authentication' (checked), 'Server' (empty), 'Port' (empty), 'Secret Code' (empty), and 'Audit Logging' (unchecked). The sidebar on the left has a 'Home' button and a list of tabs: 'Interfaces', 'System', 'General', 'Users', 'Active Directory Services', 'Unit Failover', 'SNMP', 'DHCP Server', 'Syslog', 'NetFlow', 'HostName', 'Static ARP', 'Auto Configuration', 'Auto Upgrade', and 'Secret Code'.

Figure 14.9 Central Manager Login

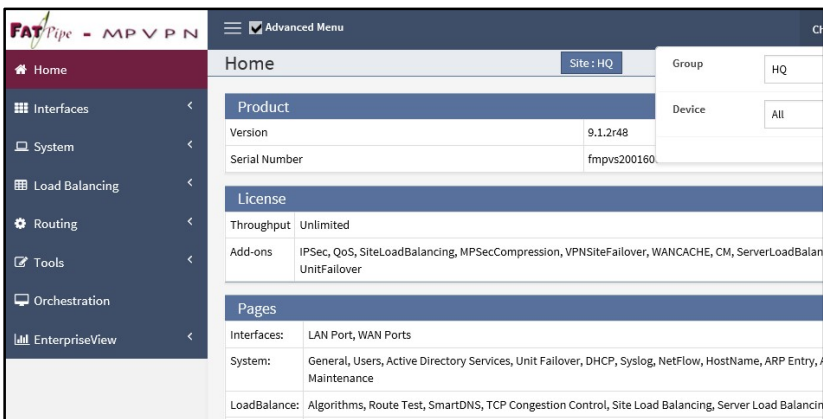


Figure 14.10

To see the status of all the devices together under a group select the group name from the central manager header panel (see figure 14.10) and select all in the devices drop down menu. A window will pop up showing the details of all the devices under the group. (See Figure 14.11)



Figure 14.11

Global Outbound Policy

Global Outbound Policy short for Global Outbound Policy Routing Rule, works the same way as the Outbound Policy Routing rule (see chapter 5: Routing for details). The Global Outbound Policy Routing rule is used when a single policy needs to be applied to one or all FatPipe devices across your network. The Global Policy Routing Rule is configured on the FatPipe MPVPN device with the CM license unit and is applied to all the devices.

This page is divided into two section one is for template and the other one is for actual rule. There are two types of rules Top and Bottom. The Top rule has the highest priority and the Bottom rule has the lowest. The Top Global Policy routing rules takes precedence over locally created Outbound Policy routing rule, and the Bottom rule takes subservience, the order being , global top rules followed by local rules followed by bottom rules.

This Global Outbound Policy Routing Rule will be visible to the branch office units but it cannot be edited or deleted

To make a Global Outbound Policy Routing Rule go to the routing page and select the Global Outbound Policy tab. Under the Template section click on add button to create a template. A single template may contain one or multiple rules. This template is applied to the branch units and all the rules under it. Give a name to the template (See figure 14.12 and figure 14.13)

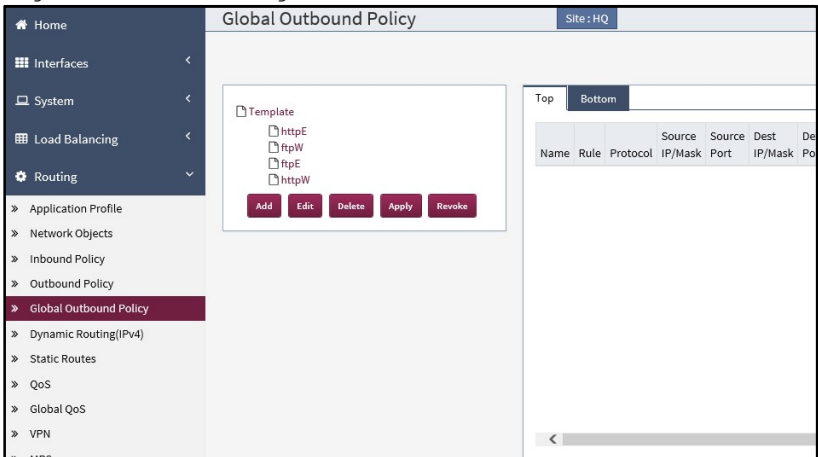


Figure 14.12 Creating an Outbound Policy Routing Rule Template.

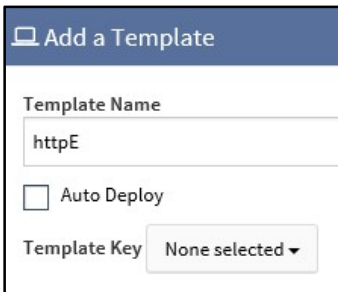


Figure 14.13 Adding a Template

Editing an Template

Click the Edit button to edit an existing template. (See figure 14.14)

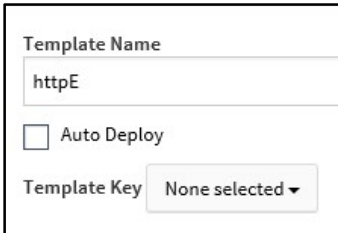


Figure 14.14 Editing a template

Creating a Top Global Outbound to create a Top Global Outbound policy rule create a template as shown in figure 14.12 and click the top tab in the Global Outbound Policy rule page section. Click the add button under this page to create an Outbound Policy Routing Rule. (See figure 14.15)

Name

SAP

Source

IP

*

Action

Allow

Protocol

TCP

Source

Port

*

QoS

None

DSCP

0

Destination

IP

10.2.0.219/32

Application Profiles

UDP Agg

Enable

Destination

Port

*

Application Rules

Equal Bandwidth Distribution

Traffic Mode:

Interface Priority

Interface Specific

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	P
WAN1	Yes	Yes			Yes		Yes			
WAN2	Yes	Yes			Yes		Yes			

Figure 14.15 Adding a Global Outbound Policy Routing Rule

Edit an Outbound Policy Routing Rule

Select the Global Outbound Policy Routing rule that needs to be edited and click the edit button (see figure 14.16)

Name

SAP

Protocol

TCP

DSCP

0

UDP Ag

Enable

Enable

Enable

Destina

Destina

Source

IP

Source

Port

Destination

IP

Port

Port

*

*

10.2.0.219/32

*

Action

Allow

QoS

None

Application Profiles

Application Rules

Add

Edit

Delete

☐ Equal Bandwidth Distribution

Traffic Mode:

☒ Interface Priority

☐ Interface Specific

Interface	NAT	Port NAT	NAT IP/Mask	NAT Port	DSCP Tagging	Value	Enable DynLoadBalOpt	Latency Threshold	Jitter Threshold	P
WAN1	Yes	Yes			Yes		Yes			
WAN2	Yes	Yes			Yes		Yes			

Figure 14.16

To apply a Template to one or multiple branch units select the template from the template section and click the Apply button, this will bring up a pop-up window where you can select to apply the rule on an entire Group, or one or multiple units. If you want to apply the rule on all the units select the check box Units. If you want to select some or all the groups then click on this will expand the list and you can select the groups you want. If the rule needs to be applied for some or specific until then expand the groups and select the units. After the units are selected click on the Proceed button. This will open another pop-up window displaying the "successful" message. If the FatPipe MPVPN device is not able to communicate with the other unit or units then the "unsuccessful" message is displayed in the pop-up. (See figure 14.17 and 14.18)

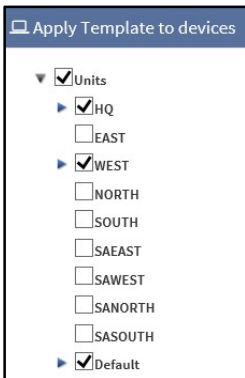


Figure 14.17 Applying Global Outbound Policy Routing Template



Figure 14.18 Success message after Template applied to remote unit

Verifying ruled applied to branch unit.

To verify that the rule has been applied to the branch unit login to the branch unit from the CM control panel by selecting the correct group and device name at the control panel and then click manage. Go to the routing page and select Global Outbound Policy Tab. You will see that the rule/s is present here the rule can only be viewed it cannot be edited or deleted. (See figure 14.19)

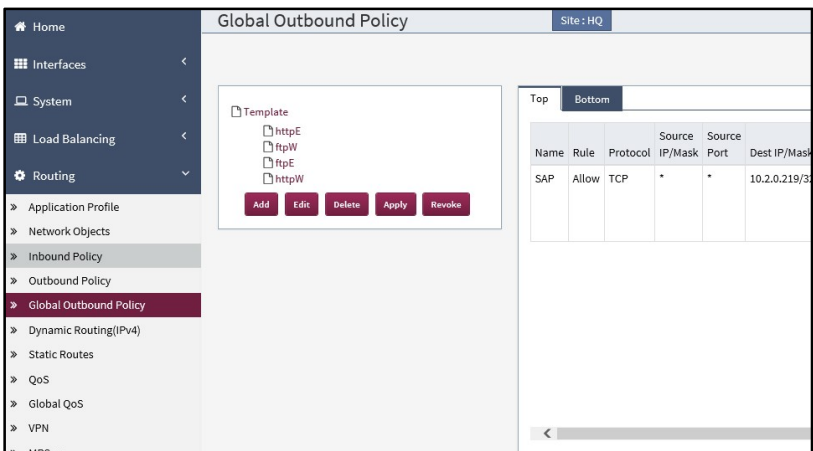


Figure 14.19 Verifying applied rule in remote unit

To edit a rule select the CM licensed unit from the CM control panel and click manage. Now goto Routing page and select the Global Outbound policy tab. Select the Template and then select the policy that needs to be edited. Click the edit button in the policy making section. This will open the outbound policy configuration window (see figure 14.20).

Name: SAP

Protocol: TCP

DSCP: 0

UDP Ag: Enable

Source: IP

Source: Port

Destination: IP

Destination: Port

*

*

10.2.0.219/32

*

Action: Allow

QoS: None

Application Profiles

Application Rules

Add Edit Delete

☐ Equal Bandwidth Distribution

Traffic Mode:
☒ Interface Priority
☐ Interface Specific

Interface	NAT	Port	NAT	NAT	DSCP	Enable	Latency	Jitter	P
			IP/Mask	Port	Tagging	Value	Threshold	Threshold	T
WAN1	Yes	Yes			Yes	Yes			
WAN2	Yes	Yes			Yes	Yes			

Figure 14.20 Editing Global Outbound Policy Rule.

Creating a Bottom Outbound Policy Rule

A Bottom Outbound Policy Rule can be created in the same template or in a new Template. To create please see figure 14.12. After creating a new template click the Bottom tab in the Global Outbound Policy making section this will open the Global Outbound Policy. (See Figure 14.21 and Figure 14.22)

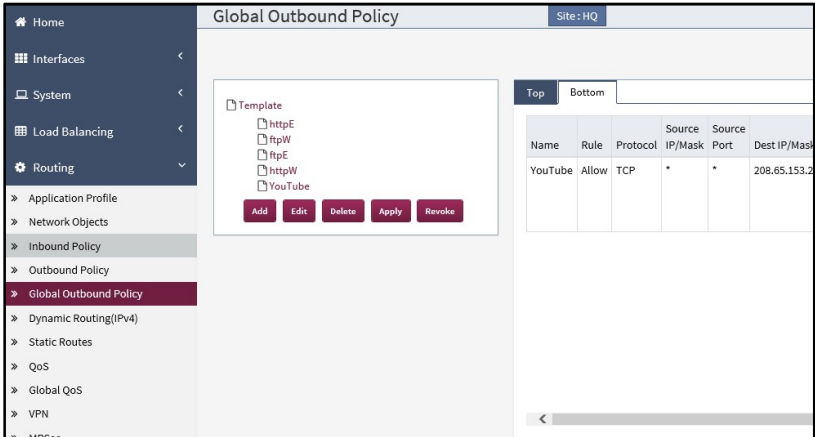


Figure 14.21 Creating a Bottom Outbound Policy Rule Template.

Name YouTube **Protocol** TCP **DSCP** 0 **UDP Ag** Enable **Enable** **Enable**

Source IP **Source** Port **Destination** IP **Port** 80

* * www.youtube.com

Action Allow **QoS** None **Application Profiles**

Application Rules

☐ Equal Bandwidth Distribution

Traffic Mode:

☒ Interface Priority ☐ Interface Specific

Interface	NAT	Port	NAT	NAT	DSCP	Value	Enable	Latency	Jitter	Pa
			IP/Mask	Port	Tagging		DynLoadBalOpt	Threshold	Threshold	Tr
WAN1	Yes	Yes			Yes		Yes			
WAN2	Yes	Yes			Yes		Yes			

Figure 14.22 Add a Bottom Outbound Policy Rule

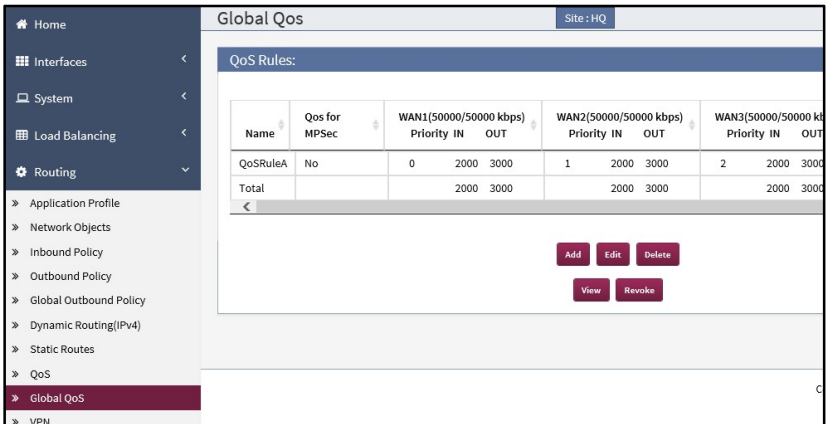


Figure 14.23

Applying Bottom Outbound Policy Rule Template to a Branch unit device see figure 14.17 and 14.18

To view Global Outbound Policy Rule/s that are applied to the CM licensed Unit click the Show HQ Rules button under Global Outbound Policy. See Figure 14.23. This will give the list of rules that are applied to the CM Licensed unit. Click the View button to see the details of the rule (see figure 14.24)

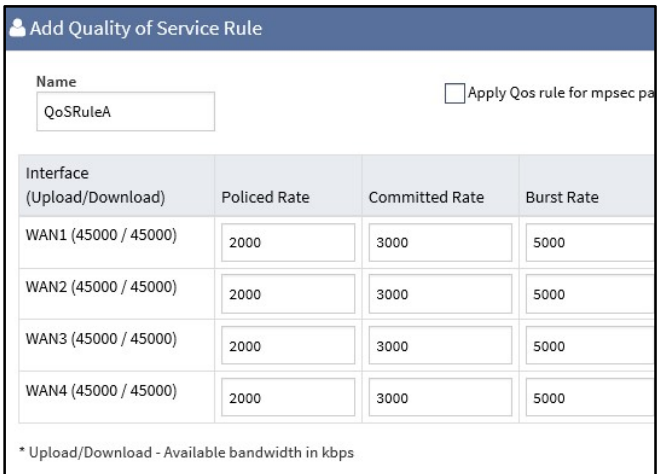


Figure 14.24 Adding Global Outbound Policy

Global Quality of Service

Global QoS is an add-on feature that comes with the CM license, Global QoS works same as QoS (please refer

chapter 7: Quality of Service for details). When applying a single QoS policy to multiple sites, Global QoS can be used to reduce the effort of repetitive configuration. Global QoS can be configured and applied to all the Branches. This global QoS will be visible to the branch office units but it cannot be edited or deleted. It can only be edited or deleted using the CM login. The global QoS can only be applied to a Global PRR.

Creating a Global QoS

To create a Global Quality of Service go to the Routing policy page and click the Global QoS tab. To add a policy click the Add button. See Figure 14.25

This will open a new Window give the policy a name enter the Policed, Committed and Burst and priority. See Figure 14.26. For details on Policy rates please refer to chapter 7: Quality of service Click Ok and then click Save.

To apply the policy to go to Routing page click Global Outbound Policy tab select the template. If a new Template needs to be created please see figure 14.12 and 14.13. If the QoS has to be applied to an existing rule select the rule click edit this would open the Outbound policy rule window. (See figure 14.26 and 14.27)

Interface (Upload/Download)	Policed Rate	Committed Rate	Burst Rate
WAN1 (45000 / 45000)	2000	3000	5000
WAN2 (45000 / 45000)	2000	3000	5000
WAN3 (45000 / 45000)	2000	3000	5000
WAN4 (45000 / 45000)	2000	3000	5000

* Upload/Download - Available bandwidth in kbps

Figure 14.25 editing a Global outbound Policy

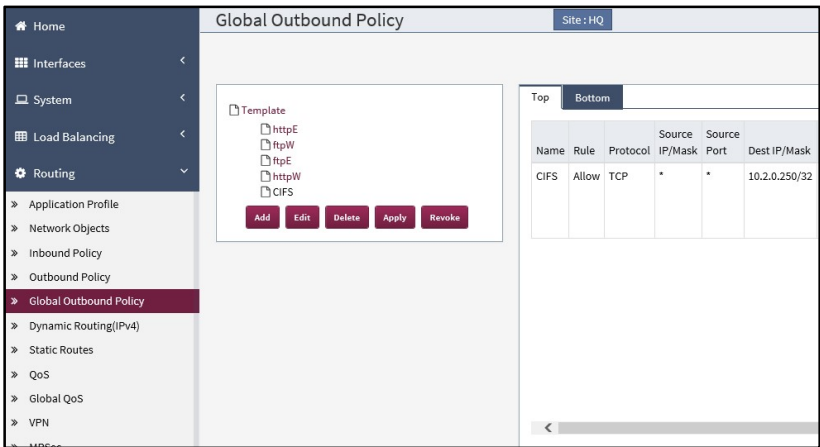


Figure 14.26 Adding a QoS to an existing Global Outbound Policy

Name: CIFS

Protocol: TCP

DSCP: 0

Source: IP

Source: Port

Destination: IP

Source: *

Destination: 10.2.0.250/32

Action: Allow

QoS: QoSRuleA

Application Profiles

Application Rules

Equal Bandwidth Distribution

Traffic Mode: Interface Priority (selected) Interface Specific

Interface	NAT	Port	NAT	NAT	DSCP	Enable	Latency	Jitter
			IP/Mask	Port	Tagging	Value	Threshold	Threshold
WAN1	Yes	Yes			Yes	Yes		
WAN2	Yes	Yes			Yes	Yes		

Figure 14.27

Click the Apply button in the Template section to apply the rule to remote FatPipe see figure 14.17 .MPVPN units. A successful message will be displayed (see Figure 14.28)

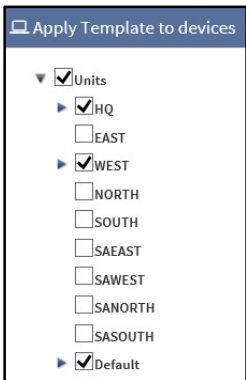


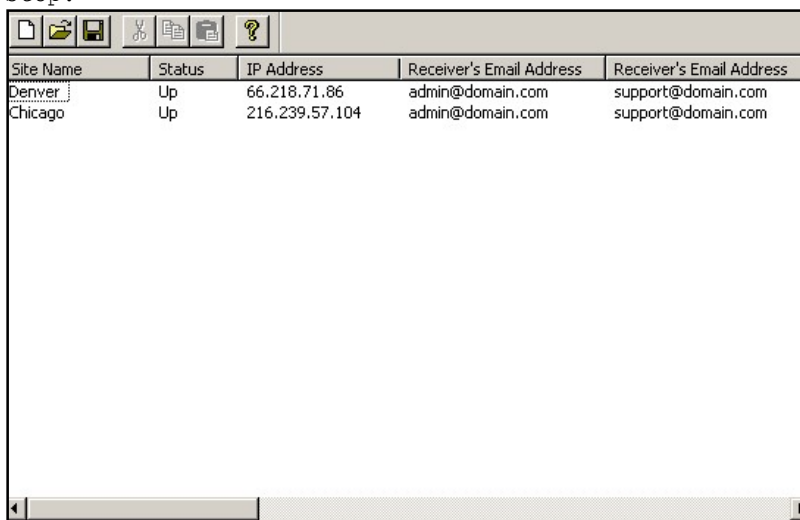
Figure 14.28

Chapter 15: Paging Software

FatPipe MPVPN comes with monitoring software that can continuously test the MPVPN unit and services going through it. The software alerts you if a WAN failure occurs. This monitoring software, called Paging Software, is available for download at <http://www.FatPipeinc.com/paging>

The Paging Software installs on any Windows[®] PC on the network (see Figure 15.1). To use the Paging Software, you should have a text mode cell phone and have e-mail paging capability.

If the status of the network is normal, the status entry in the list will display as "Up," otherwise it will display as "Down." The Paging Software will automatically perform monitoring upon startup. To stop the monitoring, click Paging on the menu and then choose Stop.



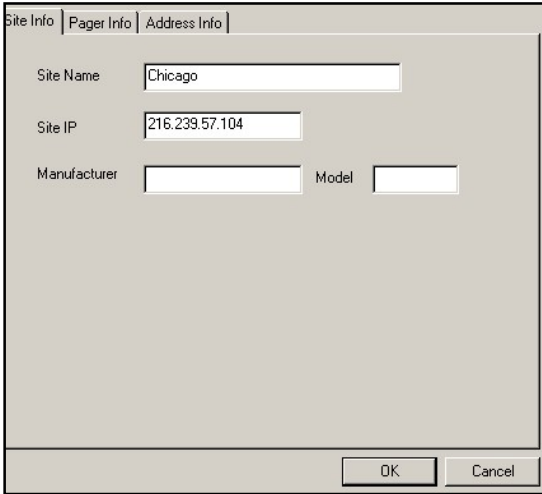
Site Name	Status	IP Address	Receiver's Email Address	Receiver's Email Address
Denver	Up	66.218.71.86	admin@domain.com	support@domain.com
Chicago	Up	216.239.57.104	admin@domain.com	support@domain.com

Figure 15.1 - Paging List

Add New Pager Information

To add new site information to the database, go to Address on the menu and then click Add. This will bring up the dialog box (see Figure 15.2). The Site Name is the place where MPVPN resides; it can be any user defined unique name. The IP Address will be any valid IP address

of the FatPipe MPVPN. The Manufacturer and Model are optional.



The image shows a Windows-style dialog box titled "Add New Site Info". It has three tabs at the top: "Site Info", "Pager Info", and "Address Info". The "Site Info" tab is currently selected. Inside the dialog, there are four text input fields. The first field is labeled "Site Name" and contains the text "Chicago". The second field is labeled "Site IP" and contains the text "216.239.57.104". The third field is labeled "Manufacturer" and is empty. The fourth field is labeled "Model" and is empty. At the bottom right of the dialog, there are two buttons: "OK" and "Cancel".

Figure 15.2 - Add New Site Info

Click the Pager Info tab to bring up a window (see Figure 15.3). The Receiver's E-mail Address1 is the destination e-mail address where information should be sent. A send receiver (Administrator) can be entered on the Receiver's E-mail Address2 (optional). The Sender's E-mail Address is the e-mail address of the sender. The user must enter the SMTP server name or IP address for the page to be sent. The fields Area Code and Pager Number also have to be entered for paging.

Site Info | **Pager Info** | Address Info

Receiver's Email Address1

Receiver's Email Address2

Sender's Email Address

Sender's SMTP Server

Area Code Pager Number

Dial Out
☐ Yes, dial out Choose a connection

Figure 15.3 - Add New Pager Info

Click the Address Info tab to bring up the window (see Figure 15.4). All fields in this window are optional. The user can enter this information for additional detail.

Site Info | Pager Info | **Address Info**

User Name

Company

Street Address

City State

Zip Code Country

Figure 15.4 - Add New Address Info

Change Existing Pager Information

To change existing site information in the database, select the site with your cursor and press the Enter key on the keyboard. Double-click the entry in the list, or go to Address on the menu and then click Edit. This will

bring up the dialog box (see Figure 15.5). All the fields can be modified in this window.

Figure 15.5 is a screenshot of the 'Edit Site Info' dialog box. It features three tabs at the top: 'Site Info' (which is the active tab), 'Pager Info', and 'Address Info'. The 'Site Info' tab contains four input fields: 'Site Name' with the text 'Denver', 'Site IP' with the text '66.218.71.86', 'Manufacturer' (empty), and 'Model' (empty). At the bottom right of the dialog are 'OK' and 'Cancel' buttons.

Figure 15.5 - Edit Site Info

Click the Pager Info tab to bring up the window (see Figure 15.6). You can modify all the fields in this window.

Figure 15.6 is a screenshot of the 'Edit Pager Info' dialog box. It features three tabs at the top: 'Site Info', 'Pager Info' (which is the active tab), and 'Address Info'. The 'Pager Info' tab contains several input fields: 'Receiver's Email Address1' (admin@domain.com), 'Receiver's Email Address2' (support@domain.com), 'Sender's Email Address' (denver-fatpipe@fatpipeinc.com), and 'Sender's SMTP Server' (12.23.113.10). Below these are 'Area Code' and 'Pager Number' fields. A 'Dial Out' section is located at the bottom left, containing a checkbox labeled 'Yes, dial out' (which is unchecked), a 'Test Connection' button, and a 'Choose a connection' dropdown menu. At the bottom right of the dialog are 'OK' and 'Cancel' buttons.

Figure 15.6 - Edit Pager Info

Click the Address Info tab to bring up the window (see Figure 15.7). You can modify all the fields in this window.

The image shows a dialog box titled "Edit Address Info". It has three tabs: "Site Info", "Pager Info", and "Address Info", with "Address Info" being the active tab. The dialog contains several text input fields for user information. The fields and their current values are: "User Name" (Administrator), "Company" (ACME), "Street Address" (301 Winchester Blvd), "City" (Miami), "State" (Florida), "Zip Code" (33133), and "Country" (USA). At the bottom right of the dialog are two buttons: "OK" and "Cancel".

User Name	Administrator				
Company	ACME				
Street Address	301 Winchester Blvd				
City	Miami	State	Florida		
Zip Code	33133	Country	USA		

Figure 15.7 - Edit Address Info

Remove Pager Entry

To remove an existing entry from the database, select the entry and press the Delete key on the keyboard. You may also go to Address on the menu and click Delete. It will bring up the dialog box (see Figure 15.8). Click on the Yes box to delete the entry or click No to cancel the operation.

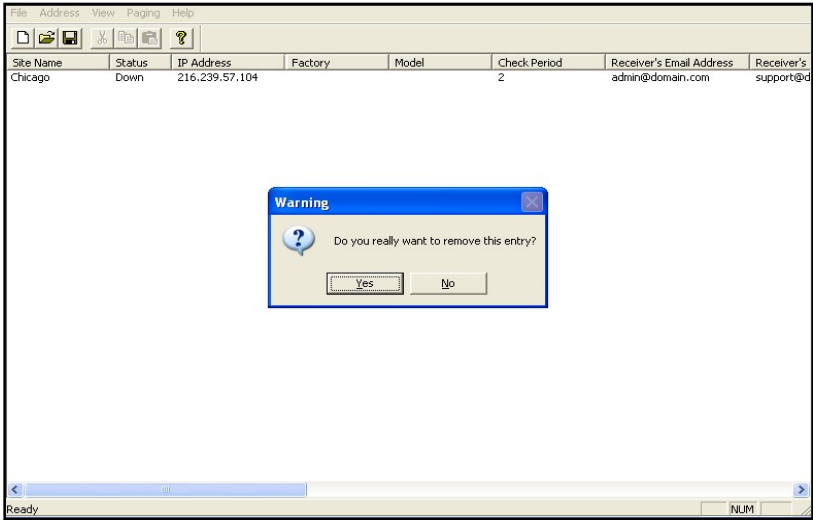


Figure 15.8 - Remove Pager Entry

Addendum A

How to Disable Default Encryption and Compression settings in Citrix

Assuming you have installed Citrix Program Neighborhood, you will need to add an ICA connection. Choose Wide Area Network or Local Area Network depending on your network configuration and click the Next button. (See Figure A1).

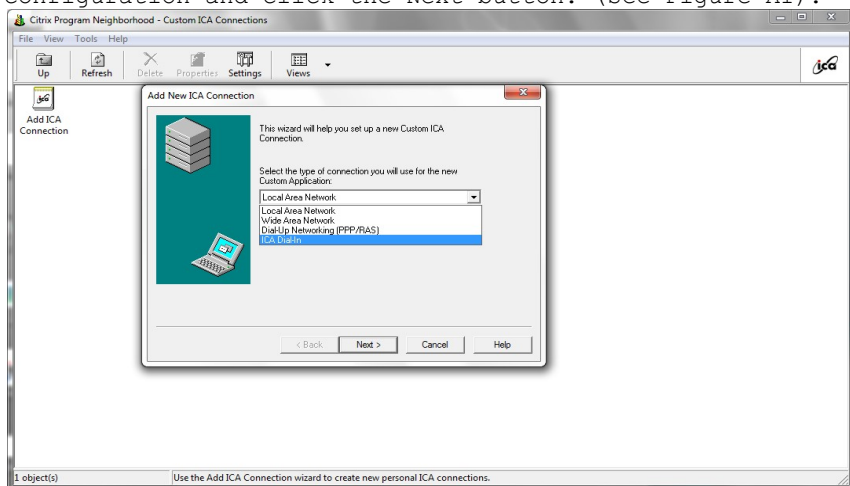


Figure A1 - Citrix Program Neighborhood - Custom ICA Connections

In the Add New ICA Connection Window, enter a description for the ICA connection, and select TCP/IP+HTTP as the network protocol.

Enter the Citrix Metaframe Presentation Server's IP address or you can click the Server Location button to locate the server. Once the server information has been entered, click next and proceed to next Window. (See Figure A2).

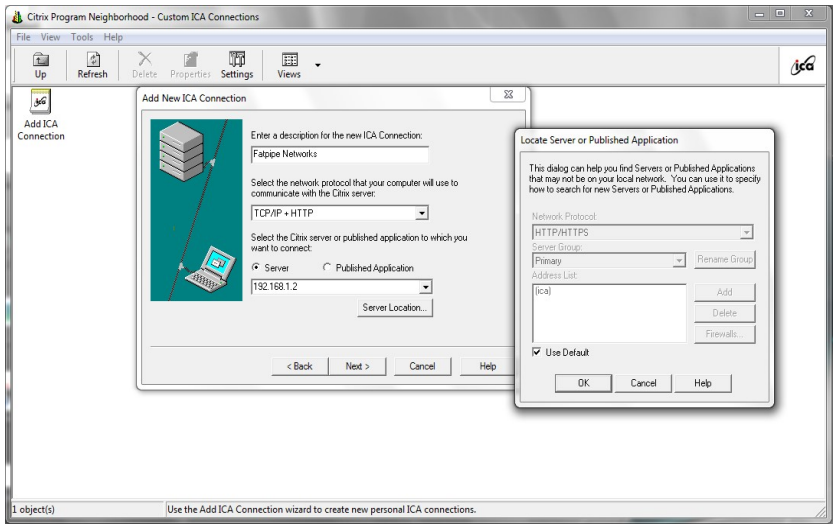


Figure A2 - New ICA Connection Window

You can provide your login credentials and domain name or you can skip this part. Select Basic under the Encryption Level dropdown box. This is required for FatPipe to save bandwidth, and click the Next button. (See Figure A3).

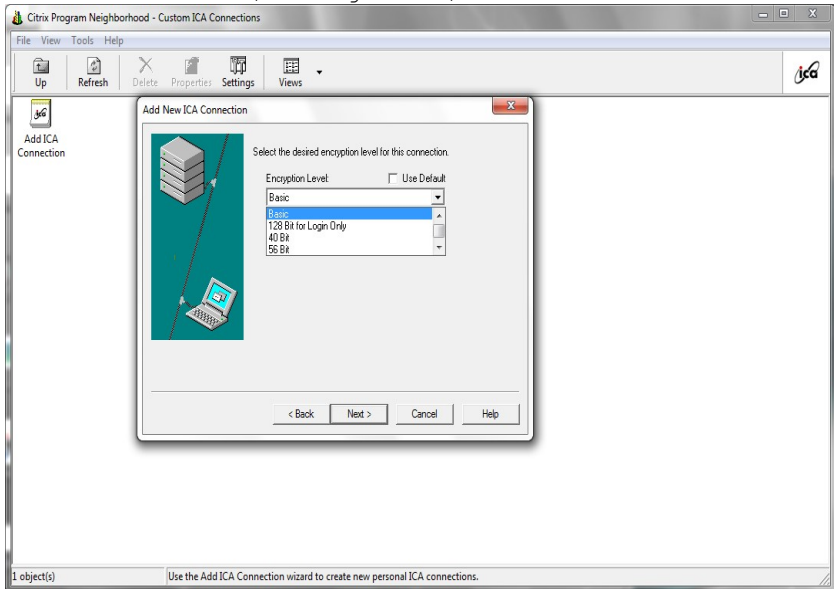


Figure A3 - Add New ICA Connection

Once you have finished adding the ICA Connection, it will show the Connection's in the Citrix Program Neighborhood. Right click on the ICA Connection icon and select Custom Connection settings. Uncheck the "Use data compression" checkbox. The Encryption Level should be set as Basic. (See Figure A4).

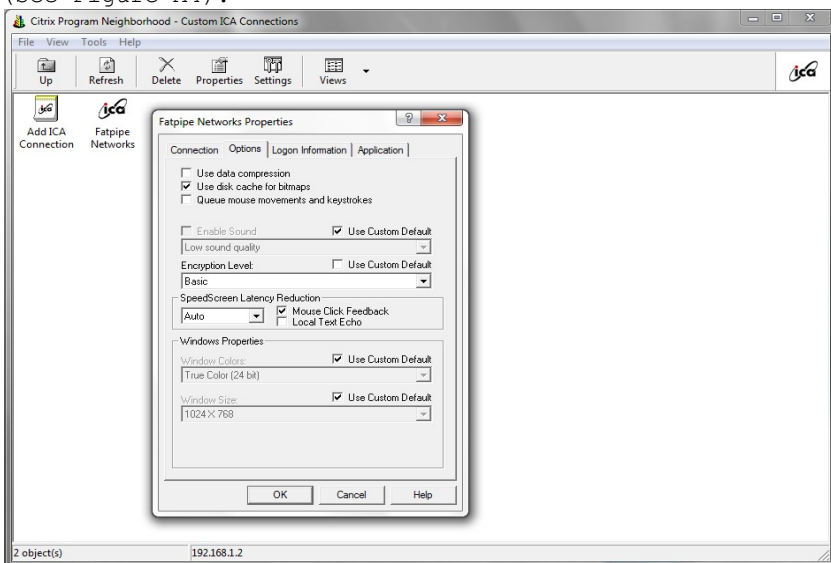


Figure A4 - ICA Connection Custom Settings Window

How to Disable Default Encryption and Compression settings in RDP

RDP Compression

By default, Microsoft Remote Desktop Protocol is compressed and encrypted. To use FatPipe for compressing and encrypting the RDP traffic, which saves bandwidth, you must disable RDP compression and encryption. The following sub-article explains the steps.

You must edit the configuration file to disable compression (the RDP client does not include an option for disabling compression in the registry).

If you have not already configured the RDP client, create the .rdp configuration file.

Select Start > Programs > Accessories > Remote Desktop Connection

Click Options and go to the General tab

Provide the details for Microsoft's Terminal Server

Click Save As in the Connection Settings area and name the file

Locate the .rdp file and open it in a text editor

Locate the following parameter:

Compression:i:1

And change it to:

Compression:i:0

Save the file and close the text editor

Double-click the .rdp file to start your RDP session

The RDP client does not allow you to completely turn off the encryption, but you can set the encryption level to "low."

Type regedit in the command window to view the registry configuration menu (See Figure A5). Follow this path:

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Terminal Server\Winstations\RDP-Tcp

Locate the following value: MinEncryptionLevel, and change its value to 1 (See Figure A6)

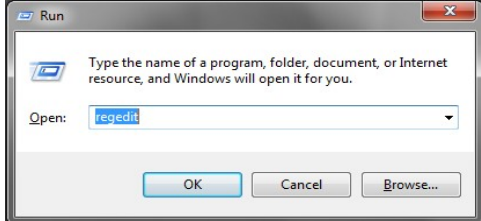


Figure A5 - Command Window

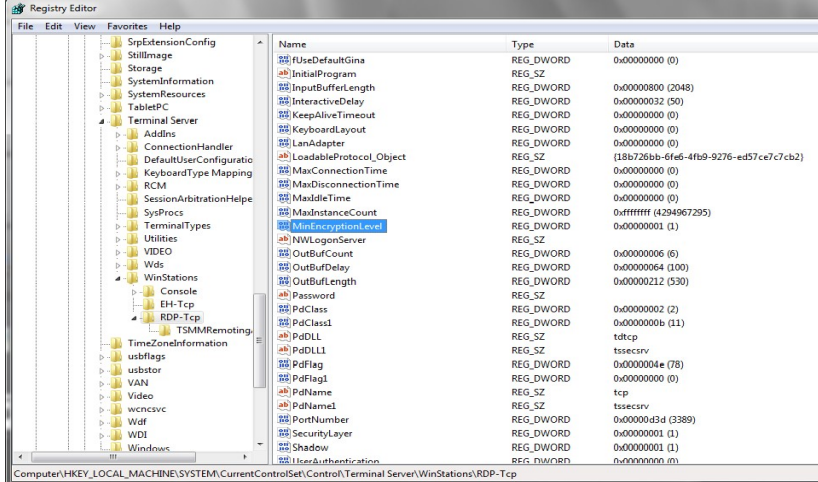


Figure A6 - Registry Configuration Menu

Note: You can disable compression via group policy if you are using Windows Server 2008 R2 and clients are running Microsoft Vista SP1 or higher.

Configuration for Windows Server 2003

Run `commandgpedit.msc`, which opens the Computer Configuration window (See Figure A7)

Follow this path:

Computer Configuration / Administrative Templates / Windows Components / Remote Desktop Services / Remote Desktop Session / Remote Session Environment (See Figures A8)

Locate the property

"Set compression algorithm for RDP data", right click and click properties. Go to setting tab and choose the disabled option (See Figure A9)

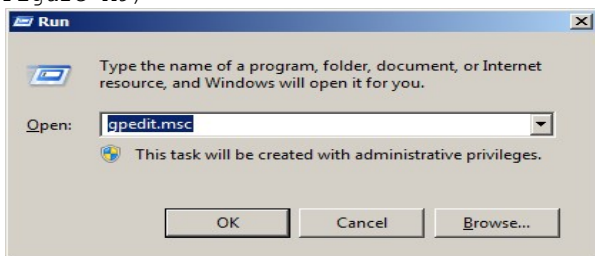


Figure A7 - Command Window

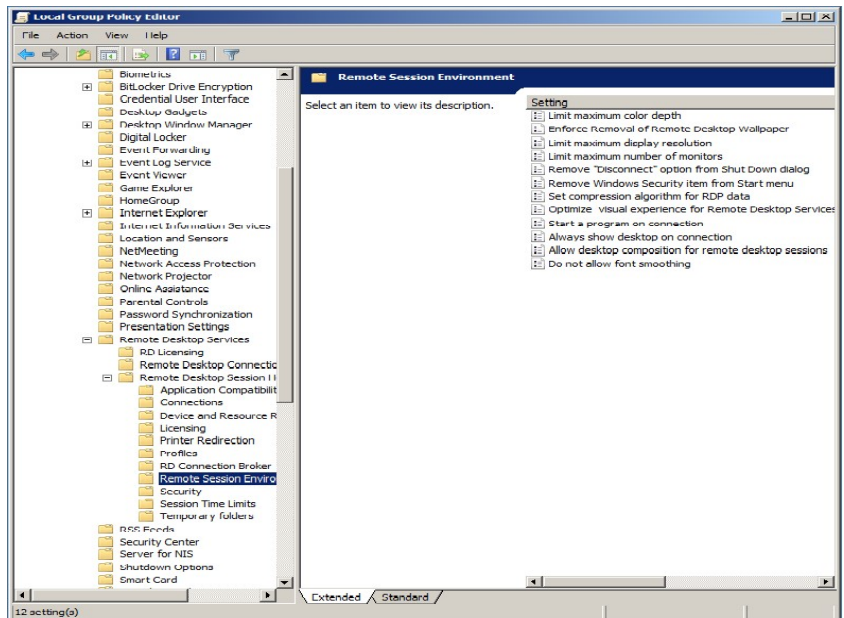


Figure A8 – Local Group Policy Editor Window

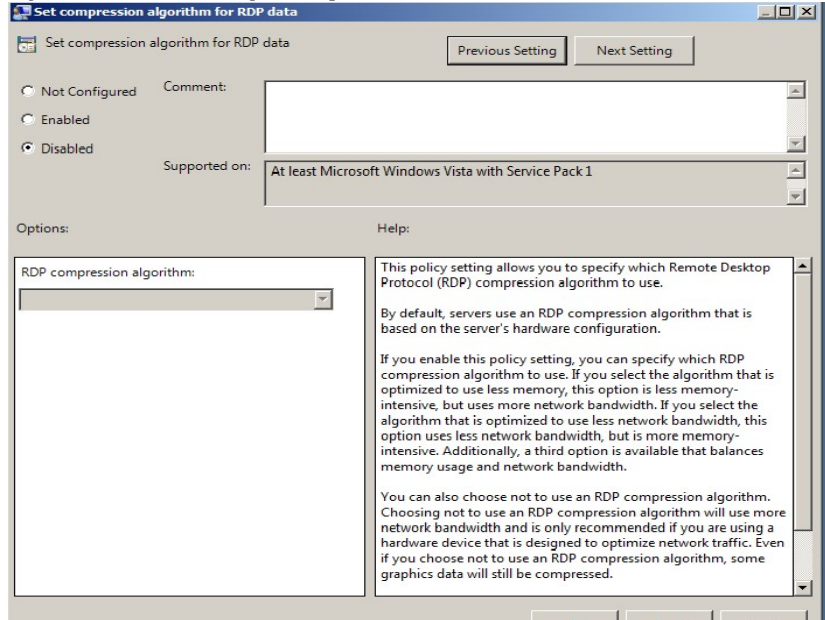


Figure A9 – Compression Algorithm for RDP Data

For disabling encryption, click Start > Remote Desktop Session Host Configuration > Administrative Tools > Remote Desktop Services. (See Figure A10).

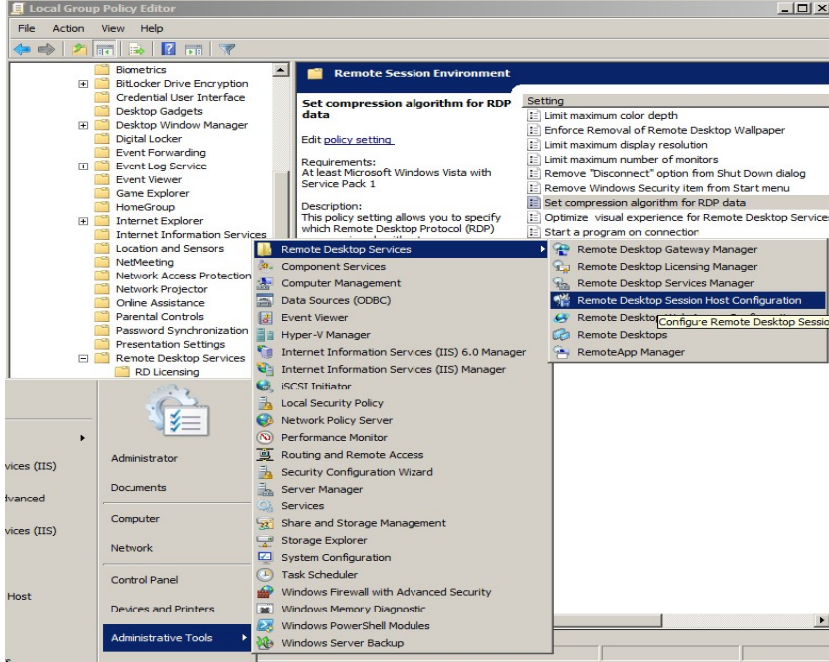


Figure A10 - Remote Desktop Session Host Configuration

It will show all the connection names. Right-click on the connection name and click properties. Under the general tab, choose RDP Security Layer for security layer setting and choose low for encryption settings. (See Figure A11).

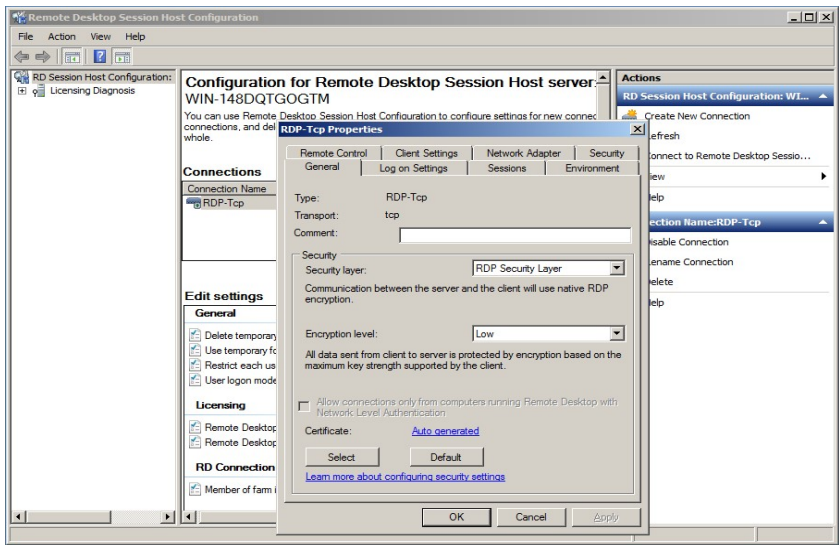


Figure A11 - RDPSecurityLayerforSecurityLayerSetting

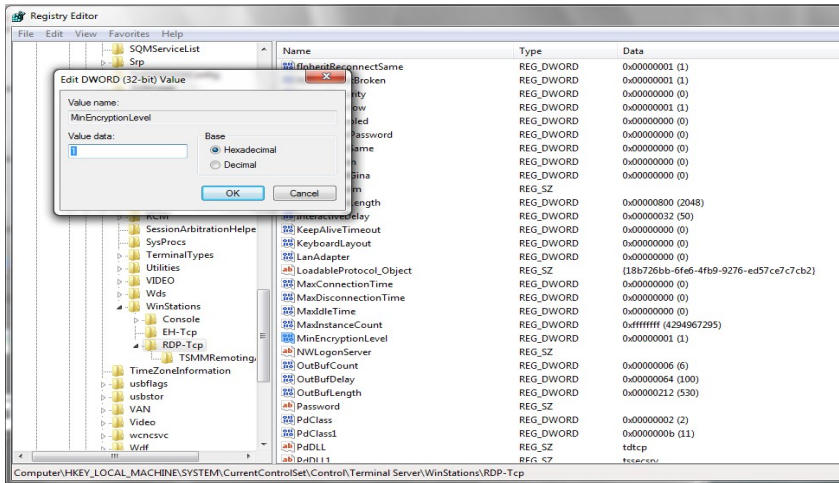


Figure A12 - Registry Configuration Menu

Addendum B

FatPipe Virtual Appliance

The FatPipe Virtual Appliance -- or "VirtFat32" -- comes as an Installer File running on a Microsoft Windows based operating systems (currently Windows 7, 32 bit version is supported).

System Requirements for FatPipe Virtual Appliance

- Windows 7, 32 bit professional
- At least four (4) GB of RAM
- Two (2) GB of RAM must be dedicated to the virtual machine
- At least one Ethernet-based network card (for WAN1 of FatPipe, additional can be added)
- At least 20 GB of dedicated hard drive space reserved for the virtual machine
- Microsoft .NET Framework 4 (Standalone Installer) (<http://www.microsoft.com/download/en/details.aspx?id=17718>)
- Microsoft Visual C++ 2010 Redistributable Package (x86) (<http://www.microsoft.com/download/en/details.aspx?id=5555>)
- CPU must support Intel® VT-x virtualization technology as FatPipe is running a 64 bit operating system on a 32 bit host operating system. Please go to the following link to a list of Intel processors that support virtualization technology: (<http://ark.intel.com/VTList.aspx>)
-

How to Install

Log in as an Administrator. (In Windows Vista or later release, all accounts that are Administrators are not full Administrators but are elevated accounts). Install the prerequisite software components first, if they are not already installed.

Double click the installation file and accept the license agreement and follow the prompts.

How to Uninstall

A shortcut named FatPipe will be created in the Start Menu. Run the Uninstall Virtnet Device shortcut first to uninstall the virtual network device driver. Click on the associated link and it will prompt you for user access

permission (in Windows 7). Next, click the Uninstall icon and follow the prompts.

Management Software

A shortcut named FatPipe will be created in the Start Menu (see Figure B1). Access that to launch the FatPipe management software. Right click on this and select Run as Administrator.

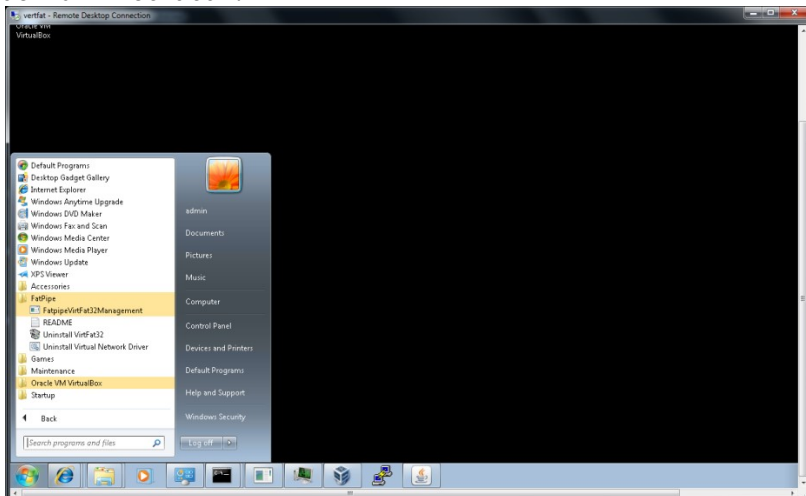


Figure B1 - Start Menu

Operating Procedure

Please run the management software "VirtFat32.exe" from the Start Menu when you are logged in as Administrator. This must be the same Administrator account you used when you installed the software.

The management software has multiple tabs. The first tab, named Info, shows information about the virtual machine's state (see figure B2).

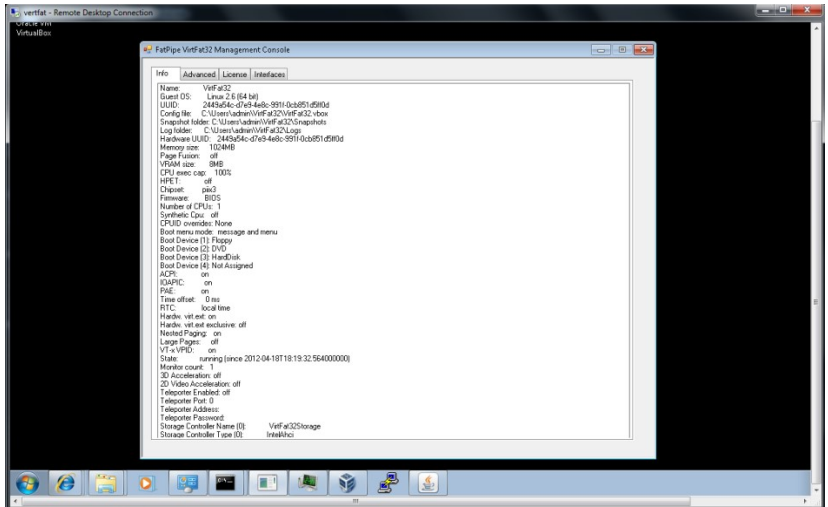


Figure B2 - The management software "VirtFat32.exe"

FatPipe uses Oracle Virtual Machine to run its virtualized platform (Oracle VirtualBox is copyright Oracle, Inc. all rights reserved to Oracle, Inc.). The "VirtualBox" binaries are GPLv2 licensed. No extension pack is used. For a copy of the license terms, please follow the link to Oracle VirtualBox license and distribution terms:

<https://www.virtualbox.org/wiki/Downloads>

A PDF document is available detailing the license terms for version 4.1.12, which FatPipe is using in its software. Any changes in license from Oracle will be reflected in newer releases of FatPipe software.

Note: Please download Adobe Acrobat Reader to read this pdf file.

Launching the Management Software

Click on the Start Menu and navigate to the FatPipe folder. Expand it and you will see the FatPipe Management Console (see Figure B3).

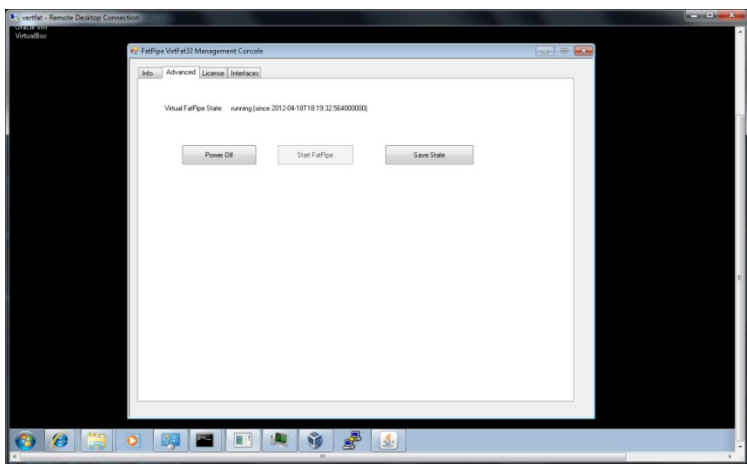


Figure B3 - Shows the state of the Virtual FatPipe Machine

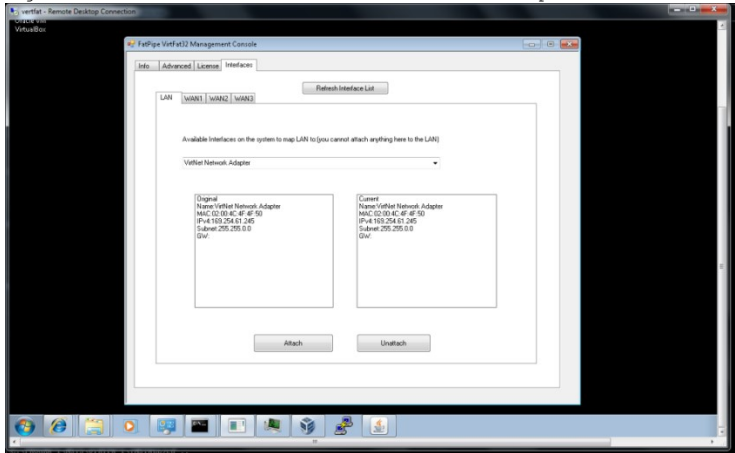


Figure B4 - Interfaces tab

System Requirements for Virtual FatPipe Disk (only)

- 64 bit or 32 bit host operating system
- Supported virtual machine platforms are "Oracle VirtualBox 4X", "VMware Player 2X", "VMware Server 2X", "VMware ESXi 5.0"
- At least 2 GB of RAM dedicated to the virtual machine
- At least 2 network cards Ethernet (preferably Intel based network controller chipsets 825X family) based (for LAN and WAN1 of FatPipe, additional can be added)
- At least 20 GB of dedicated hard drive space reserved for the virtual machine
- CPU must support Intel® VT-x virtualization technology (for 32 bit hosts only) as we are running a 64 bit Operating System on a 32 bit host operating system. Below link is a list of Intel processors that support virtualization technology.
(<http://ark.intel.com/VTList.aspx>)

•

Configuring FatPipe Virtual Machine with WAN Acceleration Using a Private WAN Address

Installing the FatPipe Virtual Machine for WAN acceleration with a public IP address is no different than installing a regular MPVPN with WAN Acceleration. However, there a slight change in configuration when the FatPipe Virtual Machine uses a private address on the WAN interface (for example, behind a DSL router).

To configure a private WAN address, go to Interfaces in the main menu, and click on the WAN interface that will use the private IP. Enter the private WAN address information under IPv4 or IPv6 section on the right hand side.

Home

Interfaces

LAN

WAN 1

WAN 2

WAN 3

WAN 4

WAN 5

System

Load Balancing

Routing

Tools

Orchestration

EnterpriseView

WAN 1

Site:

Interfaces

Line Status

DOWN

ISP Name

ISP Notes

WAN IP Settings

Obtain an IP address automatically using DHCP

Connect using PPPoE

Connect using 3G / 4G device

Specify an IP address

IPv4

IPv6

IP Address

Subnet Mask

Default Gateway

Bandwidth (Kbps)

Upload

Download

Line Type

Public

Private

Services

Ping

Remote Management

DNS

IPSEC

SNMP

Site Load Balancing

SSH

WAN Metrics*

Route Test

Perform

Always

Link Stabilizing Factor

1

Link Stabilizing Factor

Downs

1

Ethernet

MAC [aa:36:9f:54:fb:f6]

Link Speed / Duplex Mode

Auto Negotiation

Current Negotiation: NO LINK

VLAN

Enable

ID 0

Enable Bridging with LAN**

Type

Weight

1

Usage

Primary

Spillover Priority

1

Save

Refresh

*A public IP is pinged to identify latency, jitter & packet loss on that link.

**You will need to clear ARP on connected devices when enabling/disabling bridge mode.

Figure B5 - WAN Configuration using a private address

VPN Setup

When you setup the VPN between the virtual machine using a private address on the WAN interface and the FatPipe at the remote end using public addressing, the VPN setup on the remote end needs to reference the public address that sits in front of the virtual machine with the private address (see Figure B6).

To view the VPN Policy Rule page, click on the Routing button in the main menu and select the VPN Tab and then click ADD. Follow the instructions on how to setup a VPN listed in Chapter 10. When using a private IP address on the FatPipe WAN Interface, enter the private IP address from the remote FatPipe interface in the Remote FatPipe IP box listed on the bottom left hand side of the Edit VPN Policy Rule.

The public address of the device that sits in front of the remote FatPipe (such as a router) should be entered into the External IP field of the Remote Info box located on the right hand side of the Edit VPN Policy Rule page.

Add/Edit VPN Policy Rule

☐ Template

Tunnel Name

NAEAST_Rockville

Remote End

☒ Network ☐ User

Encryption

AES128

Authentication

SHA1

NAT-T

☒ Auto ☐ Forced

☐ Custom Ports

IKE Port

Numeric

Encapsulated UDP Port

Numeric

Other

TCPMSS

1372

DPD Timeout

120

DPD Delay

30

PFS

☐

Local Info

☐ Local LAN Networks

Network IP Address/Mask

192.168.0.0/24

External IP

41.1.1.2

Add

Edit

Delete

Remote Info

☐ Remote LAN Networks

Network IP Address/Mask

192.168.2.0/24

Key Management

☒ Pre-Shared Secret ☐ RSA Signature ☐ RSA Certificates

Pre-Shared Key

Figure B6 - VPN Policy Page

Technical Support

For technical support on FatPipe products, please contact FatPipe Networks directly by calling (800) 724-8521 or (801) 281-3434. Press number three (3) for Technical Support. Standard Support is available Monday through Friday, 8:00am to 6:00pm MST. Extended Support is available 24/7. You can schedule installations and upgrades outside the standard Technical Support hours with the FatPipe Technical Support team. You may visit our website, www.FatPipeinc.com, for answers to the most Frequently Asked Questions (FAQs). You can also reach support via e-mail atsupport@FatPipeinc.com.

Contact FatPipe Networks' Technical Support team for more detailed information regarding Support options. FatPipe Networks does not charge for standard Technical Support for the first 90 days from the purchase date. Feature enhancements and version upgrades are available with a support agreement package.

FatPipe					Networks
4455	South	700	East,	First	Floor
Salt Lake City, UT 84107					
Telephone: (800) 724-8521 or (801) 281-3434 ext. 3					
Fax: (801) 281-0317					
E-Mail: support@FatPipeinc.com					
Web Page: http://www.FatPipe.com					

FatPipe Product Warranty

©2000 - 2015 FatPipe Networks™, Inc. All rights reserved. Patents existing and patents pending in the U.S.A. and elsewhere. FatPipe, the FatPipe logo, Fat Pipe™, MPVPN®, and SmartDNS™ are trademarks or registered trademarks of Ragula Systems Development Company d.b.a. FatPipe Networks. Windows® is a registered trademark of Microsoft Corporation. All other companies and products names are trademarks of their respective companies. All specifications are subject to change without notice.

FatPipe Networks makes no warranty, either expressed or implied, for the hardware enclosed herein UNLESS the Warranty Registration Card, which accompanies this product, has been filled out and returned to FatPipe Networks. With the return of the Warranty Registration Card, FatPipe Networks warrants its hardware products to the original purchaser against defects in materials and workmanship for one year from shipment, as long as the product is used in its original installation.

If you discover a defect, FatPipe Networks will at its option repair or replace the purchase price of the product at no charge to you, provided it is returned during the warranty period. Transportation charges will be prepaid to FatPipe Networks.

Returns

To return a unit to FatPipe Networks for repairs, please contact the Customer Service Department at FatPipe Networks to get a Return Merchandise Authorization Number (RMA#). You must write this number on the outside of the package where it can easily be seen. No unit will be accepted without an RMA number. Also, please enclose your name, address, telephone number and a description of the problem.

Warranty Limitations

The warranty applies only to the hardware products and is not transferable. The warranty does not apply if: (1) the product has been damaged by accident, abuse, misuse or misapplication, or has not been operated in accordance with the procedures described in this and/or accompanying manuals; (2) the product has been altered or repaired by

someone other than FatPipe Networks Customer Service personnel; or (3) any serial number has been removed, defaced or in any way altered. FatPipe Networks may use remanufactured, refurbished or used parts and modules in making warranty repairs.

WARRANTIES EXCLUSIVE

IF A FATPIPE PRODUCT DOES NOT OPERATE AS WARRANTED ABOVE, CUSTOMER'S SOLE REMEDY FOR BREACH OF THAT WARRANTY SHALL BE REPAIR OR REPLACEMENT, AT FATPIPE'S OPTION. TO THE FULL EXTENT ALLOWED BY LAW, THE FOREGOING WARRANTIES AND REMEDIES ARE EXCLUSIVE AND ARE IN LIEU OF ALL OTHER WARRANTIES, TERMS, OR CONDITIONS, EXPRESSED OR IMPLIED, EITHER IN FACT OR BY OPERATION OF LAW, STATUTORY OR OTHERWISE, INCLUDING WARRANTIES, TERMS, OR CONDITIONS OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND SATISFACTORY QUALITY. FATPIPE NEITHER ASSUMES, NOR AUTHORIZES ANY OTHER PERSON TO ASSUME FOR IT, ANY OTHER LIABILITY IN CONNECTION WITH THE SALE, INSTALLATION, MAINTENANCE OR USE OF ITS PRODUCTS.

FATPIPE SHALL NOT BE LIABLE UNDER THIS WARRANTY IF ITS TESTING AND EXAMINATION DISCLOSE THAT THE ALLEGED DEFECT IN THE PRODUCT DOES NOT EXIST OR WAS CAUSED BY CUSTOMER'S OR ANY THIRD PERSON'S MISUSE, NEGLIGENCE, IMPROPER INSTALLATION OR TESTING, UNAUTHORIZED ATTEMPT TO REPAIR OR MODIFY, OR ANY OTHER CAUSE BEYOND THE RANGE OF THE INTENDED USE, OR BY ACCIDENT, FIRE, LIGHTNING, OR OTHER HAZARD.

LIMITATION OF LIABILITY

TO THE FULL EXTENT ALLOWED BY LAW, FATPIPE ALSO EXCLUDES FOR ITSELF AND ITS SUPPLIERS ANY LIABILITY, WHETHER BASED IN CONTRACT OR TORT (INCLUDING NEGLIGENCE), FOR INCIDENTAL, CONSEQUENTIAL, INDIRECT, SPECIAL, OR PUNITIVE DAMAGES OF ANY KIND, OR FOR LOSS OF REVENUE OR PROFITS, LOSS OF BUSINESS, LOSS OF INFORMATION OR DATA, OR OTHER FINANCIAL LOSS ARISING OUT OF OR IN CONNECTION WITH THE SALE, INSTALLATION, MAINTENANCE, USE, PERFORMANCE, FAILURE, OR INTERRUPTION OF ITS PRODUCTS, EVEN IF FATPIPE OR ITS AUTHORIZED RESELLER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES, AND LIMITS ITS LIABILITY TO REPAIR, REPLACEMENT, OR REFUND OF THE PRICE PAID, AT FATPIPE'S OPTION. THIS LIMITATION OF LIABILITY FOR DAMAGES WILL NOT BE AFFECTED IF ANY REMEDY PROVIDED HEREIN SHALL FAIL OF ITS ESSENTIAL PURPOSE.

DISCLAIMER

Some countries, states, or provinces do not allow the exclusion or limitation of implied warranties or the limitation of incidental or consequential damages for certain products supplied to consumers or the limitation of liability for personal injury, therefore the above limitations and exclusions may be limited in their application to you. When the implied warranties are not allowed to be excluded in their entirety, they will be limited to the remainder of the applicable written warranty. This warranty gives you specific legal rights, which may vary depending on local law.

GOVERNING LAW

This Limited Warranty shall be governed by the laws of the State of Utah, U.S.A. excluding its conflicts of laws principles and excluding the United Nations Convention on Contracts for the International Sale of Goods.

FatPipe Networks End User

Software License Agreement

IMPORTANT:Read Before Using This Product

YOU SHOULD CAREFULLY READ THE FOLLOWING TERMS AND CONDITIONS BEFORE USING THIS PRODUCT. IT CONTAINS SOFTWARE, THE USE OF WHICH IS LICENSED BY FATPIPE NETWORKS (FATPIPE) TO ITS END USERS FOR THEIR USE ONLY AS SET FORTH BELOW. IF YOU DO NOT AGREE TO THE TERMS AND CONDITIONS OF THIS AGREEMENT, DO NOT USE THE SOFTWARE. USING ANY PART OF THE SOFTWARE INDICATES THAT YOU ACCEPT THESE TERMS.

LICENSE

FatPipe grants you (Customer) a nonexclusive, nontransferable license, or in the case of Third Party software (third party owned software with which party FatPipe has a distributorship agreement), sublicense, to use the Licensed Products (FatPipe software and Third Party software) on a single authorized device for which they were acquired. Spam Police runs on a different, single authorized device.

The Licensed Products are the property of FatPipe or, in the case of Third Party software, the owner with whom FatPipe has a distributorship agreement. You agree, that you will not, unless you have the prior written permission of FatPipe: (a) attempt to recreate or modify or allow others to attempt to recreate or modify the source or object code of Licensed Products or make any changes to any accompanying documentation; (b) reverse engineer or create derivative works from the Licensed Products or related documentation; (c) copy or transfer the Licensed Products or related documentation to any other party; or (d) remove any proprietary notices, labels or marks fixed to the Licensed Products by FatPipe or its suppliers. This license does not give you any rights to patents, copyrights, trade secrets, trademarks, or any other rights to the Licensed Products except as contained herein.

TRADE SECRETS

You acknowledge and agree that the structure, sequence and organization of the Licensed Products are the valuable trade secrets of FatPipe or, in the case of Third Party software, the owner with whom FatPipe has a distributorship agreement. You agree to hold such trade secrets in confidence.

WARRANTIES

FatPipe represents and warrants that FatPipe software does not infringe any patent, copyright, trademark or trade secret rights of any third party. This warranty does not extend to any Third Party software.

FatPipe and its licensors provide Software "as is" and expressly disclaim all warranties, conditions or other terms, whether express, implied or statutory, including without limitation, warranties, conditions or other terms regarding merchantability, fitness for a particular purpose, design, condition, capacity, performance, title, and non-infringement. FatPipe does not warrant that the Software will operate uninterrupted or error-free or that all errors will be corrected. In addition, FatPipe does not warrant that the Software or any equipment, systems or network on which the Software is used will be free of vulnerability to intrusion or attack.

LIMITATION OF LIABILITY

EXCEPT FOR THE EXPRESS WARRANTIES CONTAINED ABOVE, FATPIPE MAKES NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, IN FACT OR IN LAW, INCLUDING MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR WARRANTIES THAT THE LICENSED PRODUCTS ARE ERROR FREE OR THAT THEIR USE WILL BE UNINTERRUPTED. TO THE FULL EXTENT ALLOWED BY LAW, FATPIPE ALSO EXCLUDES FOR ITSELF AND ITS SUPPLIERS ANY LIABILITY, WHETHER BASED IN CONTRACT OR TORT, FOR INCIDENTAL, CONSEQUENTIAL, INDIRECT OR SPECIAL DAMAGES OR FOR LOSS OF REVENUE OR PROFITS, LOSS OF BUSINESS, LOSS OF INFORMATION OR DATA, AND LIMITS ITS LIABILITY TO REPAIR,

REPLACEMENT, OR REFUND OF THE PURCHASE PRICE PAID, AT FATPIPE'S OPTION.

TERMINATION

Either party may terminate this license immediately upon the occurrence of any of the following events: (a) the other party has failed to cure a breach of this Agreement within thirty (30) days after receiving written notice thereof: (b) the other party institutes proceedings under bankruptcy or insolvency laws: (c) either party ceases to conduct business or to conduct the business relevant hereunder. In addition, FatPipe shall be entitled to terminate this Agreement immediately upon discovering any breach by you of any of your obligations under the License language herein.

OBLIGATIONS UPON TERMINATION

Your license to use Licensed Products is and shall be automatically and immediately revoked. You shall immediately cease use of the Licensed Products. You shall pay any current or past due invoices arising out of the performance or provision of services under this Agreement.

EXPORT RESTRICTIONS

You agree that you will not export the Licensed Products in violation of any applicable laws or regulations of the United States and/or the country where you obtained them.

EFFECT OF AGREEMENT

This Agreement embodies the entire understanding between the parties and supersedes any and all prior understandings, oral or written proposals and other communication.

ASSIGNMENT

This Agreement is binding on successors and assigns of the parties. However neither this Agreement nor any part of it shall be assigned, sublicensed, or otherwise transferred by you without FatPipe's prior written consent.

GOVERNING LAW

This Agreement shall be governed by the laws of the State of Utah, U.S.A. and subject to the jurisdiction of the courts therein.